Matter, Then On Down

One coordinate from bulk matter to the nucleus

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Abstract

Matter has no absolute size. The zero and the span of every operator description are free, and the one place scale returns is a logarithmic boundary count whose coefficient is fixed by the central charge, $A = 2c/\pi^2$ for an interval. This paper follows that boundary downward through one coordinate, $\Gamma = (z-1)/(z+1) = \tanh(\frac{1}{2}\ln z)$, which reads a ratio at every scale: occupation odds at a cut of bulk matter, successive removal energies of an atom, populations of a gas, separation energies of a nucleus, and the complex odds of opened, non-Hermitian matter. The boundary census is the gauge without its sign; the signed offset $\tau = \log\det(I-C) - \log\det C$ is its clock face. The weight between, the cross block $X_A = C_{AB}C_{BA}$, is positive, bounded by $\frac{1}{4}$, two-sided, stable under perturbation with a certified margin, bounded above by four times the Gaussian mutual information, and recovers the occupation census exactly on a pure state. The offset moves no faster than $\dot{\tau}^2 \le 4gS$, and two preparations with the same census, offset and coupling can saturate that bound in opposite directions. The same chart is a reflection chart: a step in spectral gap reflects with the peel coordinate $\tanh(\frac{1}{2}\ln\Delta_2/\Delta_1)$, a matched load sits at the centre, total reflection on the rim, and the outside record of a Hawking fermion pair at $\Gamma = \tanh(\hbar\omega/2k_BT_H)$. Helium's two thresholds give $\Gamma_0 = 0.377575$; an anchored threshold vector is an exact coordinate change of the ladder; a finite Saha gas has one neutral state, and its populations carry the peel back out once the partition functions and $n_e\lambda_e^3$ are supplied. The unbound nuclei ^{5}He, ^{5}Li and ^{8}Be sit outside the disk at $|\Gamma| = 1.074$, 1.220, 1.025, at a rim distance $2s/(S_{\text{next}} - s)$ exactly. Finite present records, from a moment list to density, flow and stress, do not fix later response, and exact witnesses show which distinction is lost and where it returns. Opened matter leaves the disk before its critical point; on the PT-symmetric chain the opened band metric is Lorentzian exactly where $E(k)^2 > 0$ and $\cos k \neq 0$, with $\det Q = -u^2w^2\cos^2 k/(16E(k)^6)$, and the exit line is $E(0)E(\pi) = u^2$, a band-averaged Petermann factor of 2 and a quarter turn of the continued endpoint offset, stated with its filter, partition and counterexamples. Every claim carries a grade and every number a script.

Grades. [T] proved here (or proved in Appendix A from the author's earlier papers); [L] checked in Lean (Appendix B); [V] verified numerically with stated controls; [R] a reproduction of a known result in this coordinate; [C] conjecture; [H] a reading; [lit] comparison with other people's published data; [S] a structural comparison between different objects that share a form, with no identification of the objects.

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1 Matter has no absolute size

Every instrument needs a zero and a span. In the operator framework of *Matter at a Scale* (v5) and *Foundational Operator Field Theory*, the zero is free and so is the span: difference observables are unchanged by $K \mapsto K + cI$, and spectral projectors, principal angles and the infrared boundary census are unchanged by $K \mapsto cK$ with $c > 0$ [T] (A.1). Within this framework only ratios are observable, and the equivalence that says so is the affine one, $K \mapsto cK + dI$ with $c > 0$, carried together with the sector it fills: a shift moves every eigenvalue together, so a fixed numerical Fermi level must move with it.

Two consequences fix how a ratio may be read. First, the ratio must be taken between quantities that have a physical zero. The eigenvalues $(1, 2)$ have ratio 2 and coordinate $\Gamma = 1/3$; shifted by $-\frac{5}{4}$ they become $(-\frac{1}{4}, \frac{3}{4})$, ratio -3 and $\Gamma = 2$, with the same gap [T]. Removal energies are measured from the continuum and separation energies from the next threshold, so their ratios mean something; a raw eigenenergy ratio does not. Second, the invariance belongs to the generator. Rescaling the generator leaves the filled projector, and so the correlation matrix C, unchanged; rescaling C itself is a different operation, under which the cross block of §2.2 scales as $X \mapsto a^2 X$ and a census taken at a fixed numerical threshold changes.

The single failure is ultraviolet. For a region A with $|\partial A|$ boundary points in a substrate of correlation length ξ and central charge c, the number N of eigenvalues of the restricted correlation matrix in $[\varepsilon, 1 - \varepsilon]$ grows as

$$N = |\partial A| \, \frac{c}{\pi^2} \, \log \frac{\xi}{a} \, \log \frac{1 - \varepsilon}{\varepsilon} + \dots \tag{1}$$

so a two-endpoint interval carries $A = 2c/\pi^2$ (A.2: an asymptotic derivation from an equally spaced entanglement ladder and the Calabrese–Cardy entropy, exact given those two inputs) [T]. The window enters as $\Lambda = \log((1 - \varepsilon)/\varepsilon)$, which becomes $\log(1/\varepsilon)$ only for small ε. The coefficient was measured at $c = 1$ as 0.202692 against $2/\pi^2 = 0.202642$. At $c = \frac{1}{2}$, on the transverse-field Ising chain, two correction families for the interval entropy coefficient give 0.1693 and 0.1586, bracketing $1/6$ from both sides, and the census coefficient reads 0.103689, 2.3% above $1/\pi^2$ and 41.8 times closer to it than to the $c = 1$ value [V] (*Matter at a Scale*, ISING-A-03 and ROBUST-01). *The Angle Between the Rails* derives the same coefficient from the spacing $\Delta = 4\pi K(\tilde{u})/K'(\tilde{u})$ of the entanglement ladder, $N = 4\Lambda/\Delta$ for two cuts [T]. A rescaling of the substrate is registered only as a running, never as a power.

The physical reading of *Matter at a Scale* is the starting point of everything below: matter is a gap wearing a boundary. A gapped substrate keeps a finite, area-law census; gas, the gapless phase, floods it.

2 The boundary keeps a ledger

2.1 Census, offset and one chart

Let C be the one-body correlation matrix of a region, with eigenvalues $\nu \in (0, 1)$. Each level has odds $z = (1 - \nu)/\nu$ and a rapidity $\varepsilon = \ln z$. Write $\Gamma = (z - 1)/(z + 1)$.

Proposition 2.1 (the bridge)**.** $\Gamma = \tanh(\varepsilon/2) = 1 - 2\nu$*, and the window* $\nu \in [\delta, 1 - \delta]$ *is the disk* $|\Gamma| \le 1 - 2\delta$*. The binary entropy integrated over the rapidity is* $\int_{-\infty}^{\infty} H(\nu(\varepsilon))\, d\varepsilon = \pi^2/3$*. [T][L]*

Proof. With $z = (1-\nu)/\nu$: $z-1 = (1-2\nu)/\nu$ and $z+1 = 1/\nu$, so $\Gamma = 1-2\nu$. Also $(e^{\varepsilon}-1)/(e^{\varepsilon}+1) = \tanh(\varepsilon/2)$. The window statement is $|1 - 2\nu| \le 1 - 2\delta$. For the integral, $\nu = 1/(1 + e^{\varepsilon})$ and

$d\nu = -\nu(1-\nu)\,d\varepsilon$, so $\int H\,d\varepsilon = \int_0^1 H(\nu)/(\nu(1-\nu))\,d\nu = 2\int_0^1 (-\ln\nu)/(1-\nu)\,d\nu = 2\zeta(2) = \pi^2/3$ (A.3). □

Checked to 1.1×10^{-16} and 6.7×10^{-15} (script `s01`). The identity $1-2\nu = \tanh(\varepsilon/2)$ is standard in entanglement-spectrum work; the reading is the author's. *The Offset Speed Limit* names the same function four ways at once: the Smith-chart reflection $(z-1)/(z+1)$, the two-level Gibbs population gap $\tanh(\beta\Delta/2)$, the asymmetry $1-2\nu$ of an occupation against its log-odds, and the peel coordinate of VOTKP. Section 4 adds a fifth: the reflection of a wave at a change of spectral gap.

Summed over the levels of the region, the rapidity is the author's signed offset,

$$\tau = \sum_\nu \varepsilon_\nu = \mathrm{Tr}[\log(I-C) - \log C] = \log\frac{\det(I-C)}{\det C} = \log\frac{P(N_A=0)}{P(N_A=|A|)}, \tag{2}$$

the log-odds of the region being empty against full (*The Offset Belongs to the Boundary*, v31, Theorem 2). The last equality holds for a number-conserving Gaussian (quasi-free) fermion state, where the reduced state is a product over the eigenmodes of C (A.4) [T]; for a general state the one-body matrix does not fix the configuration probabilities, and (2) is read as the trace formula alone. The census counts levels inside a window of $\nu(1-\nu)$, and $\nu(1-\nu) = (1-\Gamma^2)/4$ is blind to the sign of Γ. In the author's words: "The offset is the clock face; the census is the gauge without its sign." The matrices $\mathrm{diag}(0.2, 0.8)$ and $\mathrm{diag}(0.2, 0.2)$ have the same defect $C - C^2 = 0.16I$, so every census window counts them alike; their offsets are 0 and $\log 16$.

Theorem 2.2 (the Cayley domain, branch explicit)**.** *For $z \in \mathbb{C}\setminus\{-1\}$, $|z+1|^2 - |z-1|^2 = 4\,\mathrm{Re}\,z$, so $|\Gamma| < 1 \iff \mathrm{Re}\,z > 0$, $|\Gamma| = 1 \iff \mathrm{Re}\,z = 0$, $|\Gamma| > 1 \iff \mathrm{Re}\,z < 0$. For $z = e^W$ the Cayley expression is $\tanh(W/2)$, and $|\tanh(W/2)| < 1 \iff \cos(\mathrm{Im}\,W) > 0$; on the principal branch $\mathrm{Im}\,W \in (-\pi, \pi]$ this is $|\mathrm{Im}\,W| < \pi/2$. [T][L]*

Proof. Expand the squared moduli and divide their difference by $|z+1|^2 > 0$; then $\mathrm{Re}\,e^W = e^{\mathrm{Re}\,W}\cos(\mathrm{Im}\,W)$. Cosine is positive on the principal branch exactly in the open interval stated. □

Three statements are kept apart. A negative real ratio has principal phase $\mathrm{Im}\,W = \pi$, a half turn, and lies outside the disk. Only the single ratio $z = -1$ is the Cayley pole, where Γ is undefined rather than large; $z = -2$, for instance, gives the finite $\Gamma = 3$. And a complex ratio does not by itself leave the disk: $z = 1 - i\gamma$ gives $|\Gamma|^2 = \gamma^2/(4+\gamma^2) < 1$ for every γ. Leaving needs a ratio whose real part changes sign. Without the branch restriction the interval statement is false, since the criterion is periodic in $\mathrm{Im}\,W$ modulo 2π.

2.2 The weight between

Split the space into the region A and the rest B, of dimensions a and b, and let $C = C^*$ with $0 \le C \le I$. Write

$$C = \begin{pmatrix} A_C & B_C \\ B_C^* & D_C \end{pmatrix}, \qquad F_A = A_C - A_C^2, \qquad T_A = (C-C^2)_{AA}, \qquad X_A = B_C B_C^* = C_{AB}C_{BA},$$

and $X_B = B_C^* B_C$. Fix $0 < \varepsilon < \frac{1}{2}$ and $\theta = \varepsilon(1-\varepsilon)$. The *raw census* $N_\varepsilon^{\mathrm{raw}} = \mathrm{rank}\,\mathbf{1}_{[\varepsilon,1-\varepsilon]}(A_C)$ counts intermediate local occupations; the *boundary census* $N_\varepsilon^{\partial} = \mathrm{rank}\,\mathbf{1}_{[\theta,\infty)}(X_A)$ counts squared cross-block singular values at or above θ. Counts include multiplicity and equality at the threshold, and the partition is part of the definition.

Theorem 2.3 (the weight between)**.** *For every such C:*

1. *(defect split) $F_A = T_A + X_A$. This needs only block multiplication; with $C^2 = C$ it gives $X_A = A_C - A_C^2$ with no Hermiticity used. [T][L]*
2. *(order and size) $0 \le X_A \le F_A \le \frac{1}{4} I_A$, $0 \le T_A \le F_A$, and $\|B_C\| \le \frac{1}{2}$, attained by $C = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. [T][L]*
3. *(two-sided) X_A and X_B have the same positive eigenvalues with multiplicity, so $N_\varepsilon^\partial(A) = N_\varepsilon^\partial(B) \le \operatorname{rank} B_C \le \min(a, b)$ and $\operatorname{Tr} X_A = \operatorname{Tr} X_B = \|B_C\|_F^2 \le \frac{1}{4}\min(a, b)$. A finite census cannot diverge at fixed a, b; a divergent census needs a declared sequence of spaces. [T]*
4. *(comparison and recovery) $N_\varepsilon^\partial \le N_\varepsilon^{\mathrm{raw}}$, with equality when $C^2 = C$. [T]*
5. *(trace estimate) $\theta N_\varepsilon^\partial \le \operatorname{Tr} X_A = \sum_{i \in A, j \in B} |C_{ij}|^2$, and $\operatorname{Tr} X_A = 0 \iff B_C = 0$. A census of zero allows nonzero strengths below θ; it is not a certificate of no correlation. [T][L]*
6. *(local unitaries) For $U = U_A \oplus U_B$ unitary, $C \mapsto UCU^*$ conjugates A_C, X_A, T_A by U_A and leaves both censuses and every displayed spectrum and trace unchanged. A unitary that mixes the regions changes the observable. [T]*

Proof. (1) The AA block of C^2 is $A_C^2 + B_C B_C^*$, so $T_A = A_C - A_C^2 - B_C B_C^*$. (2) Compression preserves order, so $0 \le A_C \le I$; the spectral theorem gives $C - C^2 \ge 0$ (eigenvalues $c(1-c) \ge 0$), so $T_A \ge 0$; $X_A = B_C B_C^* \ge 0$; (1) gives $X_A \le F_A$ and $T_A \le F_A$; and $F_A = \frac{1}{4} I - (A_C - \frac{1}{2} I)^2 \le \frac{1}{4} I$, so $\|B_C\|^2 = \|X_A\| \le \frac{1}{4}$. (3) If $B_C B_C^* u = \lambda u$ with $\lambda > 0$ then $B_C^* u \ne 0$ and $B_C^* B_C (B_C^* u) = \lambda B_C^* u$; the reverse map is B_C/λ, so the eigenspaces have equal dimension; sum and count. (4) For $\nu \in [0, 1]$, $\nu(1-\nu) - \varepsilon(1-\varepsilon) = (\nu - \varepsilon)(1 - \varepsilon - \nu)$, so $\nu(1-\nu) \ge \theta \iff \nu \in [\varepsilon, 1-\varepsilon]$ and $\operatorname{rank} \mathbf{1}_{[\theta,\infty)}(F_A) = N_\varepsilon^{\mathrm{raw}}$. By (2) and the min–max principle $\lambda_j(X_A) \le \lambda_j(F_A)$ for each ordered eigenvalue. If $C^2 = C$ then $T_A = 0$ and $X_A = F_A$. (5) Each counted eigenvalue contributes at least θ to the trace and the rest are nonnegative; $\operatorname{Tr} B_C B_C^* = \sum |(B_C)_{ij}|^2$. (6) Block multiplication; conjugation preserves spectra and trace. □

The boundary coordinate. Each eigenvalue λ of X_A lies in $[0, \frac{1}{4}]$, so

$$\Gamma_b(\lambda) = \sqrt{1 - 4\lambda} \in [0, 1], \qquad \lambda = \frac{1 - \Gamma_b^2}{4}, \qquad \lambda \ge \varepsilon(1-\varepsilon) \iff \Gamma_b \le 1 - 2\varepsilon \tag{3}$$

is a real, unsigned coordinate [T]. On a pure state, item (4) identifies $\lambda = \nu(1-\nu)$ and $\Gamma_b = |1 - 2\nu|$: the boundary coordinate is the modulus of the census point, not the signed point; the map forgets the difference between ν and $1 - \nu$ [T][L]. For mixed states X_A need not commute with A_C, its eigenvalues cannot in general be paired level by level with $\nu(1-\nu)$, and the order comparisons of Theorem 2.3 are what remain.

One block, many jobs. The same positive square of an off-diagonal block appears wherever a space is cut [T][S] (*Beyond the Event Horizon*, ed. 10, §5.4). For an orthogonal projector $P = \begin{pmatrix} C & B \\ B^\dagger & C' \end{pmatrix}$, $C - C^2 = BB^\dagger$ and $\operatorname{Tr}(C^2 - C) = -\operatorname{Tr} BB^\dagger \le 0$: the boundary census of *Light Keeps the Ledger* and the negative quadratic form of the gravity edition. For a Hamiltonian $H = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}$, the memory kernel $K(t) = Be^{-iDt}B^\dagger$ has footprint $K(0) = BB^\dagger$ (§8), and

$PH^2P - (PHP)^2 = PHQHP = BB^\dagger$ is exit and return. Applied to the correlation matrix, the block is X_A; applied to the generator, $\|PH(1-P)\|_F^2 = E_H$ is the coupling energy of *The Weight Between*. These are the same functional on different operators and different cuts. They are not one quantity, and §2.6 shows a pair of states where they differ by a factor that grows without bound.

2.3 Stability and a certified count

Theorem 2.4 (operator-norm stability)**.** *Let $C, \tilde{C}$ be positive contractions on the same partition, $\delta = \|C - \tilde{C}\|$. Then $\|X_A(C) - X_A(\tilde{C})\| \le \delta$, and for $\delta < \theta$,*

$$\mathcal{N}_{X_A(C)}(\theta + \delta) \le N_\varepsilon^\partial(\tilde{C}) \le \mathcal{N}_{X_A(C)}(\theta - \delta),$$

where $\mathcal{N}_Y(s) = \operatorname{rank} \mathbf{1}_{[s,\infty)}(Y)$. If $\operatorname{dist}(\theta, \operatorname{spec} X_A(C)) > \delta$ the two counts are equal. [T]

Proof. Compression gives $\|B - \tilde{B}\| \le \delta$ and Theorem 2.3(2) gives $\|B\|, \|\tilde{B}\| \le \frac{1}{2}$, so $\|BB^* - \tilde{B}\tilde{B}^*\| = \|(B-\tilde{B})B^* + \tilde{B}(B^* - \tilde{B}^*)\| \le (\|B\| + \|\tilde{B}\|)\delta \le \delta$. Weyl's inequality moves each ordered eigenvalue by at most δ; the strict margin keeps every eigenvalue on its side of θ. □

A floating-point residual is not by itself an operator-norm error bound; a certified count needs a validated bound and a margin larger than it.

Theorem 2.5 (near-projector comparison)**.** *Put $\eta = \|C - C^2\| \in [0, \frac{1}{4}]$. Then $X_A \le F_A \le X_A + \eta I_A$, hence $\mathcal{N}_{F_A}(\theta+\eta) \le N_\varepsilon^\partial \le N_\varepsilon^{\mathrm{raw}}$. If $\theta + \eta \le \frac{1}{4}$ the lower count is $N_{\varepsilon_\eta}^{\mathrm{raw}}$ with $\varepsilon_\eta = \frac{1}{2}(1 - \sqrt{1 - 4(\theta+\eta)})$; otherwise it is zero. [T]*

Proof. $0 \le C - C^2 \le \eta I$ compresses to $0 \le T_A \le \eta I_A$; apply the defect split and ordered-eigenvalue monotonicity, then solve $\nu(1-\nu) \ge \theta + \eta$. □

This measures the distance between local occupation counting and cross-boundary counting without assuming that the matrices commute, and returns item (4) at $\eta = 0$.

Proposition 2.6 (the coordinate is Hölder, and the bound is Hermitian)**.** *For $\lambda, \tilde{\lambda} \in [0, \frac{1}{4}]$, $|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le 2\sqrt{|\lambda - \tilde{\lambda}|}$, and if both radii are at least $g > 0$, $|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le (2/g)|\lambda - \tilde{\lambda}|$. Combined with Theorem 2.4, ordered boundary coordinates move by at most $2\sqrt{\delta}$. The eigenvalue step does not transfer to nonnormal matrices: $J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $J_\delta = \begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}$ differ by δ in norm while their eigenvalues move from $0, 0$ to $\pm\sqrt{\delta}$. [T]*

Proof. $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$ for $x, y \ge 0$ with $x = 1 - 4\lambda$; rationalising, $|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| = 4|\lambda - \tilde{\lambda}|/(\Gamma_b(\lambda) + \Gamma_b(\tilde{\lambda}))$. □

The square-root loss at the centre is real: moving $\lambda = \frac{1}{4}$ to $\frac{1}{4} - h$ moves the radius from 0 to $2\sqrt{h}$. A computed non-Hermitian residual therefore needs a conditioning or pseudospectral analysis before it becomes a certified disk count; idempotence does not supply one (§9).

2.4 What the local record cannot say

Proposition 2.7 (same local block, different weight between)**.** *Every positive contraction A on $\mathcal{H}_A$ has the projector extension*

$$P_A = \begin{pmatrix} A & \sqrt{A - A^2} \\ \sqrt{A - A^2} & I - A \end{pmatrix}$$

on $\mathcal{H}_A \oplus \mathcal{H}_A$, with the same AA block as the disconnected $C_0 = A \oplus (I - A)$, yet $X_A(P_A) = A - A^2$ while $X_A(C_0) = 0$. [T]

Proof. The square root commutes with A. The diagonal blocks of P_A^2 are $A^2 + (A - A^2) = A$ and $(A - A^2) + (I - A)^2 = I - A$; the off-diagonal block is $A\sqrt{A - A^2} + \sqrt{A - A^2}(I - A) = \sqrt{A - A^2}$. So $P_A^2 = P_A = P_A^*$. □

The local occupation spectrum therefore does not decide how much of its defect is boundary correlation and how much is global mixing. Three further exact examples mark the edge of the census [T]:

- *Disconnected thermal state.* $C = pI$ with $\varepsilon \le p \le 1 - \varepsilon$ has $N_\varepsilon^{\mathrm{raw}} = a$, $X_A = 0$, $N_\varepsilon^{\partial} = 0$; its Gaussian state is a product with zero mutual information and local number variance $ap(1-p)$. Every boundary coordinate is 1 while every raw coordinate is $1 - 2p$.
- *Classical correlation invisible to C.* On two modes $\rho_{\mathrm{cl}} = \frac{1}{2}|10\rangle\langle 10| + \frac{1}{2}|01\rangle\langle 01|$ has $C = \frac{1}{2}I_2$, so $X_A = 0$, yet $\mathrm{Cov}(n_1, n_2) = -\frac{1}{4}$ and mutual information $\log 2$. The Gaussian state $I_4/4$ has the same C and neither.
- *Entanglement invisible to C.* On four modes split $\{1, 2\}|\{3, 4\}$, $|\Psi\rangle = (|1100\rangle + |0011\rangle)/\sqrt{2}$ has fixed particle number two, $C = \frac{1}{2}I_4$ and $X_A = 0$, and entanglement entropy $\log 2$. No one-body operator connects the two configurations.

The census is therefore a statement about the one-body record, and it reads correlation faithfully inside the quasi-free class of §2.5, not outside it.

2.5 Number variance and mutual information

Proposition 2.8 (number covariance)**.** *For a gauge-invariant quasi-free fermion state with $C_{ij} = \langle a_j^\dagger a_i \rangle$, $N_A = \sum_{i \in A} n_i$ and $N_B = \sum_{j \in B} n_j$,*

$$\mathrm{Var}(N_A) = \mathrm{Tr}(A_C - A_C^2), \qquad \mathrm{Cov}(N_A, N_B) = -\operatorname{Tr} X_A, \qquad \mathrm{Var}(N_A) = \operatorname{Tr} T_A - \mathrm{Cov}(N_A, N_B). \quad [T]$$

Proof. For $i \neq j$ Wick factorisation with no anomalous pairing gives $\langle n_i n_j \rangle = C_{ii}C_{jj} - |C_{ij}|^2$; for $i = j$, $n_i^2 = n_i$. So $\mathrm{Cov}(n_i, n_j) = \delta_{ij}C_{ii} - |C_{ij}|^2$. Sum over $A \times A$ and $A \times B$, then use the defect split. □

On a pure Slater state $\mathrm{Var}(N_A) = \operatorname{Tr} X_A$: the weight between is the number fluctuation of the region.

Theorem 2.9 (the weight between bounds the information across the cut)**.** *For the state of Proposition 2.8, with $h(x) = -x \log x - (1-x)\log(1-x)$ and $I_G(A : B) = \operatorname{Tr} h(A_C) + \operatorname{Tr} h(D_C) - \operatorname{Tr} h(C)$,*

$$4 \operatorname{Tr} X_A \le I_G(A : B),$$

and the constant 4 is sharp. If in addition $\delta I \le C \le (1 - \delta)I$ with $0 < \delta \le \frac{1}{2}$, then $I_G(A : B) \le \operatorname{Tr} X_A/(\delta(1 - \delta))$. Consequently $N_\varepsilon^{\partial} \le I_G(A : B)/(4\varepsilon(1 - \varepsilon))$, and within this class $I_G = 0 \iff X_A = 0$. [T]

Proof. Diagonalising a quasi-free covariance gives independent modes with Bernoulli occupations, so $S(\rho) = \operatorname{Tr} h(C)$, and reduced states are quasi-free with the compressed covariances. Let $S = I_A \oplus (-I_B)$ and $C_0 = A_C \oplus D_C = \frac{1}{2}(C + SCS)$. *Trace Jensen:* if f is concave on an interval containing

the spectra of Hermitian H, K, then $\operatorname{Tr} f(tH+(1-t)K) \geq t \operatorname{Tr} f(H) + (1-t)\operatorname{Tr} f(K)$; diagonalise $tH + (1-t)K$ in an orthonormal basis u_i, apply scalar concavity to $\langle u_i, Hu_i\rangle, \langle u_i, Ku_i\rangle$, then $f(\langle u_i, Hu_i\rangle) \geq \langle u_i, f(H)u_i\rangle$, and sum. *Lower bound:* $f(x) = h(x)+2x^2$ has $f'' = 4-1/(x(1-x)) \leq 0$ on $(0,1)$, so $\operatorname{Tr} f(C_0) \geq \operatorname{Tr} f(C)$ (unitary invariance of $\operatorname{Tr} f(SCS)$), i.e. $I_G \geq 2[\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2] = 4 \operatorname{Tr} B_C B_C^*$. *Upper bound:* on $[\delta, 1-\delta]$, $g(x) = h(x)+x^2/(2\delta(1-\delta))$ is convex, so $\operatorname{Tr} g(C_0) \leq \operatorname{Tr} g(C)$, which rearranges with $\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2 = 2 \operatorname{Tr} X_A$. *Sharpness:* $C = \frac{1}{2} I_2 + r\sigma_x$ has $\operatorname{Tr} X_A = r^2$ and $I_G = 2\log 2 - 2h(\frac{1}{2}+r) = 4r^2 + O(r^4)$. The census bound combines this with Theorem 2.3(5). □

Sixty-four random covariances respect both bounds with minimum margins 9.1×10^{-6} (lower) and 5.9×10^{-5} (upper); the ratio $I_G / \operatorname{Tr} X_A$ tends to 4.000000020 on the sharp family; and on pure states with an occupation $p \to 0$ the upper ratio runs 11.3, 20.4, ... without bound, so no upper constant exists without a spectral margin [V] (script `g_gaussian_information_checks`). The bound holds for fermionic Gaussian states; it does not transfer a bosonic theorem, and outside the class the counterexamples above decide.

2.6 The speed of the record

Theorem 2.10 (the offset speed limit; *The Offset Speed Limit*, Thm 1 and Cor. 3)**.** *Let $P(t)$ be a smooth family of orthogonal projectors on $\mathbb{C}^n$, E an $n \times m$ isometry onto the region, $C = E^\dagger P E$ with $0 < C < I$, $\tau = \operatorname{Tr}[\log(I-C) - \log C]$, $g = \frac{1}{2}\|\dot{P}\|_F^2$ and $S = \operatorname{Tr}[(C(I-C))^{-1}]$. Then*

$$\dot{\tau}^2 \leq 4gS, \tag{4}$$

with equality iff $\dot{P}$ is a real multiple of $[P,[P,F]]$, $F = EWE^\dagger$, $W = -(C(I-C))^{-1}$. If $\dot{P} = -i[H,P]$ then $g = E_H = \|PH(1-P)\|_F^2$ and $|\dot{\tau}| \leq 2\sqrt{E_H S}$. [T]

The proof is A.7. The generator coupling E_H is taken on the moving projector P and the generator H; the cross weight $\operatorname{Tr} X_A$ is taken on a fixed spatial cut of C. The bound is a statement about the first, and the following exact pair shows that the second cannot stand in for it.

Proposition 2.11 (the speed budget does not choose the future)**.** *Set $\hbar = 1$, $H = \omega\sigma_x$ with $\omega > 0$, and $P_\pm(0) = \frac{1}{2}(I \pm \sigma_y)$, read through the first basis vector. Both are orthogonal rank-one projectors with $\nu = \frac{1}{2}$, $\Gamma = 0$, $\tau = 0$, $X = \frac{1}{4}$ and $E_H = \omega^2$ at $t = 0$. Under the same evolution*

$$\nu_\pm(t) = \tfrac{1}{2}(1 \pm \sin 2\omega t), \qquad \tau_\pm(t) = \mp 2 \operatorname{artanh}(\sin 2\omega t),$$

and both saturate $\dot{\tau}^2 = 4gS$ at every time while their offsets move in opposite directions. At $t = 0$, $|\dot{\tau}| = 4\omega = 2\sqrt{E_H S}$, whereas $2\sqrt{\operatorname{Tr} X \cdot S} = 2$; for $\omega > \frac{1}{2}$ the cross weight would understate the speed. [T][V]

Proof. $e^{-i\omega t\sigma_x}$ rotates the Bloch vector $(0, \pm 1, 0)$ in the y–z plane at rate 2ω, so $\nu_\pm = \frac{1}{2}(1 + r_z)$ is as stated and $\tau = \log((1-\nu)/\nu)$ follows. The state stays orthogonal to the x axis, so $E_H = \omega^2$ throughout and $\|\dot{P}\|_F^2 = \frac{1}{2}|\dot{r}|^2 = 2\omega^2$, $g = \omega^2$. With $s = \sin 2\omega t$, $S = 4/(1-s^2)$ and $\dot{\tau} = \mp 4\omega \cos(2\omega t)/(1-s^2)$, so $\dot{\tau}^2 = 16\omega^2/(1-s^2) = 4gS$. □

The two preparations share census, offset, cross weight and coupling; they differ in the relative phase to the fixed generator, the same datum that decides the matched-boundary example $C_\pm = \frac{1}{2}I \pm r\sigma_y$ (B12 of the formal development; here at $r = \frac{1}{2}$). Script `s13` checks saturation to 1.4×10^{-13} over three frequencies, both signs and 31 times, and confirms $|\dot{\tau}| = 4, 8 > 2$ at $\omega = 1, 2$. If $PH(1-P) = 0$ the projector, and with it the offset, is frozen in that evolution: no weight between in the generator, no motion of the record. A frozen offset is a statement about this motion; it does not say the record holds no memory.

2.7 Warm matter: heat moves the gauge, the cut keeps its strengths

At finite temperature $C^2 \neq C$. The first term of the defect split is mixing inside the region; only X_A crosses the cut. Counting $\lambda(X_A) \geq \varepsilon(1-\varepsilon)$ instead of $\nu \in [\varepsilon, 1-\varepsilon]$ separates the two. On a ring of 128 sites (antiperiodic, $\mu = 0$, half-ring region, $\varepsilon = 0.1$) the raw count rises from 2 to 24 to 64 as T goes $0 \to 0.5 \to 2$, while the boundary count goes $2 \to 0 \to 0$; disconnected sites $C = 0.3I$ give raw 32, boundary 0 [V] (script `s07`, Figure 1, left). A boundary count of zero means every cross-cut strength is below the threshold, not that the cut is uncorrelated: $\operatorname{Tr} X_A = 0.170835$ in the warm uniform ring at $T = 0.5$, and the boundary strengths themselves change with temperature. Heat separates local mixing from cross-cut correlation; it does not leave the cut untouched. The disconnected control is the exact case $X_A = 0$, $\Gamma_b = 1$ for every level.

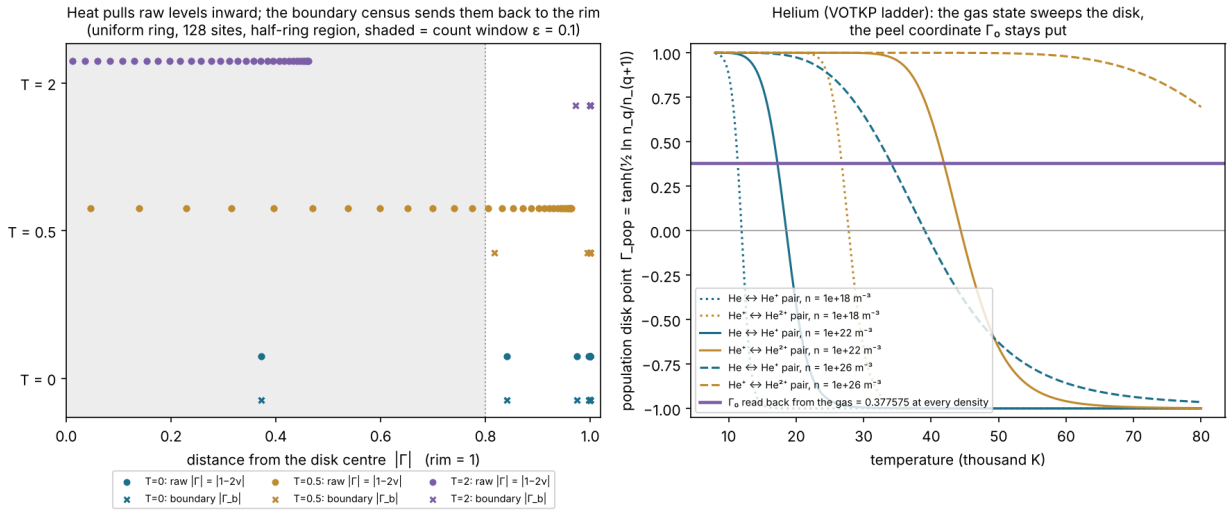


Figure 1: Left: heat pulls raw census levels toward the centre of the disk; the boundary coordinate $|\Gamma_b|$ counts only strengths that cross the cut, and a zero count at $T = 0.5$ still has $\operatorname{Tr} X_A = 0.170835$ below the window. Right: helium's charge-state pairs sweep the disk as the gas changes, while Γ_0 read back from the gas stays at 0.377575 (ideal Saha model, ground-state partitions).

3 One chart for all of it

Theorem 3.1 (peel steps add like velocities; VOTKP Theorem 15.1)**.** *For a positive ladder $I_0, \ldots, I_n$ with steps $\Gamma_k = \tanh(\frac{1}{2}\ln(I_{k+1}/I_k))$, the composite of a block of steps is $\Gamma_{j\to l} = \Gamma_j \oplus \Gamma_{j+1} \oplus \cdots \oplus \Gamma_{l-1}$, with $u \oplus v = (u+v)/(1+uv)$. [T][L]*

Proof. $\ln(I_l/I_j) = \sum_k \ln(I_{k+1}/I_k)$, and $\tanh(a+b) = \tanh a \oplus \tanh b$ (A.8). □

This is Einstein's addition of collinear velocities; the half-rapidities add. On the Smith disk read as the Poincaré disk the peel is a walk along a geodesic diameter whose hyperbolic lengths, $2\operatorname{artanh}|\Gamma|$, add (VOTKP Corollary 15.2) [T]. The composite forgets the order of its steps; the resolved ladder keeps it.

Matter's census ladder is a peel ladder. On the transverse-field Ising chain ($c = \frac{1}{2}$) the single-particle entanglement energies are odd multiples $\varepsilon_0, 3\varepsilon_0, 5\varepsilon_0, \ldots$ of one spacing (*Matter at a*

Scale, LADDER-C12-01). By Theorem 3.1 the levels then satisfy $\Gamma_{j+1} = \Gamma_j \oplus \tanh \varepsilon_0$: one constant peel step. Measured residuals are 8.0×10^{-6}, 5.5×10^{-6}, 3.2×10^{-6} at $\xi = 20.5$, 50.5, 125.5, and the ladder's own uniformity error falls 0.022, 0.011, 0.0062 [V] (script `s02`). Near the rim the disk compresses differences, so the informative number is the uniformity error in rapidity.

Figure 2 places every reading of this paper on the one disk at its signed value. Real ratios sit on the real diameter; the bound nuclei, read in VOTKP's order $z = S_{\text{next}}/S_{\text{first}}$, have $z < 1$ and so lie on the negative half, and the unbound ones lie past $+1$.

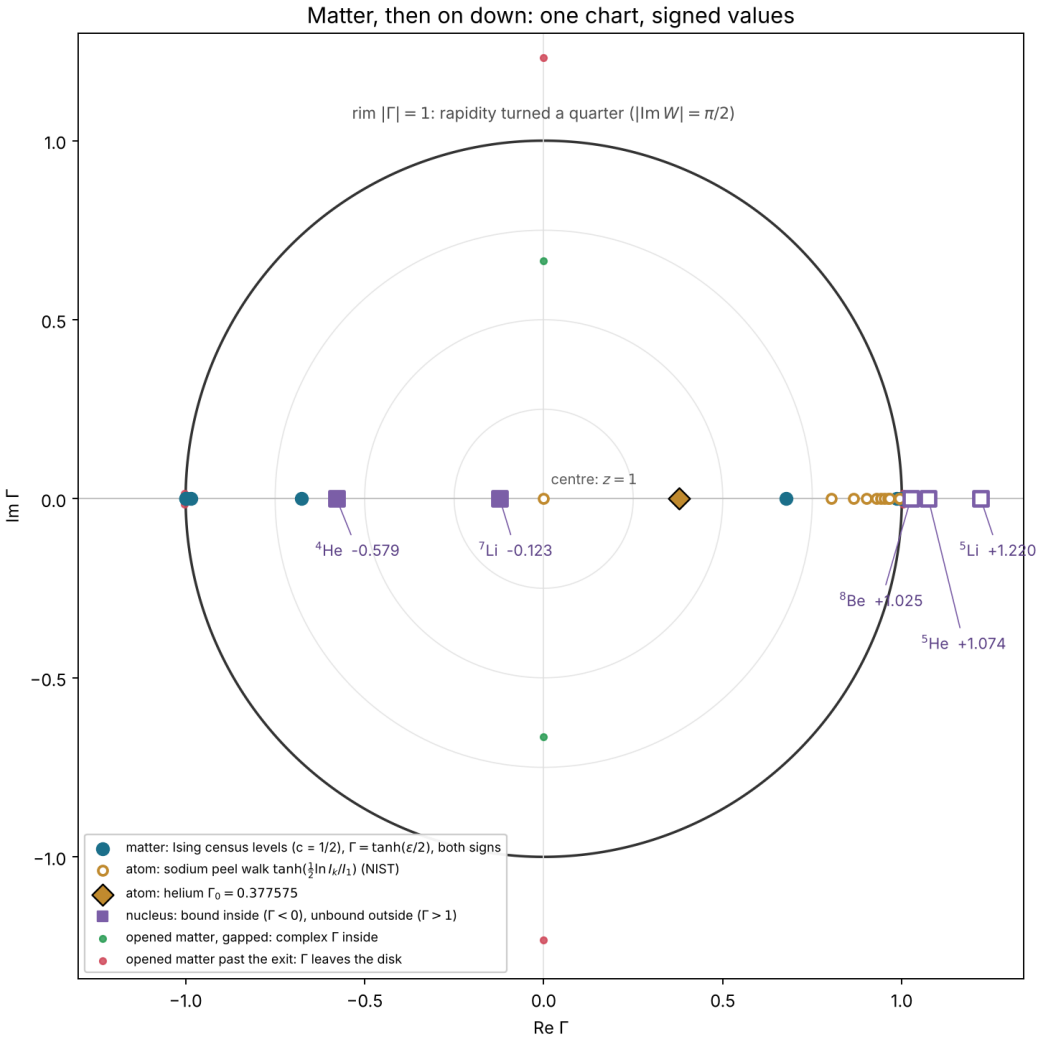


Figure 2: One chart, every scale, signed. Blue: the Ising census ladder $\Gamma = \tanh(\varepsilon/2)$ on both signs. Gold: sodium's peel walk $\tanh(\frac{1}{2} \ln I_k/I_1)$ from the NIST thresholds, ending at 0.9938, and helium's Γ_0. Purple: nuclear separation-energy peels in the order $z = S_{\text{next}}/S_{\text{first}}$ (NUBASE2020); ^{7}Li and ^{4}He inside at -0.123 and -0.579, the unbound mass-5 and mass-8 gaps outside. Green and red: census levels of the opened non-Hermitian SSH chain ($u = 0.5$) before ($v = 0.30$) and past ($v = 0.49$) its exit.

4 The disk as a reflection chart

The Smith chart was built for reflection. The black-hole paper of the author (*Beyond the Event Horizon*, ed. 10, §§5, 7, 8) places three boundary readings on the same disk, and each lends the matter coordinate a meaning it did not have from occupations alone. They are different normalised impedances on one chart; their relative positions are not a physical identification (Figure 3).

4.1 A step in spectral gap reflects with the peel coordinate

On a projector family with pullback metric $g = a(y)\,dx^2 + b(y)\,dy^2$ ($a = g_{\parallel}$, $b = g_{\perp}$), a wave $\psi = e^{ipx}u(y)$ of the Laplace–Beltrami operator at frequency ω conserves the Clairaut constant $C = a\sin^2\theta$ and carries the admittance $Y = \omega\sqrt{a - C} = \omega\sqrt{g_{\parallel}}\cos\theta$. At a step $(a_1, b_1) \to (a_2, b_2)$,

$$\Gamma = \frac{\sqrt{a_1 - C} - \sqrt{a_2 - C}}{\sqrt{a_1 - C} + \sqrt{a_2 - C}}, \qquad \sqrt{a_2 - C} \to i\sqrt{C - a_2} \text{ when } C > a_2, \tag{5}$$

so $g_{\perp}$ drops out, and $|\Gamma| = 1$ exactly beyond the geodesic turning point $a_2 = C$ [T] (ed. 10, eqs. 47–49). At normal incidence $C = 0$ and $\Gamma = \tanh(\frac{1}{2}\ln\sqrt{a_1/a_2})$. When the index is set by the spectral gap at fixed coupling, as on the IQGT family $\Sigma = \cos 3x\,\sigma_x + \sin 3x\,\sigma_y + 2.5y\,\sigma_z$ where $g_{\parallel}\Delta^2 = 9$ [T] (*Intrinsic QGT*, §8.6), $\sqrt{g_{\parallel}} \propto 1/\Delta$ and

$$\Gamma = \frac{\Delta_2 - \Delta_1}{\Delta_2 + \Delta_1} = \tanh\left(\tfrac{1}{2}\ln\frac{\Delta_2}{\Delta_1}\right): \tag{6}$$

the peel coordinate, now as the reflection of a wave at a change of gap [T]. Script `s13` solves the matching problem directly (continuity of u and u' for $u'' + \omega^2 n^2 u = 0$, $n = 1/\Delta$) on 200 random steps and recovers (6) to 3.3×10^{-16} [V]. The black-hole paper's own sharp-step matching reproduces (5) to 1.7×10^{-11} on 400 random steps, 89 beyond the turning point. Its preregistered Richardson gate for smooth steps failed (residual 7.4×10^{-4} against 10^{-5}), and the first-order limit was then tested directly with its own condition. A polar stratum is a perfect mirror there, $|r| = 1$ to 2×10^{-16} [V][S].

So the atom's peel and a gap step are one formula: successive removal thresholds are ratios of gaps measured from the continuum, and Γ is what a wave crossing from one gap to the next would see reflected.

4.2 The centre is a matched load; the inside is passive

Proposition 4.1 (Schur–Cayley boundary; ed. 10, Prop. 13)**.** *Let L be passive,* $\operatorname{Herm} L = \frac{1}{2}(L + L^*) \geq 0$*, split into boundary b and interior i with L_{ii} invertible. The boundary impedance* $Z_b = L_{bb} - L_{bi}L_{ii}^{-1}L_{ib} = W^*LW$*,* $W = \begin{pmatrix} I \\ -L_{ii}^{-1}L_{ib} \end{pmatrix}$*, has* $\operatorname{Herm} Z_b = W^*(\operatorname{Herm} L)W \geq 0$*, and* $S_b = (Z_b - I)(Z_b + I)^{-1}$ *satisfies*

$$I - S_b^*S_b = 4(WV_b)^*(\operatorname{Herm} L)(WV_b) \geq 0, \qquad V_b = (Z_b + I)^{-1},$$

so $\|S_b\| \leq 1$, with $S_b = 0$ exactly for the matched load $Z_b = I$. No normality is assumed. [T]

Proof. $W^*LW = L_{bb} - L_{bi}L_{ii}^{-1}L_{ib} - L_{bi}L_{ii}^{-1}L_{ib} + L_{bi}L_{ii}^{-1}L_{ii}L_{ii}^{-1}L_{ib} = Z_b$, and taking Hermitian parts commutes with the congruence. With $S_b = (Z_b - I)V_b$, $I - S_b^*S_b = V_b^*[(Z_b + I)^*(Z_b + I) - (Z_b - I)^*(Z_b - I)]V_b = 2V_b^*(Z_b + Z_b^*)V_b = 4V_b^*(\operatorname{Herm} Z_b)V_b$, and substitute $\operatorname{Herm} Z_b$. □

The inside of the disk is the passive boundary; the centre is the load that returns nothing; the rim is total reflection. Through the membrane paradigm the stretched horizon has surface resistivity 4π in Gaussian units, the impedance of empty space $\mu_0 c = 376.73\,\Omega$, so normalised by the vacuum impedance the horizon surface is $Z_b = 1$, $S_b = 0$: the centre of the chart [S] (sourced; ed. 10 §5.2). The match belongs to the horizon surface, not to the hole seen from far away: the potential barrier outside reflects long-wavelength radiation. Matter's unbound nuclei (§7) lie outside the rim, beyond what any passive boundary can reach.

4.3 The outside record of a Hawking fermion pair

Proposition 4.2 (the fermion record; ed. 10, eqs. 38–39)**.** *For a Dirac pair* $\cos r|00\rangle + e^{i\phi}\sin r|11\rangle$ *with* $\tan^2 r = e^{-\hbar\omega/k_B T_H}$*, the outside occupation is* $C = \sin^2 r$*, and*

$$\tau = \log\frac{\det(I-C)}{\det C} = \log\frac{P(\text{empty})}{P(\text{full})} = \frac{\hbar\omega}{k_B T_H}, \qquad \Gamma = 1 - 2C = \tanh\frac{\hbar\omega}{2k_B T_H},$$

for every pair phase ϕ*; for independent modes* $\tau = \sum_k \hbar\omega_k/k_B T_H$*. [T]*

Proof. The partial trace over the inside mode removes the coherence and leaves occupation $\sin^2 r$, independent of ϕ. Then $(1-C)/C = \cot^2 r = e^{\hbar\omega/k_B T_H}$; take logs and apply Proposition 2.1. □

Script `s13` takes the partial trace at six frequencies and seven phases and recovers both identities to 5×10^{-16} [V]. The identity is the logarithm of a Boltzmann factor; what it adds is the reading. The outside record is the matter offset of §2: it sits at $\Gamma = \tanh(\hbar\omega/2k_B T_H)$, reaches the centre where the matched horizon sits only at zero frequency, and particle–hole exchange $C \mapsto 1-C$ sends $\Gamma \mapsto -\Gamma$. It fixes only the ratio ω/T_H, never an absolute scale, which is §1's zero and span again; and the phase ϕ that decides what the pair does next never enters it, which is Proposition 2.11's lesson at a horizon [H][S]. The construction needs $0 < C < I$, so it holds for fermion modes; the bosonic pair has no full state.

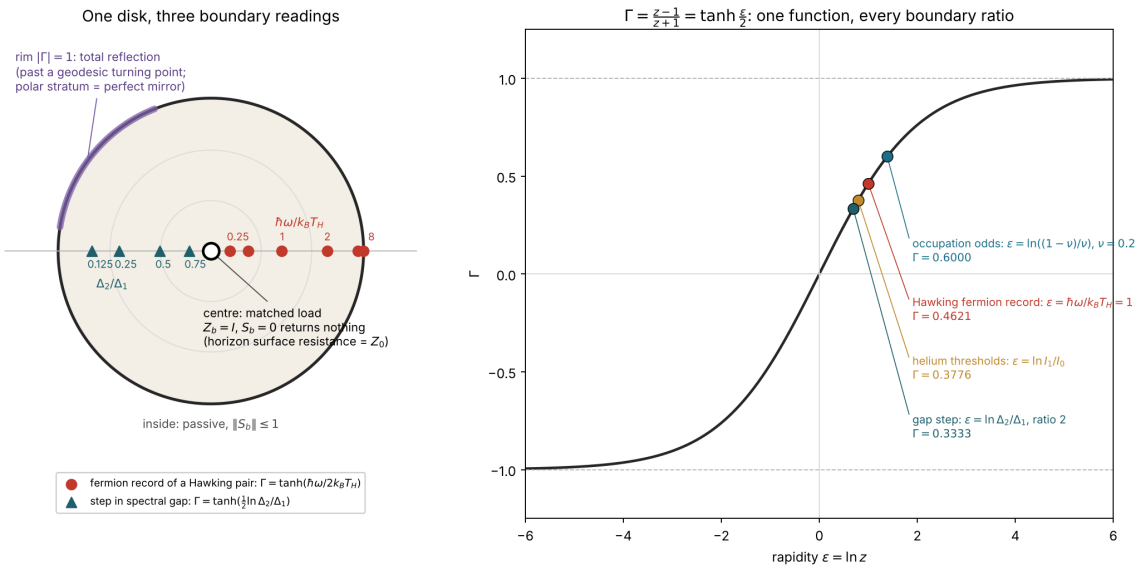


Figure 3: Left: one disk carrying three boundary readings: the matched load at the centre, total reflection on the rim, the Hawking fermion record $\tanh(\hbar\omega/2k_BT_H)$ (red, labelled by $\hbar\omega/k_BT_H$) and the reflection at a step in spectral gap $\tanh(\frac{1}{2}\ln\Delta_2/\Delta_1)$ (green, labelled by Δ_2/Δ_1). Right: the single curve $\Gamma = \tanh(\varepsilon/2)$ with four readings of the rapidity marked at their real values.

5 The atom

For an atom the ratio is between successive relaxed removal thresholds, $z_q = I_{q+1}/I_q$, and every step lies on the positive real radius (VOTKP §8, Theorem 8.1 chart). The index labels successive charge-state removals, not excitations within one fixed-charge Hamiltonian.

Theorem 5.1 (anchored coordinate equivalence)**.** *For $Z \geq 1$ the map $I = (I_0, \ldots, I_{Z-1}) \mapsto (I_0, \Gamma_0, \ldots, \Gamma_{Z-2})$ is a bijection from $(0,\infty)^Z$ onto $(0,\infty)\times(-1,1)^{Z-1}$, with inverse $I_q = I_0 \prod_{j<q}(1+\Gamma_j)/(1-\Gamma_j)$. A common scale $I \mapsto cI$ leaves every Γ_q unchanged; two ladders have the same ratio vector iff they differ by one positive scale; and the anchored vector fixes every binding sum $B_q = \sum_{j\geq q} I_j$. [T][L]*

Proof. For $a, b > 0$, $-1 < (b-a)/(b+a) < 1$ and $(1+\Gamma)/(1-\Gamma) = b/a$; adjacent ratios telescope. Conversely every factor of the inverse is positive for $\Gamma_j \in (-1,1)$. Common scaling cancels in each ratio; equal ratios give $I'_q/I_q = I'_0/I_0$ by induction. □

Any function of the complete ladder is a function of the anchored vector and conversely: the coordinate changes conditioning and representation, never information. Neither representation knows a specimen's temperature, density, partition weights, matrix elements or preparation (§6).

Helium's two thresholds, 24.587389011 and 54.4177655282 eV, give $\Gamma_0 = 0.37757506698375$ and a binding sum of 79.0051545392 eV [lit][V]. VOTKP Theorem 15.4 (scale blindness) is the atomic form of §1: multiplying every mass of a nonrelativistic Coulomb system by λ scales every energy by λ and leaves every ratio fixed, so the peel coordinate is blind to which particle carries the charge except through charges, mass ratios and statistics [T] (A.9). The coordinate needs a physical zero: removal energies are measured from the continuum, which is why the ratio means something, and why quantum-dot addition spectra, which can be moved up or down, have no peel (VOTKP §15.3).

Sodium's eleven thresholds walk from the centre to $\Gamma = 0.9938$ along one diameter (Figure 2); shell jumps are the long strides. For a helium-like ion of charge $Z \geq 2$ the last peel step is, to first order in the electron repulsion, $\Gamma = 5/(8Z-5) \to 5/(8Z)$ [T][L] (A.10). The middle of the peel follows the Thomas–Fermi law with $\Gamma_k \to \tanh(\frac{2}{3}\ln\frac{k+1}{k}) \approx 2/(3k)$, cited in VOTKP §15.11 [lit].

6 Gas

Matter at a Scale §4 makes gas the gapless phase: the census floods, directions are free and failures are non-modal. On the disk the gas state is a layer that can be varied freely while an atomic record survives it. This section proves how much of the atom the gas carries, and what the gas must supply before the atom can be read back out. The model is the ideal, nonrelativistic, thermodynamic-equilibrium Saha gas of a pure species; mixtures, external electrons and non-LTE kinetics need other closures.

6.1 One neutral state

Fix $Z \geq 1$, a nucleus density $N > 0$ and positive coefficients $S_0, \dots, S_{Z-1}$ (number-density units). The finite Saha model is $n_{q+1}/n_q = S_q/x$, $\sum_q n_q = N$, $x = \sum_q q n_q$, with $x = n_e > 0$. Set $a_0 = 1$, $a_q = \prod_{j<q} S_j$, $W(x) = \sum_q a_q x^{-q}$, $p_q = a_q x^{-q}/W$, $\mu = \sum q p_q$, $V = \sum (q-\mu)^2 p_q$. Every population is positive, $0 < \mu < Z$, $V > 0$, and neutrality is $x = N\mu(x)$.

Lemma 6.1 (exponential family)**.** *With $y = \log x$: $dp_q/dy = p_q(\mu - q)$, $d\mu/dy = -V$, and $\frac{d}{dy}\mathbb{E}[f(q)] = -\mathrm{Cov}(f(q), q)$. [T]*

Proof. $dW(e^y)/dy = -\mu W$; differentiate $\log p_q$; multiply by q or f and sum. □

Theorem 6.2 (a unique neutral gas)**.** *The model has exactly one positive solution, with $0 < x < NZ$; it is the unique positive root of $P(x) = \sum_q a_q x^{Z-q+1} - N\sum_q q a_q x^{Z-q}$, of degree $Z+1$. For helium, with $u = x/N$, $\alpha = S_0/N$, $\beta = S_1/N$: $u^3 + \alpha u^2 + (\alpha\beta - \alpha)u - 2\alpha\beta = 0$, $0 < u < 2$. [T][L](three charges)*

Proof. $F(x) = x - N\mu(x)$ has $F' = 1 + NV/x > 1$; as $x \downarrow 0$, $p_Z \to 1$ and $F \to -NZ < 0$; at $x = NZ$, $\mu < Z$ so $F > 0$. Strict monotonicity gives one root; ratios and normalisation fix the populations. □

Theorem 6.3 (how the neutral gas responds)**.** *At fixed coefficients,*

$$\frac{d\log x}{d\log N} = \frac{\mu}{\mu+V} \in (0,1), \qquad \frac{d\mu}{d\log N} = -\frac{\mu V}{\mu+V} < 0, \qquad \frac{d\log p_q}{d\log N} = (\mu - q)\frac{\mu}{\mu+V},$$

and every upper tail $\Pr(q \geq r)$ falls with density. If θ varies the coefficients with $h_q = \partial_\theta \log a_q$ and $C_h = \mathrm{Cov}_p(q, h_q)$, then $\partial_\theta \log x = C_h/(\mu+V)$, $\partial_\theta \mu = \mu C_h/(\mu+V)$, $\partial_\theta p_q = p_q[h_q - \mathbb{E}h - (q-\mu)C_h/(\mu+V)]$; raising any single S_r raises the mean charge. With T-independent partition weights, $d\log x/d\log T = \mathrm{Cov}(q, \frac{3}{2}q + E_q/k_BT)/(\mu+V) > 0$ with $E_q = \sum_{j<q} I_j$, and at fixed T, N, U, $\partial\mu/\partial I_r = -\mu\,\mathrm{Cov}(q, \mathbf{1}_{q>r})/(k_BT(\mu+V)) < 0$. [T]

Proof. Differentiate $x = N\mu(x)$: $dx/dN = \mu/(1+NV/x) = \mu^2/(\mu+V)$. Lemma 6.1 gives the populations. For an increasing indicator f, $\mathrm{Cov}(f(q), q) = \frac{1}{2}\sum_{i,j} p_i p_j (f(i) - f(j))(i-j) > 0$. At fixed x, $\partial_\theta p_q = p_q(h_q - \mathbb{E}h)$, so $\partial_\theta \mu = C_h$ at fixed x; the change through x adds $-V\partial_\theta \log x$, and neutrality gives $\partial_\theta \mu = \mu \partial_\theta \log x$. For temperature, $a_q = (2/\lambda_e^3)^q (U_q/U_0) e^{-E_q/k_BT}$ has $h_q = 3q/2 + E_q/k_BT$ with positive increments $3/2 + I_q/k_BT$. □

With temperature-dependent partitions the derivative gains $d\log(U_q/U_0)/d\log T$ and positivity needs an extra condition.

The physical coefficients are $S_q(T) = \frac{2U_{q+1}(T)}{\lambda_e(T)^3 U_q(T)} e^{-I_q/k_BT}$, $\lambda_e = h/\sqrt{2\pi m_e k_B T}$, and by Theorem 5.1

$$S_q(T) = \frac{2U_{q+1}(T)}{\lambda_e(T)^3 U_q(T)} \exp\Big[-\frac{I_0}{k_BT} \prod_{j<q} \frac{1+\Gamma_j}{1-\Gamma_j} \Big] :$$

Saha with a reconstructed threshold input, not a new ionisation law [T]. It exposes the inputs the gas needs beyond the peel: the absolute thermal ratio I_0/k_BT, the density relative to λ_e^3, internal partition ratios, and a justified equilibrium.

Theorem 6.4 (scaling and what the atom alone cannot know)**.** *(i) $S_q \mapsto cS_q$, $N \mapsto cN$ preserves every charge fraction and sends $x \mapsto cx$. (ii) $m_e \mapsto am_e$, $T \mapsto bT$, $I_q \mapsto bI_q$ with partition ratios held, and $N \mapsto (ab)^{3/2}N$, preserve all populations. (iii) No function of the complete anchored threshold record determines the mean charge for all densities at fixed T, U, m_e; and no function of the unanchored ratio vector determines it even at fixed T, N, U, m_e. [T]*

Proof. (i) Ratios S_q/x and neutrality are preserved; Theorem 6.2 gives uniqueness. (ii) $S'_q = (ab)^{3/2}S_q$, then (i). (iii) By Theorem 6.3 the mean depends strictly on N; common threshold scaling fixes the ratio vector and changes the mean through $\partial\mu/\partial I_r < 0$. □

The missing absolute anchor is physically consequential: common scaling of every threshold leaves every Γ_q fixed and changes the gas. At nucleus density $10^{22}\,\mathrm{m}^{-3}$, with helium's ground partitions $(1, 2, 1)$, the neutral fraction is 0.999374, 0.289781 and 8.73350×10^{-6} at 10^4, 2×10^4 and $5 \times 10^4\,\mathrm{K}$ while Γ_0 stays at 0.377575: the atomic record alone does not determine a gas population [V].

6.2 The gas carries the peel back

Theorem 6.5 (population combinations recover the ladder)**.** *At each sampled temperature, with $\beta = 1/k_BT$,*

$$L_q = \log\frac{n_q n_{q+2}}{n_{q+1}^2} = \log\frac{U_q U_{q+2}}{U_{q+1}^2} - \beta(I_{q+1} - I_q),$$
$$L_B = \log\frac{n_0}{n_Z(n_e\lambda_e^3)^Z} = \log\frac{U_0}{2^Z U_Z} + \beta B, \qquad B = \sum_q I_q.$$

The L_q are population-only: n_e, λ_e and N cancel. L_B is density-compensated: it needs the sample's $n_e\lambda_e^3$. With temperature-independent partitions, two temperatures give $D_q = -\Delta L_q/\Delta\beta = I_{q+1} - I_q$ and $B = \Delta L_B/\Delta\beta$, and then $I_0 = \frac{1}{Z}[B - \sum_{r=0}^{Z-2}(Z-1-r)D_r]$, $I_q = I_0 + \sum_{r<q} D_r$, absolute scale included. [T][L](cancellation)

Proof. Divide adjacent Saha relations for L_q; multiply all Z relations,

$$\frac{n_Z}{n_0} = \frac{2^Z U_Z\, e^{-\beta B}}{(n_e\lambda_e^3)^Z U_0},$$

for L_B. Summing $I_q = I_0 + \sum_{r<q} D_r$ counts each D_r exactly $Z-1-r$ times. No neutrality is used unless n_e is inferred from pure-species fractions. □

For helium with $U = (1, 2, 1)$: $\log(n_0 n_2/n_1^2) = -\beta D - \log 4$, $\log(n_0/(n_2 n_e^2 \lambda_e^6)) = \beta B - \log 4$, $\Gamma_0 = D/B$. Read back across 10^{18} to $10^{26}\,\mathrm{m}^{-3}$ at 30 000 and 32 000 K, the gas returns $\Gamma_0 = 0.37757506698375$ to 2×10^{-16} (Figure 1, right; script `s08`) [R]. This is a model identity and a code check, not an independent measurement of atomic thresholds.

Theorem 6.6 (partition correction and conditioning)**.** *If known $U_q(T)$ vary with temperature, $\tilde{L}_q = L_q - \log(U_q U_{q+2}/U_{q+1}^2)$ and $\tilde{L}_B = L_B - \log(U_0/2^Z U_Z)$ have exact two-temperature slopes D_q and B. Omitting the correction biases them by $-\Delta \log(U_q U_{q+2}/U_{q+1}^2)/\Delta\beta$ and $+\Delta \log(U_0/(2^Z U_Z))/\Delta\beta$; for finite excitation partitions $d \log U_q/d\beta = -\epsilon_q$, the thermal mean excitation, so helium reads $D_{\mathrm{app}} = D + \epsilon_0 + \epsilon_2 - 2\epsilon_1$ and $B_{\mathrm{app}} = B - \epsilon_0 + \epsilon_2$. If each corrected combination has endpoint error at most η_1, η_2, its slope error is at most $(\eta_1 + \eta_2)/|\Delta\beta|$, and with $|\hat{D} - D| \le \eta_D$, $|\hat{B} - B| \le \eta_B < B$, $|\hat{D}/\hat{B} - D/B| \le (\eta_D + |D/B|\eta_B)/(B - \eta_B)$. [T]*

Proof. Subtract the known partition terms and difference in β; for the ratio, write $\hat{D}/\hat{B} - D/B = [B(\hat{D} - D) - D(\hat{B} - B)]/(B\hat{B})$ and use $|\hat{B}| \ge B - \eta_B$. □

The registered opposite used the synthetic partition $U_1(T) = 2 + 4e^{-10\,\mathrm{eV}/k_B T}$, not a claimed helium level scheme: ignoring its temperature dependence returns $\Gamma_0 = 0.36614943350748$, a bias of -0.0114256; subtracting the known partition terms restores 0.37757506698375 to 1.4×10^{-14} [V] (script `g01_verify_formal_gas`, specified before its run). Close temperatures amplify observational error; rare populations make the logarithms ill-conditioned. Real helium excited-state partitions, continuum lowering and non-LTE populations are separate validation tasks.

7 The nucleus

A nucleus peels by separation energies. VOTKP §15.4 uses $z = S_{\mathrm{next}}/S_{\mathrm{first}}$ and the same Γ, "the form that extends to $z < 0$". A bound peel has $z > 0$ and sits inside the disk. A negative separation energy gives $z < 0$ and $|\Gamma| > 1$ (Theorem 2.2). Recomputed here from NUBASE2020 with the author's own script, with the signed coordinate for VOTKP's order:

case	S_{first} (keV)	S_{next} (keV)	Γ	$\lvert\Gamma\rvert$	half-life
^{5}He: n, then n from ^{4}He	−734.8	20577.6	+1.074062	1.074	602 ys
^{5}Li: p, then p from ^{4}He	−1966.1	19813.9	+1.220318	1.220	370 ys
^{8}Be: α (^{12}C ladder)	−91.8	7366.6	+1.025238	1.025	81.9 as
^{7}Li (bound)	7251.1	5664.4	−0.122852	0.123	stable
^{4}He (bound)	20577.6	5493.5	−0.578576	0.579	stable

The magnitudes agree with VOTKP §15.4 to every printed digit [V][lit]. Reversing a ratio flips the sign of Γ and keeps its magnitude; the ^{8}Be row follows the α chain $^{12}\mathrm{C} \to {}^8\mathrm{Be} \to 2\alpha$, with S_{first} the α separation from ^{8}Be and S_{next} that from ^{12}C.

How far outside, exactly. VOTKP §15.4 notes that the distance outside the rim is close to twice the decay energy over the neighbouring step. It is exact in this form. Write $S_{\mathrm{first}} = -s < 0$ and $a = S_{\mathrm{next}}/s > 1$, so $z = -a$ and

$$\Gamma = \frac{a+1}{a-1}, \qquad |\Gamma| - 1 = \frac{2}{a-1} = \frac{2s}{S_{\mathrm{next}} - s},$$

which is $2s/S_{\mathrm{next}}$ when $s \ll S_{\mathrm{next}}$ [T][L]. On the three gap nuclei the identity holds exactly (0.074062, 0.220318, 0.025238), against the small-s values 0.071417, 0.198457, 0.024923 (script `s12`). The form

needs $S_{\text{next}} > s$; if both separation energies are negative the ratio is positive and the point is inside. The Cayley pole $z = -1$ is $S_{\text{first}} + S_{\text{next}} = 0$; none of the three gaps is near it. The placement says no more than the sign of a separation energy, and the author's own caution stands: a resonance's leak through its barrier "is not a literal atomic or nuclear Smith reflection derived from the threshold coordinate".

The half-turn. Write $\Gamma = \tanh(W/2)$ with $W = \operatorname{Log} z$. A negative ratio has $\operatorname{Im} W = \pi$ exactly: the three unbound nuclei are peels whose rapidity has turned half a revolution, and the rim of the disk is the quarter-turn (Theorem 2.2). Random complex z (4000 draws) give zero mismatches, and $z = -28$ gives $\Gamma = 1.0741$, the ^{5}He value [V] (script `s04a`).

8 Present records and later responses

A census, an offset, a threshold ladder and a gas state are present records. This section asks which later quantities a present record fixes, and answers with exact witnesses: in each, two preparations agree on a declared record and disagree on a declared later readout, and the lost distinction is named. A complete state evolves deterministically; what fails is the sufficiency of the summary.

8.1 Retained and hidden amplitudes

Let $\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B$ be finite-dimensional, $H = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix} = H^\dagger$, $\psi = (x, y)$, $i\hbar\dot{\psi} = H\psi$. The partition is physical data; if it is an invariant spectral subspace of H then $B = 0$.

Theorem 8.1 (the reduced equation, with the hidden source)**.** *With $K(t) = Be^{-iDt/\hbar}B^\dagger$ and $J(t) = Be^{-iDt/\hbar}y(0)$,*

$$\dot{x}(t) = -\frac{i}{\hbar}Ax(t) - \frac{i}{\hbar}J(t) - \frac{1}{\hbar^2}\int_0^t K(t-s)x(s)\,ds.$$

A, K, J and $x(0)$ determine the retained evolution; a kernel without the source term does not when $y(0) \neq 0$. If E_D is the spectral measure of D, $d\mathsf{M}(\lambda) = B\,dE_D(\lambda)\,B^\dagger$ is a positive operator-valued measure with $K(t) = \int e^{-i\lambda t/\hbar}d\mathsf{M}$, $\Sigma(z) = B(z-D)^{-1}B^\dagger = \int d\mathsf{M}/(z-\lambda)$, and the retained resolvent is $G_A(z) = (z - A - \Sigma(z))^{-1}$. $K(0) = \mathsf{M}(\mathbb{R}) = BB^\dagger$ is the total channel mass, not its frequency distribution; $\operatorname{Im}\Sigma(z) \leq 0$ *for* $\operatorname{Im} z > 0$*; and* $\sum_{j,k} u_j^\dagger K(t_j - t_k)u_k = \|\sum_k e^{iDt_k/\hbar}B^\dagger u_k\|^2 \geq 0$. *[T]*

Proof. Variation of constants in the hidden equation, $y(t) = e^{-iDt/\hbar}y(0) - \frac{i}{\hbar}\int_0^t e^{-iD(t-s)/\hbar}B^\dagger x(s)ds$, substituted in $\dot{x} = -\frac{i}{\hbar}(Ax + By)$. The spectral theorem gives the integrals and the positivity of each atom $BE_D(\{\lambda\})B^\dagger$; block elimination gives the Schur inverse; $\operatorname{Im}\Sigma = -(\operatorname{Im} z)B((\operatorname{Re} z - D)^2 + (\operatorname{Im} z)^2)^{-1}B^\dagger$. □

The footprint $K(0) = BB^\dagger$ is the weight between of §2.2 for the generator. These properties constrain admissible kernels; they do not turn a scalar delay curve into a reconstructed measure.

Proposition 8.2 (equal spectrum, equal footprint, different survival)**.** *Set $\hbar = 1$, $\Lambda = \operatorname{diag}(-3, -1, 1, 3)$, $w^{(0)} = (1,1,1,1)/4$, $w^{(1)} = (3,7,1,5)/16$. For each, $s = (\sqrt{w_j})$, $v = e_0 - s$, $R_w = I - vv^T/(1 - s_0)$ (a reflection with $R_w e_0 = s$), $H_w = R_w \Lambda R_w^T$. Both Hamiltonians have spectrum $\{-3, -1, 1, 3\}$, retained block $A = 0$ and footprint $BB^T = 5$; from $\psi(0) = e_0$ their retained survivals at $t = \pi/4$ are 0 and $1/32$. [T][V]*

Proof. $v^T v = 2(1 - s_0)$ makes R_w an orthogonal reflection. The amplitude is $a_w(t) = \sum_j w_j e^{-i\lambda_j t}$. Both weight vectors have $\sum w\lambda = 0$ and $\sum w\lambda^2 = 5$, which give $A = 0$ and $BB^T = 5$. For $w^{(0)}$, $a_0 = (\cos 3t + \cos t)/2$; the difference weights $(-1, 3, -3, 1)/16$ give $a_1 - a_0 = \frac{i}{8}(3\sin t - \sin 3t) = \frac{i}{2}\sin^3 t$. At $t = \pi/4$, $a_0 = 0$, $a_1 = i\sqrt{2}/8$. □

The third moments are 0 and 3: BDB^T is 0 and 3, so $K'(0) = -iBDB^T$ differs while $K(0)$ agrees. The retained resolvents are $G_0 = (z^3 - 5z)/Q$ and $G_1 = (z^3 - 5z + 3)/Q$ with $Q = (z^2 - 1)(z^2 - 9)$, and $\Sigma_0 = (5z^2 - 9)/(z^3 - 5z)$, $\Sigma_1 = (5z^2 + 3z - 9)/(z^3 - 5z + 3)$: both begin $5/z$ at large z and part at the next coefficient.

Proposition 8.3 (same kernel, different hidden source)**.** *$H = \sigma_x$, $\psi_\pm(0) = (1, \pm i)/\sqrt{2}$: equal retained amplitude at $t = 0$, $A = D = 0$, $K(t) = 1$, yet $|x_\pm(t)|^2 = \frac{1}{2}(1 \pm \sin 2t)$, so 1 and 0 at $t = \pi/4$; the hidden sources are $J_\pm = \pm i/\sqrt{2}$. [T]*

Proof. $e^{-iHt} = \cos t\, I - i \sin t\, H$ gives $x_\pm = (\cos t \pm \sin t)/\sqrt{2}$. □

Proposition 8.3 changes the initial state; Proposition 8.2 keeps it and changes the spectral weights. They are two separate omissions, and Proposition 2.11 is the first of them seen through the offset.

8.2 A finite moment record

Theorem 8.4 (matched moments, different response)**.** *Let $m \geq 1$, $x_j = a + jh$ ($h \neq 0$, $0 \leq j \leq m$), $p_j > 0$ with $\sum p_j = 1$, $w_j = (-1)^j \binom{m}{j}$, $q_j = p_j + \delta w_j$ with $0 < |\delta| < \min_j p_j/|w_j|$. Then $p \neq q$ are positive probability vectors with equal moments of every order $r < m$, $\sum_j (q_j - p_j) x_j^m = \delta(-1)^m m! h^m$, and $F_q(t) - F_p(t) = \delta e^{-iat}(1 - e^{-iht})^m$ for $F_p(t) = \sum p_j e^{-itx_j}$. No rule of the first m moments reproduces the whole response. [T][L]($m = 3$ witness)*

Proof. $\sum w_j = (1 - 1)^m = 0$. The m-th forward difference annihilates polynomials of degree below m and returns $m!h^m$ on x^m. The exponential sum is a binomial expansion; its real zeros are the recurrence times $ht \in 2\pi\mathbb{Z}$. □

The readout matters as well. For $m = 3$ on $(-3, -1, 1, 3)$, the uniform weights and $(3, 7, 1, 5)/16$ share normalisation, mean 0 and second moment 5, and their intensities at $t = \pi/4$ are 0 and $1/32$; the reflected weights $(5, 1, 7, 3)/16$ have the complex-conjugate response and the same intensity at every real time (Figure 4, left). An intensity-only detector erases what a phase-resolved one keeps. The anchored VOTKP ladder is a different case: it is an invertible coordinate of the complete threshold list (Theorem 5.1), and nothing is lost in that map; the missing response data are the preparation and channel weights the list never claimed to hold. Exact rational checks at orders 1 through 12 hold with residuals at most 3.3×10^{-16} [V].

8.3 A gas witness: same density, flow and stress

Theorem 8.5 (kinetic witness)**.** *On a period L with $k = 2\pi m/L$, fix $\rho_* > 0$, $0 < \eta < 1$, $v_0 > 0$, $0 < a < 1/12$, $\sigma \geq 0$; $u = (-3, -1, 1, 3)$, $d = (-1, 3, -3, 1)$, $\rho(x) = \rho_*[1 + \eta \cos kx]$, g_σ the centred Gaussian of variance σ^2 (a point mass at $\sigma = 0$), $p_\pm(v) = \sum_j (\frac{1}{4} \pm a d_j) g_\sigma(v - v_0 u_j)$ and $f_\pm(x, v, 0) = \rho(x) p_\pm(v)$.*

1. *(K1) Pointwise $M_0^\pm = \rho$, $M_1^\pm = 0$, $M_2^\pm = (5v_0^2 + \sigma^2)\rho$, but $M_3^\pm = \pm 48 a v_0^3 \rho$.*

2. *(K2) Under the same free streaming $\partial_t f + v\partial_x f = 0$, with $q = kv_0 t$,*

$$n_\pm = \rho_*\big[1 + \eta e^{-\sigma^2 k^2 t^2/2}\big(\tfrac{1}{2}[\cos q + \cos 3q]\cos kx \mp 8a\sin^3 q \sin kx\big)\big],$$
$$\Delta n = -16a\eta\rho_* e^{-\sigma^2 k^2 t^2/2}\sin kx \sin^3(kv_0 t).$$

3. *(K3) $\Delta n = \partial_t \Delta n = \partial_t^2 \Delta n = 0$ at $t = 0$, while $\partial_t^3 \Delta n = -\partial_x^3 \Delta M_3 = -96a\eta\rho_* k^3 v_0^3 \sin kx$.*

4. *(K4) The pre-release reset $\mathcal{E}_s[f] = e^{-\nu s} f + (1 - e^{-\nu s})\rho p_0$, $p_0 = \frac{1}{4}\sum_j g_\sigma(v - v_0 u_j)$, preserves density, flow, stress and positivity and multiplies the later contrast by exactly $e^{-\nu s}$. [T]*

Proof. $\sum d_j = 0$ and $\min(\frac{1}{4} \pm a d_j) = \frac{1}{4} - 3a > 0$. Exact sums: $\sum d_j u_j^\ell = 0$ for $\ell \le 2$ and 48 for $\ell = 3$; convolution adds σ^2 to second moments and $3\sigma^2 v_0 u_j$ to third, whose d-weighted sum vanishes. Characteristics give $f(x - vt, v, 0)$; the Gaussian transform gives $e^{-ikv_0 u_j t} e^{-\sigma^2 k^2 t^2/2}$; $\sum d_j \cos q u_j = 0$ and $\sum d_j \sin q u_j = 2\sin 3q - 6\sin q = -8\sin^3 q$. The moment hierarchy $\partial_t M_\ell + \partial_x M_{\ell+1} = 0$ iterated three times gives K3. The reset is a convex combination of positive distributions with equal first three moments and is linear. □

At $a = 1/24$, $\eta = \frac{1}{2}$, $\rho_* = v_0 = k = 1$, $\sigma = 0$, $x = t = \pi/2$: $n_+ = 5/6$, $n_- = 7/6$, $\Delta n = -1/3$ (Figure 4, right). Stress separates first through its derivative, flow next, density at third order. The full kinetic state is Markovian; history dependence appears only after the higher velocity structure is discarded. The reset is a prescribed preparation map, not the self-consistent collision operator of an isolated gas.

control	exact prediction	what it shows
$a = 0$	$f_+ = f_-$, $\Delta n = 0$	the hidden distinction removed
$\eta = 0$	$\Delta n = 0$ always	a homogeneous density cannot read the velocity record
$\sin kx = 0$	no contrast there	placement matters
$kv_0 t \in \pi\mathbb{Z}$	no contrast then	a null readout is not equality of states
$s \to \infty$, $\nu > 0$	$\Delta n \to 0$	the reset erases what was prepared
M_3 also recorded	differ at $t = 0$	the three-moment match is no longer the experiment
$\sigma > 0$	Gaussian envelope	spread dephases this readout, states still differ

8.4 The angular gas chain

Free streaming in three dimensions, written in normalised Legendre moments of the direction cosine, is the tridiagonal chain $M_{l,l+1} = M_{l+1,l} = c_l = (l+1)/\sqrt{(2l+1)(2l+3)}$. With density-preserving relaxation at rate κ,

$$L = -ikM - \kappa(I - P_0), \qquad P_0 = e_0 e_0^T,$$

and the density response is $G(s) = \langle e_0, (s - L)^{-1} e_0 \rangle$. Four exact results follow, each a card of the angular gas.

Proposition 8.6 (ANG1, the free-streaming load)**.** *At $\kappa = 0$, $g_0(s) = \frac{1}{2}\int_{-1}^{1} \frac{d\mu}{s + ik\mu} = \frac{\arctan(k/s)}{k}$, and the load the hidden moments present to density, $Z_b(s) = 1/g_0(s) - s$, tends to $2k/\pi$ as $s \to 0^+$. [T]*

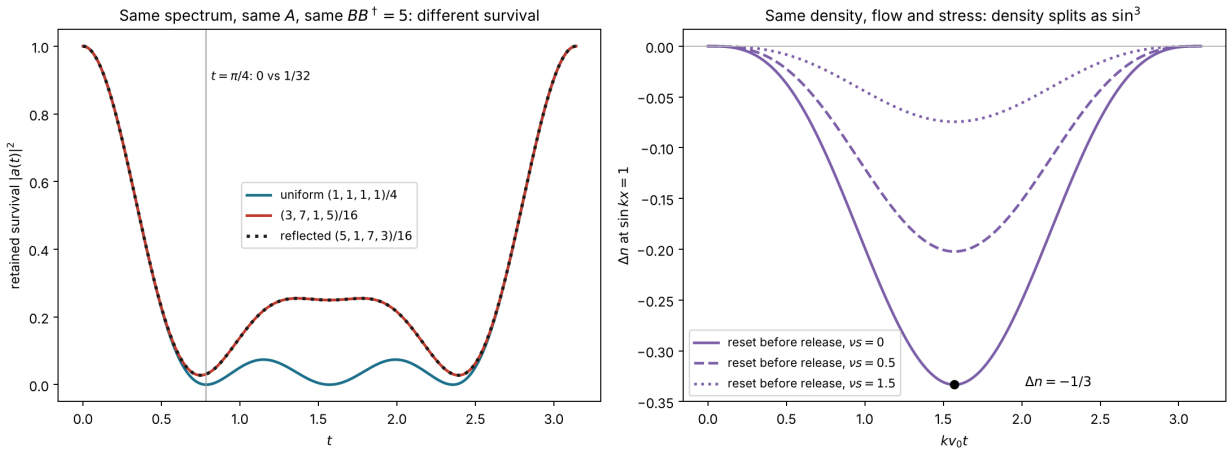


Figure 4: Equal present records, different futures. Left: same spectrum, same retained block, same footprint $BB^\dagger = 5$, and survival 0 against 1/32 at $t = \pi/4$; the reflected weights (dotted) have the same intensity as $(3, 7, 1, 5)/16$ at every time. Right: equal density, flow and stress; free streaming splits density as $-\frac{1}{3}\sin^3(kv_0t)$ at the exact instance, and a reset before release scales the split by $e^{-\nu s}$.

Proof. $\frac{1}{2}\int_{-1}^{1} \frac{s\,d\mu}{s^2+k^2\mu^2} = \frac{1}{k}\arctan\frac{k}{s}$; the odd part integrates to zero. As $s \to 0^+$, $\arctan(k/s) \to \pi/2$. By Theorem 8.1, $g_0 = (s - L_{00} - \Sigma)^{-1}$ with $L_{00} = 0$, so $Z_b = -\Sigma(s)$ is the Schur load of the moments above density. □

$Z_b(10^{-2}, 10^{-4}, 10^{-6}) = 0.63070,\ 0.636560,\ 0.6366192$ against $2/\pi = 0.6366198$ [V]. This is the declared density response and its reference load, not an electromagnetic reflectance.

Proposition 8.7 (ANG2, the relaxation pole)**.** *For $\kappa > 0$, $G = g/(1 - \kappa g)$ with $g(s) = g_0(s + \kappa)$. An isolated real pole to the right of the continuum, $s^* > -\kappa$, exists iff $\kappa > 2k/\pi$, and then $s^* = k\cot(k/\kappa) - \kappa$; at $\kappa = 2k/\pi$ it meets the branch point $s = -\kappa$. [T]*

Proof. $s - L = (s + \kappa + ikM) - \kappa P_0$, a rank-one change; Sherman–Morrison on e_0 gives $G = g/(1 - \kappa g)$ with $g = \langle e_0, (s + \kappa + ikM)^{-1} e_0\rangle = g_0(s + \kappa)$. A real pole with $w = s + \kappa > 0$ needs $\arctan(k/w) = k/\kappa$, solvable iff $k/\kappa < \pi/2$, giving $w = k\cot(k/\kappa)$; as $\kappa \downarrow 2k/\pi$, $w \downarrow 0$. □

ANG1 supplies ANG2's threshold: the pole is born where κ equals the free-streaming load $2k/\pi$ [T]. Residuals of the pole condition at $\kappa = 0.64, 0.7, 0.916, 1, 2$ are below 2×10^{-16} [V]. It is not the two-moment exceptional point of a truncated chain.

Proposition 8.8 (ANG3, first return)**.** *A difference planted in moment $l > 2$ first reaches the stress moment at order $l - 2$ in t, with Taylor coefficient $k^{l-2}\prod_{j=2}^{l-1} c_j/(l-2)!$, and first reaches density at order l, for every $\kappa \geq 0$. [T]*

Proof. $(L^q)_{2,l}$ is a sum over walks of length q from l to 2 on the path graph with loops; every off-diagonal step changes the index by one, so $q \geq l - 2$, and the unique shortest walk uses only off-diagonal steps, contributing $(-ik)^{l-2}\prod_{j=2}^{l-1} c_j$. Diagonal $-\kappa$ terms lengthen walks and enter only at higher order. The same argument gives order l for density. □

For $l = 3, 4, 5, 8$ the coefficients are 0.507092553, 0.127775313, 0.0214031702, 2.2474×10^{-5}, and every lower power vanishes to 10^{-15} [V] (Figure 5). This is the predictive quotient realised on a source model: a planted record is invisible to stress for $l - 3$ orders and to density for $l - 1$.

Proposition 8.9 (ANG4, the two books of the chain)**.** *For every truncation $N \geq 2$, $E^- = \frac{1}{8}\|[L, L^\dagger]\|_F^2 = k^2\kappa^2/3$, while $E^+ = \frac{1}{8}\|L^2 - L^{\dagger 2}\|_F^2$ grows with N. [T]*

Proof. With $L = \mathcal{A} + \mathcal{B}$, $\mathcal{A} = -ikM$ anti-Hermitian and $\mathcal{B} = -\kappa(I - P_0)$ Hermitian, $L^\dagger = -\mathcal{A} + \mathcal{B}$ and $[L, L^\dagger] = 2[\mathcal{A}, \mathcal{B}] = 2ik\kappa[M, I - P_0] = -2ik\kappa[M, P_0]$. Only $c_0 = 1/\sqrt{3}$ touches e_0, so $[M, P_0] = c_0(e_1 e_0^T - e_0 e_1^T)$ with $\|\cdot\|_F^2 = 2c_0^2 = \frac{2}{3}$. Then $E^- = \frac{1}{8} \cdot 4k^2\kappa^2 \cdot \frac{2}{3}$. □

At $k = \kappa = 1$ [V]:

N	2	5	20	100	200	400
E^-	1/3	1/3	1/3	1/3	1/3	1/3
E^+	0.333	3.444	18.49	98.50	198.50	398.50
$\|L\|_F^2 - \sum \|\lambda\|^2$	1.0000	1.0811	1.1556	1.1715	1.173488	1.174462

The exact E^-, the growth of E^+ and the numerical convergence trend of the departure are three different statuses; an infinite-dimensional limit of the departure is not claimed.

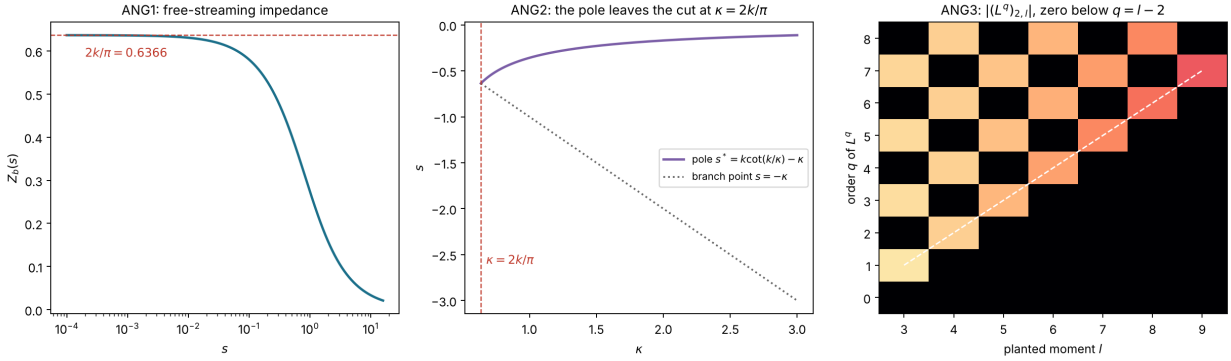


Figure 5: The angular gas chain ($k = 1$). Left: the free-streaming load $Z_b(s)$ tends to $2k/\pi$. Middle: the relaxation pole $s^* = k\cot(k/\kappa) - \kappa$ leaves the branch point at $\kappa = 2k/\pi$. Right: $|(L^q)_{2,l}|$ on a log scale; the planted moment reaches stress no sooner than order $l - 2$ (dashed).

9 Opened matter

9.1 The opened chain

Opening matter here means a non-Hermitian effective generator. Take the PT-symmetric SSH chain

$$h(k) = \begin{pmatrix} iu & v + we^{-ik} \\ v + we^{ik} & -iu \end{pmatrix} = d \cdot \sigma,$$

$$d = (v + w\cos k,\ w\sin k,\ iu), \qquad s = E(k)^2 = d \cdot d = |v + we^{ik}|^2 - u^2,$$

with the bilinear dot product. Fill the lower band with the biorthogonal projector $C = \sum_{\text{filled}} |R_k\rangle\langle L_k|$, $\langle L_j | R_k\rangle = \delta_{jk}$, and restrict to a block. The occupations are complex, and so is $\Gamma = 1 - 2\nu$.

Theorem 9.1 (what survives opening)**.** *Let* $C = \begin{pmatrix} A & B \\ E & D \end{pmatrix}$ *be any idempotent on a finite complex space with a fixed block split, and* $Y_A = BE = C_{AB}C_{BA}$*. Then* $Y_A = A - A^2$*, and its eigenvalues are* $\nu_j(1-\nu_j)$ *with algebraic multiplicity, Jordan blocks allowed. With* $R_b(\lambda) = \sqrt{|1-4\lambda|}$*,* $R_b(\nu(1-\nu)) = |1-2\nu|$*, so the radial census* $\#\{j : R_b(\lambda_j) \le r\} = \#\{j : |1-2\nu_j| \le r\}$ *survives for every* r*, and the condition is the disk* $|\lambda - \frac{1}{4}| \le r^2/4$*. Positivity does not survive:* $C = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}$ *is idempotent with* $Y_A = -2$ *and* $R_b = 3$*, and every square* A *is the compression of the idempotent* $\mathcal{P}(A) = \begin{pmatrix} A & I \\ A - A^2 & I - A \end{pmatrix}$*. [T][L]*

Proof. The AA block of $C^2 = C$ is $A^2 + BE = A$. Triangularise A; the polynomial $z - z^2$ of a triangular matrix has the polynomial of its diagonal on the diagonal. $1 - 4\nu(1-\nu) = (1-2\nu)^2$. For $\mathcal{P}(A)$ multiply blocks; every entry is a polynomial in A. □

The biorthogonal C of the chain is idempotent to 3×10^{-14}, and $|C^2 - C| \le 2.8 \times 10^{-14}$, $|X_A - C_A(1-C_A)| \le 1.3 \times 10^{-14}$ in script `s07` [V]. A principal square root of $(1-2\nu)^2$ returns $\pm(1-2\nu)$, not the modulus; the radial coordinate uses the modulus. The variance and Gaussian readings of §2 need a positive metric and do not come from idempotence; a positive reformulation would need its own inner product, adjoint and partition.

The opened band metric, exactly. The cone paper (v8) shows that at a Hermitian point the round-trip form $Q = M_{+-}M_{-+}/(E_+ - E_-)^2$, $M_{ab} = \langle L_a|\partial h|R_b\rangle$, has $Q \ge 0$ (A.13), and measures a Lorentzian cone when the system is opened. The hypothesis is a Hermitian family, or at least a Hermitian tangent: $h(\theta) = \sigma_z + i\theta\sigma_x$ is Hermitian at $\theta = 0$ yet has $Q = -\frac{1}{4}$ there. On the opened plane the sign is now exact.

Theorem 9.2 (signature of the opened plane)**.** *At fixed real* u, w*, in coordinates* (k, v) *with* $s \ne 0$*, the bilinear two-level metric* $Q_{ab} = [s(\partial_a d \cdot \partial_b d) - (d \cdot \partial_a d)(d \cdot \partial_b d)]/(4s^2)$ *is real and*

$$\det Q = -\frac{u^2 w^2 \cos^2 k}{16\, s^3}.$$

If $s > 0$ *and* $uw\cos k \ne 0$ *the metric is Lorentzian; if* $s < 0$ *and* $uw\cos k \ne 0$ *it is negative definite; it degenerates on* $u = 0$*,* $w = 0$ *or* $\cos k = 0$*; and at* $s = 0$*, the exceptional points, the formula is singular rather than zero. [T][V]*

Proof. With $\alpha = \partial_k d = (-w\sin k, w\cos k, 0)$ and $\beta = \partial_v d = (1,0,0)$, the numerator of $\det Q$ is s times the bilinear Gram determinant of (d, α, β), which is $s[d \cdot (\alpha \times \beta)]^2$ (A.16). Here $d \cdot (\alpha \times \beta) = -iuw\cos k$, so the numerator is $-su^2w^2\cos^2 k$; divide by $16s^4$. For $s < 0$, $\det Q > 0$ and $Q_{vv} = (w^2\sin^2 k - u^2)/(4s^2) < 0$ because $u^2 > |v + we^{ik}|^2 \ge w^2 \sin^2 k$. □

Sympy reduces $16s^3 \det Q + u^2w^2\cos^2 k$ to 0 identically, and 2000 random biorthogonal eigenvector evaluations agree to 4.1×10^{-14} (script `s13`) [V]. The 93% of v3's scan is the grid fraction of the region $s > 0$ off the lines $\cos k = 0$ at $u = 0.5$, here 0.9304 on a 400×400 grid: a property of the chosen window, not a constant (Figure 6). At fixed k the boundary is $s = 0$; at $k = \pi$ it is $v = w \pm u$, the critical points, and $k = \pi/2, 3\pi/2$ are separate degeneracy lines.

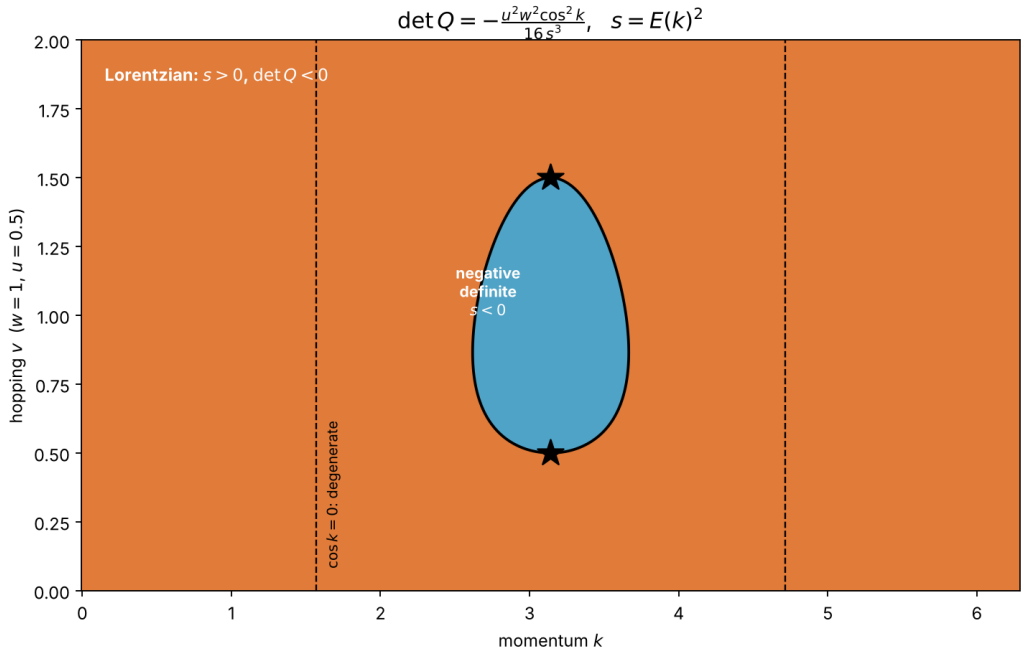


Figure 6: The opened plane's signature from the exact determinant, $u = 0.5$, $w = 1$. Orange: $s = E^2 > 0$, Lorentzian. Blue: $s < 0$, negative definite. Dashed: $\cos k = 0$, degenerate. Black: $s = 0$, the exceptional points, where the formula is singular. Stars: the critical points $k = \pi$, $v = w \mp u$.

Entropy on the opened chain. At the critical point the Rényi-2 coefficient reads $-1.79 \to -1.96$ on the $v < w$ side and $-0.88 \to -1.09$ on the $v > w$ side, where the fit has not converged; the biorthogonal entropy is complex there, with $\max|\operatorname{Im} S_2| = 6.3$ at $v < w$ [V] (script `s03b`). A negative fitted coefficient is a property of the biorthogonal entropy convention (Chang, You, Wen and Ryu), not a negative number of channels: a nonnegative census with $N(\xi) = a \log \xi + O(1)$ cannot have $a < 0$ [T].

9.2 Where opened matter leaves the disk

Approaching the critical point at fixed u, the census levels leave the disk *before* it, inside the gapped phase, at $v_{\text{exit}} = 0.468$ for $u = 0.5$ [V]. At every critical point tested ($u = 0.2, \ldots, 0.7$) a level sits at $\operatorname{Im} W = \pi$ exactly: the half-turn of the unbound nuclei.

Proposition 9.3 (the Petermann factor and the exit line)**.** *For $H = d \cdot \sigma$, $d \in \mathbb{C}^3$, $E^2 = d \cdot d \neq 0$, the band projectors $P_\pm = \frac{1}{2}(I \pm H/E)$ are rank-one idempotents with $K_\pm = \operatorname{Tr} P_\pm^* P_\pm = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2) \geq 1$. On the PT-symmetric chain with $A = v^2 + w^2 - u^2 > |2vw| = |B|$, $K(k) = 1 + u^2/E(k)^2$ and*

$$\langle K \rangle = 1 + \frac{u^2}{\sqrt{A^2 - B^2}} = 1 + \frac{u^2}{E(0)E(\pi)}, \qquad \langle K \rangle = 2 \iff E(0)E(\pi) = u^2. \quad \textit{[T][L]}$$

Proof. $H^2 = E^2 I$ and $\operatorname{Tr} H = 0$ give idempotence and unit trace; $\operatorname{Tr} H^* H = 2\sum |d_j|^2$; $|E|^2 = |\sum d_j^2| \leq \sum |d_j|^2$. Here $\sum |d_j|^2 = |q|^2 + u^2$ and $|E|^2 = |q|^2 - u^2$. The average uses $\frac{1}{2\pi}\int_0^{2\pi} \frac{dk}{A + B\cos k} = (A^2 - B^2)^{-1/2}$ and $\sqrt{A^2 - B^2} = \sqrt{A+B}\sqrt{A-B} = E(0)E(\pi)$ (A.14). □

Checked to 3.6×10^{-14}. Predicted from $\langle K \rangle = 2$ at three values of u not used before, the exits are 0.7474014, 0.5291503, 0.2357262; measured, 0.7474015, 0.5291503, 0.2357262 (differences 4×10^{-8} and below; block 40, block-independence from `s04b`) [V]. The equivalence concerns the band average; that a finite block census crosses the rim at the same parameter is a separate statement, which carries its block, boundary conditions, filter, error and limit.

The limits of the exit law. Three findings scope it, and they belong in the statement.

1. On the cone paper's band made one-dimensional, $d = (t_1 + \cos k, \sin k, m + i\gamma)$, a mass term moves the exit far earlier ($\langle K \rangle \approx 1.01$–$1.06$ at the returned exits); the law holds only at $m = 0$, the SSH class [V].

2. The returned exit depends on the endpoint filter that discards levels with ν near 0 or 1 (ring 2000, block 40, $t_1 = 0.5$, principal logarithm):

filter δ	massive γ_{exit}	massive $\langle K \rangle$	massless γ_{exit}	massless $\langle K \rangle$
10^{-6}	0.271636929	1.057972846	0.474341649	2.000000002
10^{-8}	0.271636929	1.057972846	0.474341649	2.000000002
10^{-9}	0.169169932	1.022352491	0.474341649	2.000000002
10^{-10}	0.169169932	1.022352491	0.474341467	1.999995744
10^{-12}	0.122791366	1.011745220	0.474227956	1.997345192

   The eigenpair residuals are at most 4.4×10^{-15} while the eigenvector condition numbers run 135 to 575; a small residual is not an eigenvalue error bound for a nonnormal matrix [V] (scripts `g_incoming_exit_filter`, `g_incoming_exit_residuals`, independent band-projector assembly; Figure 7). The massive exit is not yet a filter-independent threshold; the massless exit holds through 10^{-9} and shifts below.

3. With $w = 0$ and $v^2 = 2u^2$ the band average is exactly 2, yet no block level ever leaves $\{0, 1\}$ [T]: $H(k)$ is then constant, the real-space projector is a direct sum of identical cell projectors, and any union of whole cells compresses to an idempotent. The average alone does not force an exit; the coupling and the partition are part of the law.

What survives: on the coupled PT-symmetric chain with whole-cell blocks and filters through 10^{-9}, the census leaves the disk where $E(0)E(\pi) = u^2$, to 4×10^{-8} [V]; why it does is open [C].

9.3 The offset as the exit's clock

For opened matter each level's offset $\tau_\nu = \mathrm{Log}((1-\nu)/\nu)$ is complex, and the rotation used above is $|\operatorname{Im} \tau_\nu|/\pi$. The disk exit is where some level's signed offset has turned a quarter, and the critical point where one has turned a half.

The author's closed form for the endpoint offset of a gapped chain is $2\,\mathrm{artanh}(v/\sqrt{E_{\min}E_{\max}})$ (*The Offset Belongs to the Boundary*). On the opened chain write $\gamma = u/\sqrt{E(0)E(\pi)} \geq 0$, so that $\langle K \rangle - 1 = \gamma^2$ exactly (to 2.8×10^{-14} over 20 000 random parameters). The continuation that carries the opening sends the argument to $i\gamma$:

$$\tau_{\text{end}} = 2\,\mathrm{artanh}(i\gamma) = 2i \arctan \gamma, \qquad \Gamma_{\text{end}} = \tanh \frac{\tau_{\text{end}}}{2} = i\gamma, \qquad \langle K \rangle = 1 + \gamma^2 = \sec^2\left(\tfrac{1}{2} \operatorname{Im} \tau_{\text{end}}\right). \tag{7}$$

At $\langle K \rangle = 2$, $\gamma = 1$: $\tau_{\text{end}} = i\pi/2$ is finite, and $\Gamma_{\text{end}} = i$ lies on the rim [T][V]. The exit line is a quarter turn of this offset, on the imaginary axis of the disk; the real-argument expression $2\,\mathrm{artanh}\,\gamma$, which

diverges at $\gamma = 1$, belongs to a different chart and is not this continuation (A.15). The quarter-turn is not, however, a description of the census rotation away from the exit. Tested before the run with a kill condition, at $u = 0.3, 0.5, 0.7$ and three points on the approach, the measured rotation sits 4×10^{-6} to 3.4×10^{-2} above $\operatorname{Im} \tau_{\mathrm{end}}$, independent of block size; the two meet only at the quarter-turn. The level carrying the largest rotation is often one with ν within 10^{-7} of an endpoint, the ill-conditioned levels of the filter table [V] (script `s11`). Whether a biorthogonal form of $\dot{\tau}^2 \leq 4gS$ governs the rotation is open [C].

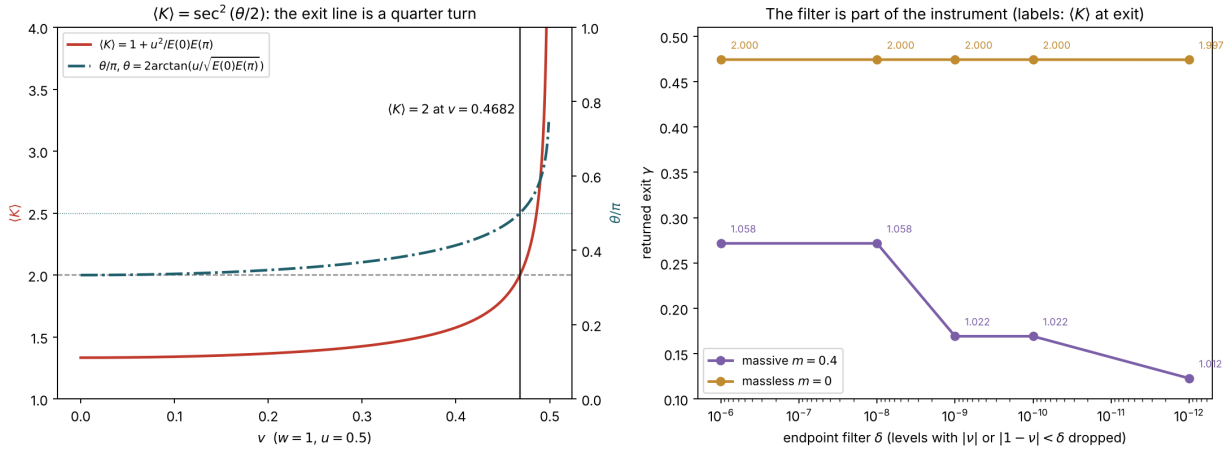


Figure 7: Left: $\langle K \rangle = 1 + u^2/E(0)E(\pi)$ and the continued endpoint phase $\theta = 2\arctan(u/\sqrt{E(0)E(\pi)})$ along v ($u = 0.5$, $w = 1$); $\langle K \rangle = 2$ exactly where $\theta = \pi/2$, at $v = 0.4682$, beside the measured census exit 0.468. Right: the exit returned at five endpoint filters; labels give $\langle K \rangle$ at the returned exit.

9.4 Two kinds of wall

The black-hole paper sorts walls by what the spectrum does along a physical path (ed. 10, §8). At a *spectral* wall a real pair turns complex: the unfolding coordinate is real on one side and imaginary on the other, and the Jordan block at the wall is a property of the operator in every frame. At a *geometric* wall the spectrum stays real on both sides, and the Jordan block lives only in a folded chart [S]. Opened matter's exit is spectral: on $\sigma_x + i\gamma\sigma_z$ the eigenvalues $\pm\sqrt{1-\gamma^2}$ turn imaginary at $\gamma = 1$, the left–right overlap collapses as $\sqrt{1-\gamma^2}$, and the Petermann factor is $(1-\gamma^2)^{-1}$. At such a point a single eigenprojector grows like $|A_{12}|/\mathrm{gap}$ while the pair projector, its contour trace (count 2) and the trace-log ledger stay exact (ed. 10 §4; in its two-mode ringdown model the single excitation factors reach 391 with nearly opposite phases while their sum stays fixed) [V][S]. The band-averaged Petermann factor of §9 is the census-side reading of the same mechanism: what grows is the non-orthogonality of single levels, and what crosses the rim is the record of the pair.

10 The even and odd halves

For an involution J ($J^2 = \mathrm{id}$) on a space of states or operators, every readout splits as $f = f_{\mathrm{even}} + f_{\mathrm{odd}}$ with $f_{\mathrm{even}} = \frac{1}{2}(f + f \circ J)$ and $f_{\mathrm{odd}} = \frac{1}{2}(f - f \circ J)$. Odd parts vanish at fixed points and under J-symmetric pairing or averaging; they do not become false, and they need not vanish merely because a surface is called a wall. Each claim below names its J.

Proposition 10.1 (complement parity)**.** *(i) Under occupation complement $\nu \mapsto 1-\nu$: $\nu(1-\nu)$, every census and every boundary coordinate Γ_b are even; $\Gamma = 1-2\nu$ and the offset τ are odd. (ii) For an orthogonal projector P and Hermitian Grassmannian tangents V, W (block off-diagonal), the map $P \mapsto I-P$, $(V,W) \mapsto (-V,-W)$ sends $Q = \mathrm{Tr}(PVW)$ to $\overline{Q}$: the quantum metric $g = \mathrm{Re}\, Q$ is even and the curvature $\Omega = 2\,\mathrm{Im}\, Q$ is odd. (iii) Under the adjoint, $R_{H^*}(z) = R_H(\bar{z})^*$ for the resolvent $R_H(z) = (z-H)^{-1}$; it is not in general $-R_H(z)$. A Bargmann invariant under complex conjugation has even real part and odd imaginary part. [T][V]*

Proof. (i) is direct. (ii) First, $\overline{Q} = \mathrm{Tr}((PVW)^*) = \mathrm{Tr}(WVP) = \mathrm{Tr}(PWV)$. Off-diagonality means $(I-P)V = (I-P)VP$ and $PW = PW(I-P)$, so $\mathrm{Tr}((I-P)VW) = \mathrm{Tr}((I-P)VPW) = \mathrm{Tr}(PW(I-P)V) = \mathrm{Tr}(PWV) = \overline{Q}$. The image of Q is $\mathrm{Tr}((I-P)(-V)(-W)) = \mathrm{Tr}((I-P)VW) = \overline{Q}$; summing the two traces also gives $\mathrm{Tr}(VW) = 2\,\mathrm{Re}\, Q$. (iii) $(\bar{z}-H)^{-1*} = (z-H^*)^{-1}$. □

Script `s13` checks (ii) on 200 random projectors to 1.5×10^{-14} and confirms that the minus-sign law for the resolvent fails on a random control (largest violation 18.9). This is where the census and the offset meet as one object: the census is the even half of the boundary log-determinant and the offset its odd half, "the gauge without its sign" and "the clock face" (atlas W9, B202, B204). Under the same complement, IQGT's metric is the even half and its Berry curvature the odd half of one Hermitian form, which is why the curvature bound $|\Omega(V,W)| \le 2\sqrt{g(V,V)g(W,W) - g(V,W)^2}$ follows from the positivity of the even part alone (A.17) [T]. The broader reading, that across the author's programme what survives a wall's involution is the even part and what is lost is the odd part, is a reading of particular cases, each with its own J, and is graded so [H]. The black-hole paper's orientation sign, invisible to the Gram observer and kept by the oriented cover, is one such case [S].

11 What is proved, what is measured, what is open

Statement	grade
Zero and span free; a ratio needs a physical zero	[T]
Census logarithm with $A = 2c/\pi^2$ (given equal spacing and Calabrese–Cardy)	[T], [V] at $c = 1, \frac{1}{2}$
$\Gamma = \tanh(\varepsilon/2) = 1-2\nu$; $\int H\, d\varepsilon = \pi^2/3$	[T][L]
Cayley domain with explicit branch; pole only at $z=-1$	[T][L]
Offset = summed rapidity = log-odds empty/full (quasi-free)	[T]
Weight between: defect split, $0 \le X \le F \le \frac{1}{4}$, two-sided, census comparison, trace estimate	[T][L](core)
$\Gamma_b = \sqrt{1-4\lambda}$; $= \lvert 1-2\nu\rvert$ on a pure state	[T][L]
Operator-norm stability with certified margin; near-projector comparison; Hölder coordinate	[T], [V]
Same local block, different weight between; three census counterexamples	[T]
Number covariance; $4\,\mathrm{Tr}\, X_A \le I_G \le \mathrm{Tr}\, X_A/\delta(1-\delta)$; census bound	[T], [V]
Offset speed limit $\dot{\tau}^2 \le 4gS$	[T]
Equal census, offset and coupling; opposite offsets, both saturating	[T][V]
Warm ring: boundary count $2 \to 0$ with $\mathrm{Tr}\, X_A = 0.170835 > 0$	[V]
Peel steps add like velocities; Ising census ladder = constant peel step	[T][L], [V]
Gap step reflects with the peel coordinate	[T][V]
Schur–Cayley boundary: passive inside, matched load at centre	[T]; horizon [S]
Hawking fermion record $\tau = \hbar\omega/k_BT_H$, $\Gamma = \tanh(\hbar\omega/2k_BT_H)$	[T][V]; reading [H]
Anchored coordinate equivalence; helium $\Gamma_0 = 0.377575$	[T][L], [lit]

Statement	grade
Scale blindness; last peel step $5/(8Z-5)$, $Z \geq 2$	[T][L]
Unique neutral Saha state; responses; scaling; atomic record insufficiency	[T][L](three charges)
Gas carries the peel back; partition correction; conditioning	[T], [R], [V]
Unbound nuclei outside; rim distance $2s/(S_{\text{next}} - s)$; half turn	[T][L], [V][lit]
Reduced equation with hidden source; channel measure	[T]
Equal spectrum and footprint, different survival; same kernel, different source	[T][V]
Finite moments do not fix response; readout dependence	[T][L]($m = 3$), [V]
Kinetic witness: same density, flow, stress; $\Delta n \propto \sin^3$; reset factor	[T]
ANG1–ANG3 exact; ANG4 $E^- = k^2\kappa^2/3$	[T], [V]
Opened idempotent: $Y_A = A - A^2$, radial census survives, positivity does not	[T][L]
$\det Q = -u^2w^2 \cos^2 k/16s^3$: Lorentzian iff $s > 0$, $\cos k \neq 0$	[T][V]
Petermann factor; $\langle K \rangle = 1 + u^2/E(0)E(\pi)$	[T][L]
Continued endpoint offset $2i \arctan \gamma$; $\Gamma_{\text{end}} = i$ on the rim at $\langle K \rangle = 2$	[T]
Opened matter exits before critical at $E(0)E(\pi) = u^2$ (PT class, whole cells, filters to 10^{-9})	[V]
Complement parity: census even, offset odd; metric even, curvature odd	[T][V]
What survives a wall is the even half	[H]

Open: why the census rotation meets the offset quarter-turn only at the exit, which needs conditioned eigenvalues for the near-endpoint levels; a proof that the exit happens at $\langle K \rangle = 2$ on the coupled chain; a definition of the massive exit with a controlled limit in filter, ring and block size and the order of limits stated, using conditioning and error bounds rather than an assumed error expansion; a biorthogonal speed limit for the complex offset; convergence on the $v > w$ side; the infinite-dimensional limit of the chain's Henrici departure; placing nuclear resonances with their widths on the chart; a second non-unitary model of a different kind; an empirical gas inverse on a real specimen with measured populations, n_e and partitions; a typed map from the speed limit to the record of a physical system beyond the projector constructions here.

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A Proofs of every result used

This appendix makes the paper standalone. Each result used in the text and not already proved where it is stated is restated and proved here, including those taken from the author's earlier papers. Where a proof rests on an outside input, the input is named and is the only thing assumed. Items marked [Lean] are also checked by machine (Appendix B).

A.1 Zero and span are free (§1). *Statement.* For a Hermitian generator K, a shift $K \mapsto K + cI$ leaves every eigenvector and every eigenvalue difference unchanged, and a positive rescaling $K \mapsto cK$, $c > 0$, leaves every eigenvector, the order of the eigenvalues, every spectral projector $\mathbf{1}[K < 0]$, and so the ground-state correlation matrix and its census, unchanged. *Proof.* $K + cI$ commutes with K and has eigenvalues $\lambda_i + c$, so $\lambda_i - \lambda_j$ is unchanged. cK has the same eigenvectors with eigenvalues $c\lambda_i$; since $c > 0$, $c\lambda_i < 0 \iff \lambda_i < 0$, so the filled projector $P = \mathbf{1}[K < 0]$ is the same operator. The regional correlation matrix $C_A = E^\dagger P E$ and every function of its spectrum, including the census, are unchanged; so are principal angles between spectral subspaces, which depend only on the subspaces. Under a shift the filled projector is $\mathbf{1}[K < -c]$ for the same Fermi level, which is why the sector must be carried with the shift. □

A.2 The census coefficient $A = 2c/\pi^2$ (§1, eq. 1). *Inputs.* (i) The single-particle entanglement spectrum at one cut is equally spaced, $\varepsilon_j = (j + \frac{1}{2})\Delta$ on both signs, as measured on the Ising chain (script `s02`) and derived in *The Angle Between the Rails*. (ii) The entanglement entropy of one cut in a gapped chain near criticality is $S_{\text{cut}} = \frac{c}{6}\ln(\xi/a) + O(1)$ (Calabrese–Cardy). *Derivation.* A level is in the window $\nu \in [\epsilon, 1-\epsilon]$ iff $|\varepsilon| \le \Lambda = \ln((1-\epsilon)/\epsilon)$ (Proposition 2.1). With spacing Δ on both signs, one cut has $N_{\text{cut}} = 2\Lambda/\Delta + O(1)$ levels in the window. Its entropy is $S_{\text{cut}} = \sum_j H(\nu(\varepsilon_j)) = \Delta^{-1}\int H\, d\varepsilon + O(\Delta) = \pi^2/(3\Delta) + O(\Delta)$ by Proposition 2.1 (midpoint rule). Equating with (ii), $1/\Delta = (c/2\pi^2)\ln(\xi/a)$, so $N_{\text{cut}} = (c/\pi^2)\ln(\xi/a)\,\Lambda$, and an interval with two cuts has $A = 2c/\pi^2$. □ The derivation is exact given (i) and (ii); its corrections are those of the two inputs, and the level-counting convention (one complex mode per level) must match the entropy convention used in (ii).

A.3 The bridge, and $\int H\, d\varepsilon = \pi^2/3$ (Proposition 2.1). Proved in the text. The last step uses $\int_0^1 (-\ln\nu)/(1-\nu)\, d\nu = \sum_{n\ge 0}\int_0^1 \nu^n(-\ln\nu)\, d\nu = \sum_{n\ge 1} n^{-2} = \pi^2/6$, and $\int_0^1 H/(\nu(1-\nu))\, d\nu = 2\int_0^1 (-\ln\nu)/(1-\nu)\, d\nu$ by the symmetry $\nu \leftrightarrow 1-\nu$ and $H/(\nu(1-\nu)) = -\ln\nu/(1-\nu) - \ln(1-\nu)/\nu$. The identities $\Gamma = 1 - 2\nu$ and $\nu(1-\nu) = (1-\Gamma^2)/4$ [Lean: `bridge`, `census_square`].

A.4 The offset is the log-odds of empty against full (§2; *The Offset Belongs to the Boundary*, Thm 2). *Statement.* For a number-conserving Gaussian fermion state with $0 < C_A < I$, $\tau = \mathrm{Tr}[\log(I - C_A) - \log C_A] = \log\det(I - C_A) - \log\det C_A = \log P(N_A = 0)/P(N_A = |A|)$. *Proof.* The reduced state of a Gaussian state is Gaussian, $\rho_A \propto \exp(-\sum_k \varepsilon_k b_k^\dagger b_k)$ in the

eigenmodes b_k of C_A, a product over modes with occupation $\nu_k = 1/(1+e^{\varepsilon_k})$. The region is empty iff every mode is empty and full iff every mode is filled, so $P(0) = \prod_k (1-\nu_k) = \det(I - C_A)$ and $P(|A|) = \prod_k \nu_k = \det C_A$. Taking logs, $\tau = \sum_k \ln((1-\nu_k)/\nu_k) = \sum_k \varepsilon_k$. The example in the text: $\mathrm{diag}(0.2, 0.8)$ gives $\ln 4 + \ln \frac{1}{4} = 0$, $\mathrm{diag}(0.2, 0.2)$ gives $2 \ln 4 = \ln 16$, while both have $C - C^2 = 0.16 I$. □

A.5 Defect split for the correlation matrix (Theorem 2.3(1)). Block multiplication gives $(C^2)_{AA} = C_A^2 + C_{AB} C_{BA}$ [Lean: `topleft_sq`; formal suite: `block_defect_identity`]; subtracting it from C_A gives the split. If $C^2 = C$ the first term vanishes [Lean: `idempotent_cross`]. For Hermitian C, $C_{BA} = C_{AB}^\dagger$, so $\mathrm{Tr}\, C_{AB} C_{AB}^\dagger = \|C_{AB}\|_F^2$, and $P_A C (1 - P_A)$ is C_{AB} padded with zeros. The same functional applied to a generator H and a moving projector P is $E_H = \|PH(1-P)\|_F^2$. □

A.6 The boundary coordinate is the modulus, and it survives opening (§2.2, §9). *Statement.* If $C^2 = C$, each eigenvalue λ of X_A is $\nu(1-\nu)$ for an eigenvalue ν of C_A, and $\sqrt{|1 - 4\lambda|} = |1 - 2\nu|$. This holds for the biorthogonal projector $C = \sum_{k\,\text{filled}} |R_k\rangle\langle L_k|$ with $\langle L_j | R_k \rangle = \delta_{jk}$. *Proof.* By A.5, $X_A = C_A - C_A^2 = f(C_A)$ with $f(x) = x - x^2$, so by the spectral mapping theorem (valid for any square matrix, via the Jordan form) its eigenvalues are $f(\nu)$. Then $1 - 4f(\nu) = (1-2\nu)^2$ [Lean: `census_square`], whose modulus has square root $|1 - 2\nu|$ for complex ν too. For the biorthogonal projector, $C^2 = \sum_{j,k} |R_j\rangle\langle L_j | R_k\rangle\langle L_k| = C$. □

A.7 The Offset Speed Limit (Theorem 2.10; *The Offset Speed Limit*, Thm 1 and Cor. 3). *Proof. Step 1 (trace derivative).* $\frac{d}{dt} \mathrm{Tr} \log(I - C) = -\mathrm{Tr}[(I-C)^{-1}\dot{C}]$ and $\frac{d}{dt} \mathrm{Tr} \log C = \mathrm{Tr}[C^{-1}\dot{C}]$; since $(I-C)^{-1} + C^{-1} = (C(I-C))^{-1}$, $\dot{\tau} = \mathrm{Tr}[W\dot{C}]$. *Step 2 (lift).* $C = E^\dagger P E$, so $\dot{\tau} = \mathrm{Tr}[EWE^\dagger \dot{P}] = \mathrm{Tr}[F\dot{P}]$, with F Hermitian. *Step 3 (only the off-diagonal part counts).* Differentiating $P^2 = P$ gives $\dot{P}P + P\dot{P} = \dot{P}$, hence $P\dot{P}P = 0$ and $(1-P)\dot{P}(1-P) = 0$: $\dot{P}$ is block off-diagonal. So $\mathrm{Tr}[F\dot{P}] = \mathrm{Tr}[F_{\text{off}}\dot{P}]$ with $F_{\text{off}} = PF(1-P) + (1-P)FP$, and expanding, $[P,[P,F]] = PF + FP - 2PFP = F_{\text{off}} = T$. *Step 4 (Cauchy–Schwarz).* In the Hilbert–Schmidt inner product, $\dot{\tau}^2 = \langle T, \dot{P}\rangle^2 \le \|T\|_F^2 \|\dot{P}\|_F^2 = 2g\|T\|_F^2$, with equality iff $\dot{P} = \lambda T$, $\lambda \in \mathbb{R}$. *Step 5 ($\|T\|_F^2 = 2S$).* $\|T\|_F^2 = 2\,\mathrm{Tr}[PF(1-P)FP] = 2\,\mathrm{Tr}[W\,E^\dagger(1-P)E\,W\,E^\dagger P E] = 2\,\mathrm{Tr}[W(I - C)WC] = 2\,\mathrm{Tr}[W^2 C(I-C)] = 2S$, because W, C and $I - C$ commute. Hence $\dot{\tau}^2 \le 4gS$. For the corollary, $[H,P] = (1-P)HP - PH(1-P)$, whose two blocks are adjoint up to sign, so $\|\dot{P}\|_F^2 = 2\|PH(1-P)\|_F^2$ and $g = E_H$. □

A.8 Peel steps add like velocities (Theorem 3.1; VOTKP Thm 15.1). Proved in the text. In the division-free form used by the Lean file, $g(z_1) + g(z_2) = g(z_1 z_2)\,(1 + g(z_1) g(z_2))$ for $z_1, z_2 > 0$, with $g(z) = (z-1)/(z+1)$ [Lean: `rapidities_add`]. Since $1 + g_1 g_2 > 0$ for $|g_i| < 1$, this is $g(z_1 z_2) = g_1 \oplus g_2$. The ladder reconstruction is the inverse $z = (1+\Gamma)/(1-\Gamma)$ applied step by step [formal suite: `gamma_inverse`, `gamma_anchor_reconstruction`].

A.9 Scale blindness (§5; VOTKP Thm 15.4). *Statement.* For a nonrelativistic Coulomb system $H_\lambda = \sum_i p_i^2/(2\lambda m_i) + \sum_{i<j} q_i q_j / |r_i - r_j|$, multiplying every mass by $\lambda > 0$ multiplies every energy by λ, so every ratio of removal energies, and every Γ, is unchanged. *Proof.* Let U be the unitary dilation $(U\psi)(r) = \lambda^{3N/2}\psi(\lambda r)$. Then $U^\dagger r_i U = r_i/\lambda$ and $U^\dagger p_i U = \lambda p_i$, so $U^\dagger H_\lambda U = \sum_i \lambda^2 p_i^2/(2\lambda m_i) + \lambda \sum_{i<j} q_i q_j/|r_i - r_j| = \lambda H_1$. Unitarily equivalent operators have the same spectrum, so $\mathrm{spec}\, H_\lambda = \lambda\, \mathrm{spec}\, H_1$, and so do the thresholds of every ion, which are bottoms of spectra of subsystems. Ratios cancel λ. □ Mass *ratios* still enter, which is why the ^{1}H/^{2}H isotope shift is visible (VOTKP §15.3).

A.10 The last peel step at first order (§5; VOTKP §15.10). *Statement.* To first order in the electron repulsion, a helium-like ion of charge $Z \geq 2$ has removal threshold $I_{\mathrm{He}} = Z^2/2 - 5Z/8$ Hartree, the hydrogen-like ion $I_{\mathrm{H}} = Z^2/2$, and the last step has $\Gamma = 5/(8Z-5) \to 5/(8Z)$. *Proof.* The hydrogen-like ground energy is $-Z^2/2$. Two non-interacting $1s$ electrons have $-Z^2$. The first-order repulsion is $J = \iint \rho(r_1)\rho(r_2)/|r_1 - r_2|$ with $\rho = (Z^3/\pi)e^{-2Zr}$. By Gauss's law the potential of ρ is $\phi(r) = 1/r - (Z + 1/r)e^{-2Zr}$, so

$$J = 4Z^3 \int_0^\infty r^2 e^{-2Zr}\phi(r)\, dr = 4Z^3\Big[\tfrac{1}{4Z^2} - \tfrac{2Z}{(4Z)^3} - \tfrac{1}{(4Z)^2}\Big] = 4Z^3 \cdot \tfrac{10}{64Z^2} = \tfrac{5}{8}Z$$

(Unsöld 1927). So $E_{\mathrm{He}} = -Z^2 + 5Z/8$ and $I_{\mathrm{He}} = -Z^2/2 - E_{\mathrm{He}} = Z^2/2 - 5Z/8$, positive for $Z \geq 2$. With $z = I_{\mathrm{H}}/I_{\mathrm{He}}$, $\Gamma = (I_{\mathrm{H}} - I_{\mathrm{He}})/(I_{\mathrm{H}} + I_{\mathrm{He}}) = (5Z/8)/(Z^2 - 5Z/8) = 5/(8Z-5)$ [Lean: `last_step`]. □ The higher terms of the $1/Z$ series (Hylleraas; Scherr–Knight) and the Thomas–Fermi middle of the peel are cited inputs in VOTKP and are not used here.

A.11 Ideal Saha gas: the electron density cancels (§6). *Input.* The ideal Saha equation $n_{i+1}n_e/n_i = (2g_{i+1}/g_i)\lambda_e^{-3}e^{-\beta I_i}$ with constant statistical weights. *Statement.* $n_0 n_2/n_1^2 = (g_0 g_2/g_1^2)\, e^{-\beta(I_1 - I_0)}$ and $n_0/(n_2 n_e^2 \lambda_e^6) = (g_0/4g_2)\, e^{\beta(I_0 + I_1)}$; their β-slopes return $I_1 - I_0$ and $I_0 + I_1$, hence $\Gamma_0 = (I_1 - I_0)/(I_1 + I_0)$. *Proof.* Divide the $i = 1$ equation by the $i = 0$ one: n_e and λ_e cancel [Lean: `saha_ratio`]. Multiply the two equations for the second form, which needs $n_e\lambda_e^3$ per sample. The ratio of the two slopes gives Γ_0. □ This is the $Z = 2$ case of Theorem 6.5.

A.12 Outside the rim, exactly, and the Cayley pole (§7). For $S > s > 0$ and $z = S/(-s)$: $\Gamma = (S+s)/(S-s)$ and $\Gamma - 1 = 2s/(S-s)$; $S_2/S_1 = -1 \iff S_1 + S_2 = 0$ [Lean: `outside_rim`, `rim_distance`, `cayley_pole`]. The half-turn statement is Theorem 2.2.

A.13 No cone along a Hermitian family (§9; cone paper v8, Thm 1). *Statement.* For a two-level family $h(\theta)$ that is Hermitian along the direction of differentiation, with $E_+ \neq E_-$, $Q = M_{+-}M_{-+}/(E_+ - E_-)^2 \geq 0$, where $M_{ab} = \langle L_a|\partial_\theta h|R_b\rangle$. *Proof.* At a Hermitian point $L_a = R_a$; with $\partial_\theta h$ Hermitian, $M_{-+} = \overline{M_{+-}}$, and $E_\pm$ are real. Hence $Q = |M_{+-}|^2/(E_+ - E_-)^2 \geq 0$. □ The hypothesis on the tangent is needed: $h(\theta) = \sigma_z + i\theta\sigma_x$ is Hermitian at $\theta = 0$, $\partial_\theta h = i\sigma_x$ is not, and $Q = (i)(i)/4 = -\frac{1}{4}$.

A.14 The Petermann factor and the exit line (Proposition 9.3). The formula $K = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2)$ follows from $P = \frac{1}{2}(I - h/E)$ and $\operatorname{Tr}\sigma_j\sigma_k = 2\delta_{jk}$: $\operatorname{Tr} P^\dagger P = \frac{1}{4}\operatorname{Tr}[(I - \bar{d}\cdot\sigma/\bar{E})(I - d\cdot\sigma/E)] = \frac{1}{2}(1 + \bar{d}\cdot d/|E|^2)$, the cross terms being traceless. The final equivalence $1 + u^2/p = 2 \iff p = u^2$ for $p > 0$ [Lean: `exit_line`]. The integral $\frac{1}{2\pi}\int_0^{2\pi} dk/(A + B\cos k) = (A^2 - B^2)^{-1/2}$ for $A > |B|$ is the residue at the root of $Bz^2 + 2Az + B$ inside the unit circle, or the substitution $t = \tan(k/2)$ on the two half-intervals.

A.15 The quarter-turn and its branch (§9, eq. 7). Let $p = E(0)E(\pi) > 0$ and $\gamma = u/\sqrt{p}$. For real y, $\operatorname{artanh}(iy) = i\arctan y$, because $\tanh(i\phi) = i\tan\phi$. So $\tau_{\mathrm{end}} = 2i\arctan\gamma$, $\operatorname{Im}\tau_{\mathrm{end}} = \theta = 2\arctan\gamma \in (0,\pi)$, and $\Gamma_{\mathrm{end}} = \tanh(i\arctan\gamma) = i\gamma$. Then $\tan^2(\theta/2) = \gamma^2 = u^2/p$, so $\sec^2(\theta/2) = 1 + u^2/p = \langle K\rangle$, and $\langle K\rangle = 2 \iff \gamma = 1 \iff \theta = \pi/2 \iff \Gamma_{\mathrm{end}} = i$. The continuation of the closed form to the opened chain is a definition; §9 tests it and finds it does not describe the census rotation away from the exit. □

A.16 The bilinear Gram determinant (Theorem 9.2). For complex vectors $d, \alpha, \beta \in \mathbb{C}^3$ with the bilinear product, the Gram matrix G of (d, α, β) is $M^T M$ with $M = (d \,|\, \alpha \,|\, \beta)$, so $\det G = (\det M)^2 = [d \cdot (\alpha \times \beta)]^2$. Expanding $\det G$ along its first row and using $s = d \cdot d$,

$$\det G = s\big[(\alpha\cdot\alpha)(\beta\cdot\beta) - (\alpha\cdot\beta)^2\big] - (d\cdot\alpha)\big[(d\cdot\alpha)(\beta\cdot\beta) - (\alpha\cdot\beta)(d\cdot\beta)\big] + (d\cdot\beta)\big[(d\cdot\alpha)(\alpha\cdot\beta) - (\alpha\cdot\alpha)(d\cdot\beta)\big].$$

The numerator of $\det Q$ is $N_{kk}N_{vv} - N_{kv}^2$ with $N_{ab} = s(\partial_a d \cdot \partial_b d) - (d \cdot \partial_a d)(d \cdot \partial_b d)$; expanding it term by term gives exactly $s \det G$. Hence $16 s^4 \det Q = s[d \cdot (\alpha \times \beta)]^2$. □

A.17 Curvature bounded by the metric (§10; IQGT). For an orthogonal projector P and Hermitian tangents, $Q(V, W) = \mathrm{Tr}(PVW) = \mathrm{Tr}((VP)^\dagger (WP))$, a Frobenius inner product of VP and WP. Its Gram matrix on (V, W) is positive semidefinite, so $|Q(V, W)|^2 \le Q(V, V)Q(W, W)$, with $Q(V, V) = g(V, V)$ real. Writing $Q(V, W) = g(V, W) + \frac{i}{2}\Omega(V, W)$, $g(V, W)^2 + \frac{1}{4}\Omega(V, W)^2 \le g(V, V)g(W, W)$, which is the bound. If $g(V, V) = 0$ then $\Omega(V, W) = 0$. □ A biorthogonal continuation of Q is a different form and need not keep this positivity (Theorem 9.2).

What this appendix does not prove. The numerical statements graded [V] are measurements, and their evidence is the scripts of Appendix C. The two inputs of A.2, the Saha equation, the $1/Z$ coefficients and the Thomas–Fermi law cited in VOTKP, the IQGT relation $g_\parallel \Delta^2 = 9$ of the gap-step family, the membrane resistivity of the horizon, and the NUBASE2020 and NIST data are outside inputs. Why the census leaves the disk exactly at $\langle K \rangle = 2$ on the coupled chain is open (§11).

B The Lean files

Two separately built suites check the algebraic core.

VOTKP.MatterDisk. Module `VOTKP.MatterDisk` of the repository `dicipler-pixel/votkp-lean` (Lean 4 with Mathlib v4.34.1). Its twelve theorems are: `bridge`, `census_square`, `census_complement` (A.3, A.6); `rapidities_add` (A.8); `topleft_sq`, `idempotent_cross` (A.5, A.6); `exit_line` (A.14); `outside_rim`, `rim_distance`, `cayley_pole` (A.12); `saha_ratio` (A.11); `last_step` (A.10). The build record is GitHub Actions run 41729732315 at commit `0e5792a` (152 theorems across the repository; the axiom audit finds only `propext`, `Classical.choice` and `Quot.sound`). False controls live in `FalseControls/`, outside the positive build globs, and are run separately, where each must fail on a mathematical goal.

```
/-
Matter, Then On Down (Jeromie Beasley, draft v3, 3 November 2026).
The algebraic core of the paper, machine-checked.

* The bridge: for an occupation `0 < ν< 1` with odds `z = (1 -ν)/ν`, `(z -1)/(z + 1) = 1 -2ν`.
* The census is the gauge without its sign: `ν(1 -ν) = (1 -(1 -2ν)^2)/4`.
* Peel steps add like velocities: with `g z = (z -1)/(z + 1)`,
  `g z₁ + g z₂ = g (z₁ z₂) ·(1 + g z₁ ·g z₂)` for `z₁, z₂ > 0`.
* Defect split: the top-left block of `C^2` is `A^2 + B E`, so `A -A^2 = (A -(C^2)₁₁) + B E`.
* Idempotence survives opening: if `C^2 = C` then `B E = A -A^2`, with no Hermiticity assumed.
* The exit line: for `p > 0`, `1 + u^2/p = 2 ↔p = u^2`.
* The nucleus: for `S > s > 0` and `z = S/(-s)`, `Γ = (S + s)/(S -s)`, so `Γ -1 = 2s/(S -s)`;
  the Cayley pole `z = -1` is `S_first + S_next = 0`.
* The ideal gas: two Saha ratios give `n₀ n₂ / n₁^2 = S₁/S₀` with the electron density cancelled.
* The last peel step at first order: `z = (Z^2/2)/(Z^2/2 -5Z/8)` gives `Γ = 5/(8Z -5)`.
-/
import Mathlib
```

```
namespace VOTKP.MatterDisk

/-- The Smith coordinate of a ratio. -/
noncomputable def g (z : ℝ) : ℝ:= (z - 1) / (z + 1)

/-- **The bridge.** ‘Γ = 1 -2ν‘ for the odds ‘z = (1 -ν)/ν‘. -/
theorem bridge (ν : ℝ) (h0 : 0 < ν) (h1 : ν< 1) : g ((1 - ν) / ν) = 1 - 2 * ν:= by
  have hν : ν≠0 := h0.ne'
  unfold g
  rw [div_sub_one hν, div_add_one hν, div_div_div_cancel_right₀ hν]
  have h2 : (1 - ν+ ν) = 1 := by ring
  rw [h2, div_one]
  ring

/-- **The census is the gauge without its sign.** -/
theorem census_square (ν : ℝ) : ν* (1 - ν) = (1 - (1 - 2 * ν) ^ 2) / 4 := by ring

/-- The census of a level and of its complement agree. -/
theorem census_complement (ν : ℝ) : ν* (1 - ν) = (1 - ν) * (1 - (1 - ν)) := by ring

/-- **Peel steps add like velocities**, written without dividing by ‘1 + g₁ g₂‘. -/
theorem rapidities_add (z₁ z₂ : ℝ) (h₁ : 0 < z₁) (h₂ : 0 < z₂) :
    g z₁ + g z₂ = g (z₁ * z₂) * (1 + g z₁ * g z₂) := by
  have a : z₁ + 1 ≠0 := by linarith
  have b : z₂ + 1 ≠0 := by linarith
  have c : z₁ * z₂ + 1 ≠0 := by nlinarith
  unfold g
  field_simp
  ring

/-- **Defect split.** The top-left block of ‘C²‘ is ‘A² + BE‘. -/
theorem topleft_sq {m n : Type*} [Fintype m] [Fintype n] [DecidableEq m] [DecidableEq n]
    (A : Matrix m m ℂ) (B : Matrix m n ℂ) (E : Matrix n m ℂ) (D : Matrix n n ℂ) :
    (Matrix.fromBlocks A B E D * Matrix.fromBlocks A B E D).toBlocks₁₁ = A * A + B * E := by
  rw [Matrix.fromBlocks_multiply, Matrix.toBlocks_fromBlocks₁₁]

/-- **Idempotence survives opening.** If ‘C² = C‘ then ‘BE = A -A²‘; no Hermiticity is used,
so this holds for the biorthogonal projector of a non-Hermitian band. -/
theorem idempotent_cross {m n : Type*} [Fintype m] [Fintype n] [DecidableEq m] [DecidableEq n]
    (A : Matrix m m ℂ) (B : Matrix m n ℂ) (E : Matrix n m ℂ) (D : Matrix n n ℂ)
    (hC : Matrix.fromBlocks A B E D * Matrix.fromBlocks A B E D = Matrix.fromBlocks A B E D) :
    B * E = A - A * A := by
  have h := congrArg Matrix.toBlocks₁₁ hC
  rw [topleft_sq, Matrix.toBlocks_fromBlocks₁₁] at h
  calc B * E = (A * A + B * E) - A * A := by abel
    _ = A - A * A := by rw [h]

/-- **The exit line.** With ‘⟨K⟩ = 1 + u²/p‘ and ‘p = E(0)E(π) > 0‘, ‘⟨K⟩ = 2 ↔p = u²‘. -/
theorem exit_line (u p : ℝ) (hp : 0 < p) : 1 + u ^ 2 / p = 2 ↔p = u ^ 2 := by
  have hp' : p ≠0 := hp.ne'
  constructor
  ·intro h
    have : u ^ 2 / p = 1 := by linarith
    rw [div_eq_one_iff_eq hp'] at this
    exact this.symm
  ·intro h
    rw [h, div_self (by rw [← h]; exact hp')]
    norm_num

/-- **The nucleus outside the rim.** For a negative first separation energy ‘-s‘ and a positive
next one ‘S > s‘, ‘Γ = (S + s)/(S -s)‘. -/
theorem outside_rim (S s : ℝ) (hs : 0 < s) (hS : s < S) :
    g (S / (-s)) = (S + s) / (S - s) := by
  have h1 : (-s) ≠0 := by linarith
  have h2 : S - s ≠0 := by linarith
  have h3 : S / (-s) + 1 ≠0 := by
    rw [div_add_one h1]
```

```
    exact div_ne_zero (by linarith) h1
  unfold g
  rw [div_sub_one h1, div_add_one h1, div_div_div_cancel_right₀ h1]
  rw [div_eq_div_iff (by linarith) h2]
  ring

/-- **The rim distance, exactly.** `Γ -1 = 2s/(S -s)`. -/
theorem rim_distance (S s : ℝ) (hs : 0 < s) (hS : s < S) :
    g (S / (-s)) - 1 = 2 * s / (S - s) := by
  have h2 : S - s ≠0 := by linarith
  rw [outside_rim S s hs hS, div_sub_one h2]
  ring_nf

/-- **The Cayley pole.** For `S₁ ≠0`, the ratio `S₂/S₁` is `-1` exactly when `S₁ + S₂ = 0`. -/
theorem cayley_pole (S₁ S₂ : ℝ) (h : S₁ ≠0) : S₂ / S₁ = -1 ↔S₁ + S₂ = 0 := by
  rw [div_eq_iff h]
  constructor <;> intro hh <;> linarith

/-- **The ideal gas.** Two Saha ratios share the electron density, which cancels. -/
theorem saha_ratio (n₀ n₁ n₂ x S₀ S₁ : ℝ) (h₀ : n₀ ≠0) (h₁ : n₁ ≠0) (hx : x ≠0)
    (hS : S₀ ≠0) (e₀ : n₁ / n₀ = S₀ / x) (e₁ : n₂ / n₁ = S₁ / x) :
    n₀ * n₂ / n₁ ^ 2 = S₁ / S₀ := by
  have a : n₁ = S₀ / x * n₀ := by rw [← e₀]; field_simp
  have b : n₂ = S₁ / x * n₁ := by rw [← e₁]; field_simp
  rw [b, a]
  field_simp

/-- **The last peel step at first order.** With `I_H = Z²/2` and `I_He = Z²/2 -5Z/8`
(Hartree), the ratio `z = I_H / I_He` has `Γ = 5/(8Z -5)`. -/
theorem last_step (Z : ℝ) (hZ : 2 ≤Z) :
    g ((Z ^ 2 / 2) / (Z ^ 2 / 2 - 5 * Z / 8)) = 5 / (8 * Z - 5) := by
  have p1 : 0 < Z ^ 2 / 2 - 5 * Z / 8 := by nlinarith
  have h1 : Z ^ 2 / 2 - 5 * Z / 8 ≠0 := p1.ne'
  have h2 : 8 * Z - 5 ≠0 := by linarith
  have p3 : 0 < Z ^ 2 / 2 + (Z ^ 2 / 2 - 5 * Z / 8) := by nlinarith
  unfold g
  rw [div_sub_one h1, div_add_one h1, div_div_div_cancel_right₀ h1]
  rw [div_eq_div_iff p3.ne' h2]
  ring

end VOTKP.MatterDisk
```

MatterGas. The formal suite of the 3 November development: 34 positive declarations compiled under Lean 4.34.1 and Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`, with no `sorry`, `admit`, custom axiom or `native_decide`; five false controls (a common scale changes the Cayley ratio; disconnected thermal variance is zero; the maximum covariance defect is at most 1/5; the first three moments fix the third; mean charge rises with electron density) are rejected on unsolved mathematical goals. Coverage: `Boundary` gives the defect split, positivity, $0 \le BB^* \le A - A^2 \le I/4$, the projector reduction, the disconnected zero and the trace-zero criterion (Theorem 2.3), and the complex identity $1 - 4\nu(1-\nu) = (1-2\nu)^2$ with the idempotent embedding $\mathcal{P}(A)$ (Theorem 9.1); `Scalar` gives Cayley scale invariance, bounds, inverse and anchored reconstruction (Theorem 5.1); `Saha` gives three-charge uniqueness, monotone mean charge and the population cancellations (Theorems 6.2, 6.5); `Witness` gives the $m = 3$ moment witness, with weights $(7, 11, 5, 9)/32$, a different amplitude of the same construction as the text's $(3, 7, 1, 5)/16$. Ordered spectral comparisons, stability, the information bounds, general-Z Saha, the response laws, the kinetic witness and the Petermann statements are proved by hand in this paper and are not machine-checked.

```
import MatterGas.Scalar
import MatterGas.Boundary
```

```
import MatterGas.Saha
import MatterGas.Witness
```

```
/- Finite matrix core. Standard matrix algebra, specialized to the proposed boundary diagnostic.
   No empirical interpretation is encoded in these theorems. -/
import Mathlib.Analysis.Matrix.Order
import Mathlib.Data.Matrix.Block
import Mathlib.Tactic.NoncommRing
import Mathlib.Tactic.Module
import Mathlib.Tactic.NormNum

set_option linter.unusedSectionVars false

namespace MatterGas
open Matrix
open scoped MatrixOrder Matrix.Norms.L2Operator ComplexOrder

variable {m n : Type*} [Fintype m] [Fintype n] [DecidableEq m] [DecidableEq n]

/-- Principal A block of a finite matrix on A⊕B. -/
abbrev localBlock (C : Matrix (m ⊕n) (m ⊕n) ℂ) : Matrix m m ℂ:= C.toBlocks₁₁
/-- Off-diagonal cross block. -/
abbrev crossBlock (C : Matrix (m ⊕n) (m ⊕n) ℂ) : Matrix m n ℂ:= C.toBlocks₁₂
/-- Cross-boundary one-body matrix, independent of Gaussian/physical interpretation. -/
def boundary (C : Matrix (m ⊕n) (m ⊕n) ℂ) : Matrix m m ℂ:= crossBlock C * (crossBlock C)ᴴ
/-- Global failure of idempotency, restricted to A. -/
def thermalDefect (C : Matrix (m ⊕n) (m ⊕n) ℂ) : Matrix m m ℂ:= (C-C*C).toBlocks₁₁

/-- Exact block identity for a Hermitian finite matrix. -/
theorem block_defect_identity (C : Matrix (m ⊕n) (m ⊕n) ℂ) (hC : C.IsHermitian) :
    localBlock C - localBlock C * localBlock C = thermalDefect C + boundary C := by
  ext i j
  simp only [localBlock, thermalDefect, boundary, crossBlock, Matrix.toBlocks₁₁,
    Matrix.toBlocks₁₂, Matrix.of_apply, Matrix.sub_apply, Matrix.add_apply,
    Matrix.mul_apply, Matrix.conjTranspose_apply, Fintype.sum_sum_type]
  have hh (k : n) : star (C (Sum.inl j) (Sum.inr k)) = C (Sum.inr k) (Sum.inl j) := by
    exact congrFun (congrFun hC.eq (Sum.inr k)) (Sum.inl j)
  simp only [hh]
  ring

/-- The cross-boundary matrix is always positive semidefinite. -/
theorem boundary_posSemidef (C : Matrix (m ⊕n) (m ⊕n) ℂ) :
    (boundary C).PosSemidef := Matrix.posSemidef_self_mul_conjTranspose _

/-- A positive Hermitian contraction has a positive idempotency defect. -/
theorem contraction_defect_posSemidef (C : Matrix (m ⊕n) (m ⊕n) ℂ)
    (h0 : 0 ≤C) (h1 : C ≤1) : (C-C*C).PosSemidef := by
  have hc : Commute C (1-C) := by
    show C*(1-C) = (1-C)*C
    noncomm_ring
  have hp := Commute.mul_nonneg h0 (sub_nonneg.mpr h1) hc
  simpa only [mul_sub, mul_one] using hp.posSemidef

/-- Restricting the positive defect preserves positivity. -/
theorem thermalDefect_posSemidef (C : Matrix (m ⊕n) (m ⊕n) ℂ)
    (h0 : 0 ≤C) (h1 : C ≤1) : (thermalDefect C).PosSemidef := by
  exact (contraction_defect_posSemidef C h0 h1).submatrix Sum.inl

/-- The boundary matrix is bounded by the local covariance polynomial. -/
theorem boundary_le_local_defect (C : Matrix (m ⊕n) (m ⊕n) ℂ)
    (h0 : 0 ≤C) (h1 : C ≤1) :
    boundary C ≤localBlock C - localBlock C * localBlock C := by
  rw [block_defect_identity C h0.posSemidef.isHermitian]
  exact le_add_of_nonneg_left (thermalDefect_posSemidef C h0 h1).nonneg

/-- Idempotency kills the thermal term exactly. -/
theorem thermalDefect_projector (C : Matrix (m ⊕n) (m ⊕n) ℂ) (hp : C*C=C) :
```

```
    thermalDefect C = 0 := by
  ext i j
  simp [thermalDefect, hp, Matrix.toBlocks₁₁]

/-- Pure-projector reduction of the boundary construction. -/
theorem boundary_projector (C : Matrix (m ⊕n) (m ⊕n) ℂ)
    (hC : C.IsHermitian) (hp : C*C=C) :
    boundary C = localBlock C - localBlock C * localBlock C := by
  simpa [thermalDefect_projector C hp] using (block_defect_identity C hC).symm

/-- Disconnected blocks have no cross-boundary signal. -/
theorem boundary_disconnected (A : Matrix m m ℂ) (D : Matrix n n ℂ) :
    boundary (Matrix.fromBlocks A 0 0 D) = 0 := by
  simp [boundary, crossBlock]

/-- Vanishing cross-boundary trace is equivalent to a vanishing cross block. -/
theorem boundary_trace_zero_iff (C : Matrix (m ⊕n) (m ⊕n) ℂ) :
    (boundary C).trace = 0 ↔crossBlock C = 0 := by
  exact Matrix.trace_mul_conjTranspose_self_eq_zero_iff

/-- The local covariance polynomial is bounded by one quarter of the identity.
This holds for every Hermitian A, independently of a contraction hypothesis. -/
theorem local_defect_le_quarter (A : Matrix m m ℂ) (hA : A.IsHermitian) :
    A-A*A ≤(1/4:ℝ) •(1 : Matrix m m ℂ) := by
  let S : Matrix m m ℂ:= (1/2:ℝ) •(1 : Matrix m m ℂ) - A
  have hS : S^H = S := by simp [S, hA.eq]
  have hpos := Matrix.posSemidef_conjTranspose_mul_self S
  rw [hS] at hpos
  rw [Matrix.le_iff]
  convert hpos using 1
  dsimp [S]
  simp only [sub_mul, mul_sub, smul_mul_assoc, mul_smul_comm,
    one_mul, mul_one]
  module

/-- Full finite-matrix boundary bound for every Hermitian contraction. -/
theorem boundary_bounds (C : Matrix (m ⊕n) (m ⊕n) ℂ)
    (h0 : 0 ≤C) (h1 : C ≤1) :
    0 ≤boundary C ∧boundary C ≤(1/4:ℝ) •(1 : Matrix m m ℂ) := by
  refine ⟨(boundary_posSemidef C).nonneg, ?_⟩
  exact (boundary_le_local_defect C h0 h1).trans
    (local_defect_le_quarter (localBlock C) (h0.posSemidef.isHermitian.submatrix Sum.inl))

/-- Exact coordinate change in the square-and-offset disk construction. -/
theorem square_offset_identity (nu : ℂ) :
    1-4*(nu*(1-nu)) = (1-2*nu)^2 := by ring

/-- Any finite square matrix embeds as the A block of an idempotent.
The embedding need not be Hermitian; positivity must not be inferred from idempotency alone. -/
theorem arbitrary_block_idempotent (A : Matrix m m ℂ) :
    let C := Matrix.fromBlocks A (1 : Matrix m m ℂ) (A-A*A) (1-A)
    C*C=C := by
  dsimp
  rw [Matrix.fromBlocks_multiply]
  congr 1 <;> noncomm_ring

end MatterGas
```

```
/-
Finite scalar core for Matter, Gases and VOTKP.
Prepared for Jeromie Beasley, 2026-11-03, with AI assistance.
All statements below are algebra/order facts, not empirical laws.
-/
import Mathlib.Basic.Real.Basic
import Mathlib.Tactic.Linarith
import Mathlib.Tactic.FieldSimp
import Mathlib.Tactic.Ring
```

```
import Mathlib.Tactic.NormNum
import Mathlib.Algebra.Order.BigOperators.Group.Finset

namespace MatterGas

/-- The positive-threshold Cayley coordinate. -/
noncomputable def gamma (a b : ℝ) : ℝ:= (b - a) / (b + a)

/-- Positive common rescaling leaves the coordinate unchanged. -/
theorem gamma_scale (a b c : ℝ) (hc : c ≠0) :
    gamma (c*a) (c*b) = gamma a b := by
  unfold gamma
  rw [← mul_sub, ←mul_add, mul_div_mul_left _ _ hc]

/-- A positive threshold pair has a coordinate strictly between -1 and 1. -/
theorem gamma_bounds (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    -1 < gamma a b ∧gamma a b < 1 := by
  have hden : 0 < b + a := by linarith
  unfold gamma
  constructor
  ·apply (lt_div_iff₀ hden).2
    linarith
  ·apply (div_lt_iff₀ hden).2
    linarith

/-- The coordinate together with its positive absolute anchor reconstructs b. -/
theorem gamma_inverse (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    (1 + gamma a b) / (1 - gamma a b) = b / a := by
  have ha0 : a ≠0 := ne_of_gt ha
  have hden : b+a ≠0 := ne_of_gt (add_pos hb ha)
  have hm : 1 - gamma a b ≠0 := ne_of_gt (sub_pos.mpr (gamma_bounds a b ha hb).2)
  unfold gamma at hm ⊢
  field_simp [ha0, hden, hm]
  ring_nf
  calc
    a*b*a⁻¹ = b*(a*a⁻¹) := by ring
    _ = b := by rw [mul_inv_cancel₀ ha0, mul_one]

/-- The ratio coordinate alone loses an absolute common scale. -/
theorem gamma_anchor_reconstruction (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    a * ((1 + gamma a b) / (1 - gamma a b)) = b := by
  rw [gamma_inverse a b ha hb]
  field_simp [ne_of_gt ha]

/-- Scalar polynomial behind the pure-projector census equivalence. -/
theorem census_factorization (nu eps : ℝ) :
    nu*(1-nu) - eps*(1-eps) = (nu-eps)*(1-eps-nu) := by ring

/-- Threshold equivalence, including endpoints. No spectral theorem is assumed here. -/
theorem census_threshold_iff (nu eps : ℝ) (he : eps < (1:ℝ)/2) :
    eps*(1-eps) ≤nu*(1-nu) ↔eps ≤nu ∧nu ≤1-eps := by
  constructor
  ·intro h
    constructor
    ·by_contra hn
      have hn' : nu < eps := lt_of_not_ge hn
      have hp : 0 < 1-eps-nu := by linarith
      have hnprod := mul_neg_of_neg_of_pos (sub_neg.mpr hn') hp
      nlinarith [census_factorization nu eps]
    ·by_contra hn
      have hn' : 1-eps < nu := lt_of_not_ge hn
      have hp : 0 < nu-eps := by linarith
      have hnprod := mul_neg_of_pos_of_neg hp (by linarith : 1-eps-nu < 0)
      nlinarith [census_factorization nu eps]
  ·rintro ⟨hlo,hhi⟩
    have h := mul_nonneg (sub_nonneg.mpr hlo) (by linarith : 0 ≤1-eps-nu)
    nlinarith [census_factorization nu eps]
```

```
/-- On [0,1] the scalar covariance defect lies in [0,1/4]. -/
theorem scalar_defect_bounds (nu : ℝ) (hlo : 0 ≤nu) (hhi : nu ≤1) :
    0 ≤nu*(1-nu) ∧nu*(1-nu) ≤(1:ℝ)/4 := by
  constructor
  ·exact mul_nonneg hlo (sub_nonneg.mpr hhi)
  ·nlinarith [sq_nonneg (nu - (1:ℝ)/2)]

/-- A thresholded count is bounded by the sum, for any finite nonnegative list.
This is the scalar counting ingredient; spectral identification is separate. -/
theorem threshold_count_trace_bound {ι : Type*} [Fintype ι]
    (lambda : ι→ℝ) (h : ∀i, 0 ≤lambda i) (theta : ℝ) (_ht : 0 < theta) :
    theta * ((Finset.univ.filter (fun i => theta ≤lambda i)).card : ℝ)
      ≤∑i, lambda i := by
  classical
  calc
    theta * ((Finset.univ.filter (fun i => theta ≤lambda i)).card : ℝ)
      = ∑i ∈Finset.univ.filter (fun i => theta ≤lambda i), theta := by simp [mul_comm]
    _ ≤∑i ∈Finset.univ.filter (fun i => theta ≤lambda i), lambda i := by
      apply Finset.sum_le_sum
      intro i hi
      exact (Finset.mem_filter.mp hi).2
    _ ≤∑i, lambda i := by
      apply Finset.sum_le_sum_of_subset_of_nonneg
      ·exact Finset.filter_subset _ _
      ·intro i _ _
        exact h i

/-- Raw mixed occupation persists in a disconnected example. -/
theorem disconnected_occupation_witness :
    (0:ℝ) < (3/10:ℝ)*(1-3/10) ∧(3/10:ℝ)*(1-3/10) = 21/100 := by norm_num

end MatterGas
```

```
/- Exact uniqueness core for the three-charge ideal-equilibrium reduction.
   a and b are positive equilibrium coefficient products, not fitted claims. -/
import Mathlib.Basic.Real.Basic
import Mathlib.Tactic.Linarith
import Mathlib.Tactic.FieldSimp
import Mathlib.Tactic.Ring
import Mathlib.Tactic.Positivity

namespace MatterGas
noncomputable def meanCharge (a b x : ℝ) : ℝ:= (a*x+2*b)/(x^2+a*x+b)

/-- The denominator of the three-charge population model is strictly positive. -/
theorem meanCharge_den_pos (a b x : ℝ) (ha : 0<a) (hb : 0<b) (hx : 0<x) :
    0 < x^2+a*x+b := by positivity

/-- The mean charge decreases strictly with electron density in this model. -/
theorem meanCharge_strictAnti (a b x y : ℝ) (ha : 0<a) (hb : 0<b)
    (hx : 0<x) (hxy : x<y) : meanCharge a b y < meanCharge a b x := by
  have hy : 0<y := lt_trans hx hxy
  unfold meanCharge
  apply (div_lt_div_iff₀ (meanCharge_den_pos a b y ha hb hy)
    (meanCharge_den_pos a b x ha hb hx)).2
  have hprod : 0 < (y-x)*(a*x*y + 2*b*(x+y)+a*b) := by positivity
  have hfactor : (a*x+2*b)*(y^2+a*y+b) - (a*y+2*b)*(x^2+a*x+b) =
    (y-x)*(a*x*y+2*b*(x+y)+a*b) := by ring
  linarith

/-- At most one positive charge-neutral electron density exists.
Existence and a full Saha/LTE derivation are separate manuscript claims. -/
theorem saha_neutrality_unique (a b N x y : ℝ) (ha : 0<a) (hb : 0<b) (hN : 0<N)
    (hx : 0<x) (hy : 0<y) (hxeq : x = N*meanCharge a b x)
    (hyeq : y = N*meanCharge a b y) : x=y := by
  rcases lt_trichotomy x y with hxy | heq | hyx
  ·have h := mul_lt_mul_of_pos_left (meanCharge_strictAnti a b x y ha hb hx hxy) hN
```

```
    linarith
  ·exact heq
  ·have h := mul_lt_mul_of_pos_left (meanCharge_strictAnti a b y x ha hb hy hyx) hN
    linarith

/-- At any positive neutral solution the mean charge is strictly between zero and two. -/
theorem meanCharge_bounds (a b x : ℝ) (ha : 0<a) (hb : 0<b) (hx : 0<x) :
    0 < meanCharge a b x ∧meanCharge a b x < 2 := by
  have hd := meanCharge_den_pos a b x ha hb hx
  unfold meanCharge
  constructor
  ·positivity
  ·apply (div_lt_iff₀ hd).2
    have hx2 : 0 < x^2 := sq_pos_of_pos hx
    have hax := mul_pos ha hx
    nlinarith

/-- Adjacent population ratios cancel electron density in their quotient. -/
theorem saha_ratio_cancellation (n0 n1 n2 x S0 S1 : ℝ)
    (hn0 : n0 ≠0) (hn1 : n1 ≠0) (hx : x ≠0) (hS0 : S0 ≠0)
    (h01 : n1/n0 = S0/x) (h12 : n2/n1 = S1/x) :
    n0*n2/n1^2 = S1/S0 := by
  have h01' := (div_eq_iff hn0).mp h01
  have h12' := (div_eq_iff hn1).mp h12
  rw [h12', h01']
  field_simp [hn0, hx, hS0]

/-- The product of adjacent population ratios supplies the absolute product channel. -/
theorem saha_product_cancellation (n0 n1 n2 x S0 S1 : ℝ)
    (hn0 : n0 ≠0) (hn1 : n1 ≠0) (hx : x ≠0) (hS0 : S0 ≠0) (hS1 : S1 ≠0)
    (h01 : n1/n0 = S0/x) (h12 : n2/n1 = S1/x) :
    n0/(n2*x^2) = 1/(S0*S1) := by
  have h01' := (div_eq_iff hn0).mp h01
  have h12' := (div_eq_iff hn1).mp h12
  rw [h12', h01']
  field_simp [hn0, hx, hS0, hS1]

end MatterGas
```

```
/- An exact positive four-velocity witness. It proves finite moment statements,
   not a kinetic equation or an experimental prediction. -/
import Mathlib.Basic.Real.Basic
import Mathlib.Tactic.Ring
import Mathlib.Tactic.NormNum

namespace MatterGas

/-- Four-velocity moment with support (-3,-1,1,3). -/
def fourMoment (w0 w1 w2 w3 : ℝ) (k : ℕ) : ℝ:=
  w0*(-3)^k + w1*(-1)^k + w2*1^k + w3*3^k

/-- The finite-difference null vector annihilates every quadratic polynomial. -/
theorem quadratic_moment_null (a b c : ℝ) :
    -(a*(-3)^2+b*(-3)+c) + 3*(a*(-1)^2+b*(-1)+c)
      - 3*(a*1^2+b*1+c) + (a*3^2+b*3+c) = 0 := by ring

/-- Both distributions have mass one and strictly positive weights. -/
theorem four_moment_weights :
    (1/4:ℝ)+(1/4)+(1/4)+(1/4)=1 ∧
    (7/32:ℝ)+(11/32)+(5/32)+(9/32)=1 ∧
    (0:ℝ)<7/32 ∧(0:ℝ)<11/32 ∧(0:ℝ)<5/32 ∧(0:ℝ)<9/32 := by norm_num

/-- Exact equal zeroth moment. -/
theorem four_moment_zero :
    fourMoment (1/4) (1/4) (1/4) (1/4) 0 =
    fourMoment (7/32) (11/32) (5/32) (9/32) 0 := by norm_num [fourMoment]

/-- Exact equal first moment (zero mean velocity). -/
```

```
theorem four_moment_one :
    fourMoment (1/4) (1/4) (1/4) (1/4) 1 =
    fourMoment (7/32) (11/32) (5/32) (9/32) 1 := by norm_num [fourMoment]

/-- Exact equal second moment (the value is five). -/
theorem four_moment_two :
    fourMoment (1/4) (1/4) (1/4) (1/4) 2 = 5 ∧
    fourMoment (7/32) (11/32) (5/32) (9/32) 2 = 5 := by norm_num [fourMoment]

/-- The equal low-moment records do not determine the third moment. -/
theorem four_moment_three_separation :
    fourMoment (7/32) (11/32) (5/32) (9/32) 3 -
    fourMoment (1/4) (1/4) (1/4) (1/4) 3 = (3/2:ℝ) := by norm_num [fourMoment]

end MatterGas
```

C Scripts and outputs

Every script below is printed whole, followed by its verbatim output. They run with Python 3, numpy, scipy, sympy and matplotlib; the nuclear script also needs pynucastro (for the NUBASE2020 file). Repository: `dicipler-pixel/votkp-lean`, folder `matter-disk/`. Scripts `s06`, `s07`, `s08`, `s11`, `s12` and `s13` carry a kill condition written before their run; the others are observational scans or figure builders and carry none. `s03c` is the raw metric scan that `s03d` repeats with a rank check. The `g` scripts are the formal development's own checks, rerun for this edition from `external/gpt_formal_20261103`; the gas checks follow a protocol fixed before their execution. `s09` reads the formal package from a local checkout path and its printed output is the record of that run.

`s01_bridge.py`

```
# SCRIPT: MATTER-PEEL-01
# Joins Matter at a Scale (census ladder), VOTKP (Smith/peel coordinate) and the Cone (indefinite NH metric).
import numpy as np
from scipy.integrate import quad
print('='*78); print('MATTER-PEEL-01'); print('='*78)
# (1) identity Gamma = 1-2nu = tanh(eps/2), eps = ln((1-nu)/nu)
nus=np.array([1e-6,0.01,0.2,0.37,0.5,0.73,0.999])
z=(1-nus)/nus; G=(z-1)/(z+1)
print('(1) max|Gamma-(1-2nu)| =',np.max(abs(G-(1-2*nus))),' max|Gamma-tanh(ln z/2)| =',np.max(abs(G-np.tanh(np
    .log(z)/2))))
# (2) entropy over peel rapidity
H=lambda e: np.log(1+np.exp(e))-e*np.exp(e)/(1+np.exp(e)) if e<30 else np.exp(-e)*(1+e)
def Hg(e):
    g=np.tanh(e/2); a,b=(1+g)/2,(1-g)/2
    return -sum(p*np.log(p) for p in (a,b) if p>0)
val,_=quad(Hg,-80,80,limit=500,epsabs=1e-13)
print('(2) int H d(eps) over rapidity =',val,' pi^2/3 =',np.pi**2/3,' diff',abs(val-np.pi**2/3))
```

Output:

```
==============================================================================
MATTER-PEEL-01
==============================================================================
(1) max|Gamma-(1-2nu)| = 1.1102230246251565e-16 max|Gamma-tanh(ln z/2)| = 1.1102230246251565e-16
(2) int H d(eps) over rapidity = 3.2898681336964595 pi^2/3 = 3.289868133696453 diff 6.661338147750939e-15

real 0m0.982s
user 0m0.557s
sys 0m0.121s
```

s02_ladder.py

```
# SCRIPT: MATTER-PEEL-02A constant peel step on the measured c=1/2 entanglement ladder
import numpy as np
def binary_H(x):
    x=np.clip(x,1e-15,1-1e-15); return -(x*np.log(x)+(1-x)*np.log(1-x))
def ladder(h,N,mult=6):
    xi=1/np.log(h); L=int(mult*xi)
    k=2*np.pi*(np.arange(N)+0.5)/N
    e=np.sqrt((h-np.cos(k))**2+np.sin(k)**2); ex=((h-np.cos(k))-1j*np.sin(k))/e
    l=np.arange(-L,L+1); gr=np.array([(ex*np.exp(1j*k*ll)).sum().real/N for ll in l])
    G=np.zeros((2*L,2*L)); ar=np.arange(L)
    for m in range(L):
        G[2*m,1::2]=gr[L-m:L-m+L]; G[2*m+1,0::2]=-gr[L+m-ar]
    nu=np.sort(np.abs(np.linalg.eigvalsh(1j*G)))[::2][::-1]
    p=np.clip((1+nu)/2,0.5,1-1e-15); em=np.sort(np.log(p/(1-p))); em=em[em>1e-10]
    return xi,em
op=lambda u,v:(u+v)/(1+u*v)
for h,N in [(1.05,8192),(1.02,8192),(1.008,16384)]:
    xi,em=ladder(h,N)
    lev=em[::2][:4] # distinct levels (endpoint doublets): eps0,3eps0,5eps0,7eps0
    e0=lev[0]
    unif=np.max(np.abs(lev/e0-np.array([1,3,5,7])))
    Gm=np.tanh(lev/2); step=np.tanh(e0)
    res=[abs(Gm[j+1]-op(Gm[j],step)) for j in range(3)]
    print(f'xi={xi:7.1f} eps0={e0:.4f} Gamma levels={np.round(Gm,5)} peel step tanh(eps0)={step:.5f}')
    print(f' ladder uniformity dev {unif:.4f} | max |Gamma_j+1 - (Gamma_j (+) step)| = {max(res):.2e} (rel to
        step {max(res)/step:.2e})')
```

Output:

```
xi= 20.5 eps0=1.9346 Gamma levels=[0.74751 0.99398 0.99987 1. ] peel step tanh(eps0)=0.95910
  ladder uniformity dev 0.0221 | max |Gamma_j+1 - (Gamma_j (+) step)| = 7.99e-06 (rel to step 8.33e-06)
xi= 50.5 eps0=1.6440 Gamma levels=[0.67615 0.98567 0.99946 0.99998] peel step tanh(eps0)=0.92803
  ladder uniformity dev 0.0113 | max |Gamma_j+1 - (Gamma_j (+) step)| = 5.45e-06 (rel to step 5.87e-06)
xi= 125.5 eps0=1.4275 Gamma levels=[0.61302 0.97276 0.99841 0.99991] peel step tanh(eps0)=0.89115
  ladder uniformity dev 0.0062 | max |Gamma_j+1 - (Gamma_j (+) step)| = 3.23e-06 (rel to step 3.63e-06)

real 0m3.388s
user 0m3.255s
sys 0m0.052s
```

nhssh_lib_and_s03a_occupations.py

```
# SCRIPT: MATTER-PEEL-02B opened matter: NH SSH h(k)=[[iu, v+w e^{-ik}],[v+w e^{ik}, -iu]]
import numpy as np
def bands(v,w,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T # columns: left vectors with <L_m|R_n>=delta
    return ev,R,L
def corr(v,w,u,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N
    C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=bands(v,w,u,k)
        P=np.outer(R[:,0],L[:,0].conj()) # occupied lower band, biorthogonal projector
        ph=np.exp(1j*k*np.arange(Lc))
        C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def S_of(nu):
    nu=nu.astype(complex); return -np.sum(nu*np.log(nu)+(1-nu)*np.log(1-nu))
w=1.0
print('B1/B2: biorthogonal occupations -> Gamma=1-2nu, entropy slope')
```

```
for (v,u,tag) in [(0.5,0.0,'Hermitian gapped control'),(1.0,0.0,'Hermitian critical control (c=1)'),(0.5,0.3,'
     NH gapped (|v-w|>u)'),(0.5,0.5,'NH critical |v-w|=u (EP)')]:
    N=2000; Ls=[20,40,80,160]; Ss=[]; worst=0; maxim=0
    for Lc in Ls:
        nu=np.linalg.eigvals(corr(v,w,u,N,Lc)); G=1-2*nu
        worst=max(worst,np.max(np.abs(G))); maxim=max(maxim,np.max(np.abs(G.imag)))
        Ss.append(S_of(nu))
    x=np.log(N/np.pi*np.sin(np.pi*np.array(Ls)/N))
    sl=np.polyfit(x,np.real(Ss),1)[0]
    print(f' {tag:36s} max|Gamma|={worst:9.3e} max|Im Gamma|={maxim:8.2e} c=3*slope={3*sl:+.3f} S(Im)~{np.max(
         np.abs(np.imag(Ss))):.1e}')
```

Output:

```
B1/B2: biorthogonal occupations -> Gamma=1-2nu, entropy slope
  Hermitian gapped control max|Gamma|=1.000e+00 max|Im Gamma|=6.08e-16 c=3*slope=-0.000 S(Im)~9.7e-13
  Hermitian critical control (c=1) max|Gamma|=1.000e+00 max|Im Gamma|=5.69e-16 c=3*slope=+1.000 S(Im)~1.4e-13
  NH gapped (|v-w|>u) max|Gamma|=1.000e+00 max|Im Gamma|=4.20e-01 c=3*slope=+0.000 S(Im)~3.1e-13
  NH critical |v-w|=u (EP) max|Gamma|=3.057e+00 max|Im Gamma|=3.06e+00 c=3*slope=+1.321 S(Im)~2.6e-01

real 0m15.243s
user 0m15.037s
sys 0m0.036s
```

s03b_renyi_central_charge.py

```
# SCRIPT: MATTER-PEEL-02B2 branch-safe Renyi-2 at the NH SSH EP critical point
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split("w=1.0")[0])
w=1.0
def run(v,u,N,Ls):
    S2=[];Sv=[]
    for Lc in Ls:
        nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
        S2.append(-np.sum(np.log(nu**2+(1-nu)**2)))
    x=np.log(N/np.pi*np.sin(np.pi*np.array(Ls)/N))
    S2=np.array(S2); sl=np.polyfit(x,S2.real,1)[0]
    return 4*sl, np.max(np.abs(S2.imag)), S2.real
for (v,u,tag) in [(1.0,0.0,'Hermitian critical control'),(0.5,0.5,'EP critical, v<w'),(1.5,0.5,'EP critical, v
     >w'),(0.5,0.3,'NH gapped control')]:
    for N in (1000,2000):
        c,im,S=run(v,u,N,[10,20,40,80,160])
        print(f'{tag:28s} N={N}: c from Renyi-2 = {c:+.3f} max|Im S2|={im:.1e} S2={np.round(S,3)}')
```

Output:

```
Hermitian critical control N=1000: c from Renyi-2 = +1.007 max|Im S2|=7.7e-15 S2=[1.321 1.497 1.671 1.843
     2.008]
Hermitian critical control N=2000: c from Renyi-2 = +1.007 max|Im S2|=2.2e-15 S2=[1.321 1.497 1.671 1.845
     2.016]
EP critical, v<w N=1000: c from Renyi-2 = -1.792 max|Im S2|=6.3e+00 S2=[-1.942 -2.351 -2.69 -2.963 -3.166]
EP critical, v<w N=2000: c from Renyi-2 = -1.956 max|Im S2|=6.3e+00 S2=[-2.209 -2.633 -2.997 -3.307 -3.56 ]
EP critical, v>w N=1000: c from Renyi-2 = -0.876 max|Im S2|=3.7e-14 S2=[-0.901 -1.074 -1.238 -1.383 -1.495]
EP critical, v>w N=2000: c from Renyi-2 = -1.089 max|Im S2|=1.1e-13 S2=[-1.055 -1.257 -1.455 -1.641 -1.803]
NH gapped control N=1000: c from Renyi-2 = +0.000 max|Im S2|=2.7e-15 S2=[1.778 1.778 1.778 1.778 1.778]
NH gapped control N=2000: c from Renyi-2 = +0.000 max|Im S2|=1.6e-14 S2=[1.778 1.778 1.778 1.778 1.778]

real 0m26.878s
user 0m26.613s
sys 0m0.024s
```

s03c_metric_raw.py

```
# SCRIPT: MATTER-PEEL-02B3 signature of the round-trip form on the (k, v) plane, NH SSH
import numpy as np
w=1.0
def Hm(v,u,k): return np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
def form(v,u,k):
    H=Hm(v,u,k); ev,R=np.linalg.eig(H); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T
    dk=np.array([[0,-1j*w*np.exp(-1j*k)],[1j*w*np.exp(1j*k),0]]); dv=np.array([[0,1],[1,0]],complex)
    def Q(V): return (L[:,1].conj()@V@R[:,0])*(L[:,0].conj()@V@R[:,1])/(ev[1]-ev[0])**2
    a=Q(dk).real; c=Q(dv).real; b=Q((dk+dv)/np.sqrt(2)).real-(a+c)/2
    return a,b,c
def scan(u):
    ks=np.linspace(0.02,2*np.pi-0.02,180); vs=np.linspace(0.05,1.95,96)
    lor=np.zeros((len(vs),len(ks)),bool); defin=np.zeros_like(lor)
    for i,v in enumerate(vs):
        for j,k in enumerate(ks):
            if abs(v+w*np.exp(1j*k))**2-u**2 < 1e-6: continue # skip exceptional points themselves
            a,b,c=form(v,u,k); d=a*c-b*b
            lor[i,j]=d<0; defin[i,j]=d>0
    return ks,vs,lor,defin
for u in (0.0,0.5):
    ks,vs,lor,defin=scan(u)
    print(f'u={u}: Lorentzian fraction {lor.mean():.3f} definite fraction {defin.mean():.3f}')
    if lor.any():
        idx=np.argwhere(lor); kk=ks[idx[:,1]]; vv=vs[idx[:,0]]
        dist=np.min(np.hypot(kk-np.pi, vv-0.5))
        print(f' Lorentzian cells span k in [{kk.min():.2f},{kk.max():.2f}], v in [{vv.min():.2f},{vv.max():.2f
            }]; nearest to EP (pi,0.5): {dist:.3f}')
        # PT-broken region |v+w e^{ik}|<u
        brk=np.array([[abs(v+w*np.exp(1j*k))<u for k in ks] for v in vs])
        print(f' Lorentzian cells inside PT-broken zone: {np.mean(brk[lor]):.3f}; PT-broken cells that are
            Lorentzian: {np.mean(lor[brk]) if brk.any() else float("nan"):.3f}')
        np.save('lor_u05.npy',lor); np.save('brk_u05.npy',brk)
```

Output:

```
u=0.0: Lorentzian fraction 0.338 definite fraction 0.525
  Lorentzian cells span k in [0.02,6.26], v in [0.05,1.95]; nearest to EP (pi,0.5): 0.020
  Lorentzian cells inside PT-broken zone: 0.000; PT-broken cells that are Lorentzian: nan
u=0.5: Lorentzian fraction 0.932 definite fraction 0.000
  Lorentzian cells span k in [0.02,6.26], v in [0.05,1.95]; nearest to EP (pi,0.5): 0.020
  Lorentzian cells inside PT-broken zone: 0.000; PT-broken cells that are Lorentzian: 0.000

real 0m2.668s
user 0m2.656s
sys 0m0.008s
```

s03d_metric_rank_checked.py

```
# SCRIPT: MATTER-PEEL-02B3b signature map with the rank pre-check (degenerate-plane artifact gated)
import numpy as np
exec(open('s03c_metric_raw.py').read().split('def scan')[0])
def classify(v,u,k,tol=1e-9):
    a,b,c=form(v,u,k); d=a*c-b*b; s=a*a+c*c+2*b*b
    if abs(d)<=tol*s: return 0 # rank-one / degenerate: no verdict
    return -1 if d<0 else 1
ks=np.linspace(0.02,2*np.pi-0.02,180); vs=np.linspace(0.05,1.95,96)
for u in (0.0,0.5):
    M=np.array([[classify(v,u,k) if abs(abs(v+w*np.exp(1j*k))**2-u**2)>1e-6 else 0 for k in ks] for v in vs])
    brk=np.array([[abs(v+w*np.exp(1j*k))<u for k in ks] for v in vs])
    print(f'u={u}: Lorentzian {np.mean(M==-1):.3f} definite {np.mean(M==1):.3f} degenerate {np.mean(M==0):.3f
        }')
    if (M==-1).any():
        idx=np.argwhere(M==-1); kk=ks[idx[:,1]]; vv=vs[idx[:,0]]
```

```
      print(f' Lorentzian: k in [{kk.min():.2f},{kk.max():.2f}] v in [{vv.min():.2f},{vv.max():.2f}]; nearest
           to EP {np.min(np.hypot(kk-np.pi,vv-0.5)):.3f}')
      print(f' fraction of Lorentzian cells in PT-broken zone {np.mean(brk[M==-1]):.3f}; PT-broken cells
          Lorentzian {np.mean((M==-1)[brk]) if brk.any() else float("nan"):.3f}')
   np.save(f'sig_u{u}.npy',M)
# radial look at the EP: signature along k=pi as v crosses 0.5, u=0.5
print('along k=pi:')
for v in [0.2,0.4,0.45,0.49,0.51,0.55,0.6,0.8,1.2,1.49,1.51,1.8]:
   a,b,c=form(v,0.5,np.pi); print(f' v={v:4.2f} a={a:+.3e} b={b:+.3e} c={c:+.3e} det={a*c-b*b:+.3e}')
```

Output:

```
u=0.0: Lorentzian 0.000 definite 0.000 degenerate 1.000
u=0.5: Lorentzian 0.932 definite 0.068 degenerate 0.000
   Lorentzian: k in [0.02,6.26] v in [0.05,1.95]; nearest to EP 0.020
   fraction of Lorentzian cells in PT-broken zone 0.000; PT-broken cells Lorentzian 0.000
along k=pi:
  v=0.20 a=+6.410e-01 b=-3.608e-16 c=-4.109e-01 det=-2.634e-01
  v=0.40 a=+2.273e+00 b=-2.220e-16 c=-5.165e+00 det=-1.174e+01
  v=0.45 a=+4.762e+00 b=+3.553e-15 c=-2.268e+01 det=-1.080e+02
  v=0.49 a=+2.475e+01 b=-5.684e-14 c=-6.127e+02 det=-1.517e+04
  v=0.51 a=-2.525e+01 b=-5.684e-14 c=-6.377e+02 det=+1.610e+04
  v=0.55 a=-5.263e+00 b=+7.105e-15 c=-2.770e+01 det=+1.458e+02
  v=0.60 a=-2.778e+00 b=+0.000e+00 c=-7.716e+00 det=+2.143e+01
  v=0.80 a=-1.190e+00 b=+2.220e-16 c=-1.417e+00 det=+1.687e+00
  v=1.20 a=-1.190e+00 b=+4.441e-16 c=-1.417e+00 det=+1.687e+00
  v=1.49 a=-2.525e+01 b=+2.842e-13 c=-6.377e+02 det=+1.610e+04
  v=1.51 a=+2.475e+01 b=+2.274e-13 c=-6.127e+02 det=-1.517e+04
  v=1.80 a=+6.410e-01 b=+2.776e-17 c=-4.109e-01 det=-2.634e-01

real 0m3.778s
user 0m3.739s
sys 0m0.016s
```

s04a_rotation_unwindowed.py

```
# SCRIPT: MATTER-PEEL-03 rapidity across the Cayley pole: rotation part of census levels on approach to the
     critical wall
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
# exact algebra check: |tanh(W/2)|<1 iff |Im W|<pi/2 (principal branch)
rng=np.random.default_rng(1); Z=rng.normal(size=4000)+1j*rng.normal(size=4000); W=np.log(Z)
print('theorem check, mismatches:',np.sum((np.abs(np.tanh(W/2))<1)!=(np.abs(W.imag)<np.pi/2)))
for zN in [-28.0]:
   print(' negative ratio z=',zN,' Im W=',np.log(complex(zN)).imag/np.pi,'pi Gamma=',np.tanh(np.log(complex(
        zN))/2))
def census_rot(v,u,N=2000,Lc=80):
   nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
   z=(1-nu)/nu; W=np.log(z); G=1-2*nu
   return np.max(np.abs(W.imag))/np.pi, np.max(np.abs(G))
u=0.5
print('approach from v<w side (critical at v=0.5): v, max|Im W|/pi, max|Gamma|')
for v in [0.0,0.1,0.2,0.3,0.4,0.45,0.48,0.49,0.495,0.5]:
   r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
print('opposite: approach from v>w side (critical at v=1.5)')
for v in [2.5,2.0,1.7,1.6,1.55,1.52,1.51,1.505,1.5]:
   r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
print('inside the PT-broken zone (0.5<v<1.5):')
for v in [0.6,1.0,1.4]:
   r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
```

Output:

```
theorem check, mismatches: 0
 negative ratio z= -28.0 Im W= 1.0 pi Gamma= (1.0740740740740737+9.407437688923424e-18j)
```

```
approach from v<w side (critical at v=0.5): v, max|Im W|/pi, max|Gamma|
  v=0.000 rot=0.9998 pi |Gamma|max=1.0000 in
  v=0.100 rot=0.9993 pi |Gamma|max=1.0000 in
  v=0.200 rot=0.9981 pi |Gamma|max=1.0000 in
  v=0.300 rot=1.0000 pi |Gamma|max=1.0000 in
  v=0.400 rot=0.9995 pi |Gamma|max=1.0000 in
  v=0.450 rot=0.9997 pi |Gamma|max=1.0000 in
  v=0.480 rot=1.0000 pi |Gamma|max=1.0927 OUT
  v=0.490 rot=0.9978 pi |Gamma|max=1.2326 OUT
  v=0.495 rot=0.9954 pi |Gamma|max=1.3724 OUT
  v=0.500 rot=1.0000 pi |Gamma|max=2.9595 OUT
opposite: approach from v>w side (critical at v=1.5)
  v=2.500 rot=1.0000 pi |Gamma|max=1.0000 in
  v=2.000 rot=0.9994 pi |Gamma|max=1.0000 in
  v=1.700 rot=1.0000 pi |Gamma|max=1.0000 in
  v=1.600 rot=0.9998 pi |Gamma|max=1.0000 in
  v=1.550 rot=0.9990 pi |Gamma|max=1.0000 in
  v=1.520 rot=0.9993 pi |Gamma|max=1.0000 in
  v=1.510 rot=1.0000 pi |Gamma|max=1.0009 OUT
  v=1.505 rot=1.0000 pi |Gamma|max=1.0301 OUT
  v=1.500 rot=1.0000 pi |Gamma|max=1.4681 OUT
inside the PT-broken zone (0.5<v<1.5):
  v=0.600 rot=0.9995 pi |Gamma|max=1.0000 in
  v=1.000 rot=0.9982 pi |Gamma|max=1.0000 in
  v=1.400 rot=0.9992 pi |Gamma|max=1.0000 in

real 0m17.028s
user 0m16.805s
sys 0m0.020s
```

s04b_rotation_windowed.py

```
# SCRIPT: MATTER-PEEL-03b windowed census rotation and disk exit vs block size
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
def census(v,u,N,Lc,d=1e-9):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    W=np.log((1-nu)/nu); G=1-2*nu
    return (np.max(np.abs(W.imag))/np.pi if len(nu) else 0.0), np.max(np.abs(G)) if len(nu) else 0.0
u=0.5
for (N,Lc) in [(2000,40),(2000,80),(4000,160)]:
    print(f'block Lc={Lc}, ring N={N}')
    for v in [0.3,0.4,0.45,0.47,0.48,0.49,0.495,0.5,1.505,1.51,1.52,1.55]:
        r,g=census(v,u,N,Lc)
        print(f' v={v:5.3f} max rotation={r:.4f} pi max|Gamma|={g:.4f} {"OUT" if g>1+1e-6 else "in"}')
```

Output:

```
block Lc=40, ring N=2000
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5686 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.6105 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=2.8380 OUT
   v=1.505 max rotation=0.5468 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5014 pi max|Gamma|=1.0008 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in
block Lc=80, ring N=2000
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
```

```
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5661 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.5992 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=2.9595 OUT
   v=1.505 max rotation=0.5376 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5012 pi max|Gamma|=1.0009 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in
block Lc=160, ring N=4000
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5661 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.5991 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=3.2318 OUT
   v=1.505 max rotation=0.5357 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5012 pi max|Gamma|=1.0009 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in

real 1m21.279s
user 1m20.663s
sys 0m0.049s
```

s05a_exit_line.py

```
# SCRIPT: MATTER-PEEL-03c locate the disk-exit line v_exit(u) and the rotation at the critical point
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
def census(v,u,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi
def vexit(u):
    lo,hi=0.0,1-u # rotation <1/2 at v=0 (check), critical at v=1-u
    if census(lo,u)>=0.5: return None
    for _ in range(30):
        m=(lo+hi)/2
        if census(m,u)<0.5: lo=m
        else: hi=m
    return (lo+hi)/2
print(' u v_exit crit=1-u gap at exit |v-w|-u ratio u/(w-v_exit) rot at critical')
for u in [0.2,0.3,0.4,0.5,0.6,0.7]:
    ve=vexit(u); rc=census(1-u,u)
    if ve is None: print(f' {u:.2f} rotation already >= 1/2 at v=0; rot at critical {rc:.4f}'); continue
    print(f' {u:.2f} {ve:.6f} {1-u:.3f} {abs(ve-w)-u:.6f} {u/(w-ve):.6f} {rc:.5f} pi')
```

Output:

```
 u v_exit crit=1-u gap at exit |v-w|-u ratio u/(w-v_exit) rot at critical
 0.20 0.798752 0.800 0.001248 0.993801 1.00000 pi
 0.30 0.695189 0.700 0.004811 0.984216 1.00000 pi
 0.40 0.586649 0.600 0.013351 0.967700 1.00000 pi
 0.50 0.468213 0.500 0.031787 0.940226 1.00000 pi
 0.60 0.327358 0.400 0.072642 0.892005 1.00000 pi
 0.70 0.082016 0.300 0.217984 0.762540 1.00000 pi

real 1m11.323s
user 1m10.492s
sys 0m0.024s
```

s05b_predict_exit.py

```
# SCRIPT: MATTER-PEEL-03d predict the disk-exit line from mean Petermann = 2, then measure
import numpy as np
exec(open('s05a_exit_line.py').read().split("print(' u ")[0])
def P(v,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); R=R[:,o]; L=np.linalg.inv(R).conj().T
    return np.outer(R[:,0],L[:,0].conj())
ks=2*np.pi*(np.arange(4000)+0.5)/4000
def meanK(v,u): return np.mean([np.linalg.norm(P(v,u,k),2)**2 for k in ks])
def vpred(u):
    lo,hi=0.0,1-u-1e-9
    for _ in range(40):
        m=(lo+hi)/2
        if meanK(m,u)<2: lo=m
        else: hi=m
    return (lo+hi)/2
print(' u predicted (mean K = 2) measured census exit difference')
for u in [0.25,0.45,0.65]:
    vp=vpred(u); vm=vexit(u)
    print(f' {u:.2f} {vp:.7f} {vm:.7f} {vm-vp:+.2e}')
```

Output:

```
 u predicted (mean K = 2) measured census exit difference
 0.25 0.7474014 0.7474015 +4.04e-08
 0.45 0.5291503 0.5291503 -1.53e-09
 0.65 0.2357262 0.2357262 -3.70e-08

real 1m14.012s
user 1m13.157s
sys 0m0.042s
```

s06_second_model_exit_law.py

```
# SCRIPT: MATTER-DISK-S06 is the disk-exit law universal? second model: the cone paper's band in 1D
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# Section 5 found, on the PT-symmetric SSH chain, that the census levels leave the Smith disk exactly
# where the band-averaged Petermann factor <K> equals 2. For any two-band h(k) = d(k).sigma the
# Petermann factor of the lower band is K(k) = (1 + |d|^2/|E|^2)/2 with E^2 = d.d (checked below).
# A law that belongs to the SSH chain alone would fail on a different band. Here the band is the cone
# paper's family made one-dimensional, d = (t1 + cos k, sin k, m + i*gamma): a mass term, no PT symmetry.
# Path: start closed (gamma = 0, census real, inside the disk) and raise gamma until the census leaves
# the disk (largest rotation |Im ln z| reaches pi/2, census window 1e-9 < nu < 1-1e-9).
# KILL (written before the run): at any of the four (t1, m) settings, <K> at the measured exit differs
# from 2 by more than 1e-4. Control: gamma = 0 must give a real census inside the disk.
import numpy as np
sx=np.array([[0,1],[1,0]],complex); sy=np.array([[0,-1j],[1j,0]]); sz=np.array([[1,0],[0,-1]],complex)
def H(t1,m,g,k): return (t1+np.cos(k))*sx+np.sin(k)*sy+(m+1j*g)*sz
def lower(t1,m,g,k):
    ev,R=np.linalg.eig(H(t1,m,g,k)); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T; return ev,R,L
def corr(t1,m,g,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N; C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=lower(t1,m,g,k); P=np.outer(R[:,0],L[:,0].conj())
        ph=np.exp(1j*k*np.arange(Lc)); C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def rotation(t1,m,g,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(t1,m,g,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    if len(nu)==0: return 0.0, 0.0
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi, np.max(np.abs(1-2*nu))
ks=2*np.pi*(np.arange(4000)+0.5)/4000
```

```
def meanK(t1,m,g):
    out=[]
    for k in ks:
        ev,R,L=lower(t1,m,g,k); out.append(np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real)
    return np.mean(out)
# check the general Petermann formula on random points
rng=np.random.default_rng(5); worst=0
for _ in range(300):
    t1,m,g,k=rng.uniform(0.2,2),rng.uniform(-1,1),rng.uniform(0,1.5),rng.uniform(0,2*np.pi)
    ev,R,L=lower(t1,m,g,k); d=np.array([t1+np.cos(k),np.sin(k),m+1j*g]); E2=np.sum(d*d)
    if abs(E2)<1e-3: continue
    Kn=np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real; Kf=(1+np.sum(abs(d)**2)/abs(E2))/2
    worst=max(worst,abs(Kn-Kf))
print('general Petermann formula K=(1+|d|^2/|E|^2)/2, worst error on 300 random points:',worst)
print('settings (t1, m): control at gamma=0, measured exit gamma, <K> at exit')
ok=True
for t1,m in [(0.5,0.4),(1.5,0.4),(0.5,0.0),(1.5,0.8)]:
    r0,G0=rotation(t1,m,0.0)
    lo,hi=0.0,None; g=0.05
    while hi is None and g<4:
        if rotation(t1,m,g)[0]>=0.5: hi=g
        else: lo=g; g*=1.4
    if hi is None: print(f' ({t1},{m}): census never left the disk up to gamma=4'); ok=False; continue
    for _ in range(28):
        mid=(lo+hi)/2
        if rotation(t1,m,mid)[0]<0.5: lo=mid
        else: hi=mid
    ge=(lo+hi)/2; K=meanK(t1,m,ge)
    ok&=abs(K-2)<1e-4
    print(f' ({t1},{m}): control rot={r0:.3f} pi max|Gamma|={G0:.4f} | exit gamma={ge:.7f} | <K>={K:.6f} {"
         PASS" if abs(K-2)<1e-4 else "FAIL"}')
print('VERDICT:', 'exit law <K> = 2 holds on the second model' if ok else 'KILLED on the second model at the
     stated bar')
```

Output:

```
general Petermann formula K=(1+|d|^2/|E|^2)/2, worst error on 300 random points: 1.4921397450962104e-13
settings (t1, m): control at gamma=0, measured exit gamma, <K> at exit
  (0.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1691699 | <K>=1.022352 FAIL
  (1.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1956728 | <K>=1.018838 FAIL
  (0.5,0.0): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.4743416 | <K>=2.000000 PASS
  (1.5,0.8): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.2518871 | <K>=1.015041 FAIL
VERDICT: KILLED on the second model at the stated bar

real 0m57.253s
user 0m56.558s
sys 0m0.044s
block 40, ring 2000: exit gamma = 0.169170
block 80, ring 2000: exit gamma = 0.169170
block 160, ring 4000: exit gamma = 0.169170
```

s07_boundary_census_disk.py

```
# SCRIPT: MATTER-DISK-S07 the boundary census on the Smith disk, closed, warm and opened
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# For a one-body correlation matrix C split into a region A and the rest B, block multiplication gives
# C_A - C_A^2 = (C - C^2)_A + X_A, X_A = C_AB C_BA.
# In a pure state C^2 = C, so X_A = C_A(1 - C_A): its eigenvalues are nu(1 - nu) = (1 - Gamma^2)/4 with
# Gamma = 1 - 2nu, the section-1 disk coordinate. So the boundary coordinate Gamma_b = sqrt(1 - 4 lambda(X_A))
# is the same point on the disk as |Gamma|, and the count lambda >= eps(1-eps) is the disk window
# |Gamma_b| <= 1 - 2eps. In a warm state C^2 != C: the local mixing term (C - C^2)_A pulls raw Gamma toward
# the centre without any correlation across the cut; X_A keeps only what crosses the cut.
# KILL (written before the run):
# (a) the identity fails by more than 1e-12 on 32 random Hermitian contractions;
# (b) disconnected sites C = 0.3 I give a nonzero boundary count;
```

```
# (c) at T = 0 the boundary count differs from the raw count, or |Gamma_b| differs from |Gamma| by > 1e-8;
# (d) the 128-site ring table (region 64, eps = 0.1) differs from the 3 Nov review table by more than
# 1e-5 in Tr X_A or mutual information, or by any integer in the counts;
# (e) on the opened non-Hermitian SSH chain (biorthogonal C) C^2 = C or the identity fails by > 1e-8.
import numpy as np
from numpy.linalg import eigvalsh, eigvals, eigh
rng=np.random.default_rng(7)
def blocks(C,nA): return C[:nA,:nA],C[:nA,nA:],C[nA:,:nA]
def ident_err(C,nA):
    CA,CAB,CBA=blocks(C,nA); CC=C@C
    return np.max(np.abs((CA-CA@CA)-((C-CC)[:nA,:nA]+CAB@CBA)))
# (a) random Hermitian contractions 0 <= C <= I
worst=0
for s in range(32):
    n=24; M=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(M)
    C=U@np.diag(rng.uniform(0,1,n))@U.conj().T; worst=max(worst,ident_err(C,10))
print('(a) block identity, worst error on 32 random Hermitian contractions:',f'{worst:.2e}', 'PASS' if worst<1
     e-12 else 'FAIL')
eps=0.1; thr=eps*(1-eps)
def counts(C,nA):
    CA,CAB,CBA=blocks(C,nA); nu=eigvalsh(CA); X=CAB@CBA; lam=eigvalsh((X+X.conj().T)/2)
    raw=int(np.sum((nu>=eps)&(nu<=1-eps))); bnd=int(np.sum(lam>=thr))
    return raw,bnd,np.trace(X).real,nu,lam
# (b) disconnected sites
C=0.3*np.eye(64); raw,bnd,tx,_,_=counts(C,32)
print(f'(b) disconnected C=0.3 I, region 32: raw count {raw}, boundary count {bnd}, Tr X_A {tx:.3f}',
      'PASS' if bnd==0 else 'FAIL')
# (c,d) the 128-site antiperiodic ring
def ring(tA,tB,N=128):
    H=np.zeros((N,N))
    for j in range(N):
        t=tA if j%2==0 else tB; s=-1.0 if j==N-1 else 1.0
        H[j,(j+1)%N]=H[(j+1)%N,j]=-t*s
    return H
def thermal(H,T):
    e,V=eigh(H)
    f=(e<0).astype(float) if T==0 else 1/(1+np.exp(e/T))
    return (V*f)@V.T
def Sg(nu):
    nu=np.clip(nu,1e-15,1-1e-15); return -np.sum(nu*np.log(nu)+(1-nu)*np.log(1-nu))
review={('uniform',0):(2,2,0.605657,3.923714),('uniform',0.5):(24,0,0.170835,0.949562),('uniform',2)
     :(64,0,0.027779,0.116547),
       ('gapped alternating',0):(0,0,0.029017,0.345837),('gapped alternating',0.5):(10,0,0.024935,0.188891),('
            gapped alternating',2):(64,0,0.006484,0.028357)}
okd=True; okc=True
print('(d) ring N=128 antiperiodic, mu=0, region 64, eps=0.1 [ours | 3 Nov review]')
for name,(tA,tB) in [('uniform',(1,1)),('gapped alternating',(1.5,0.5))]:
    H=ring(tA,tB)
    for T in [0,0.5,2]:
        C=thermal(H,T); raw,bnd,tx,nu,lam=counts(C,64)
        MI=Sg(eigvalsh(C[:64,:64]))+Sg(eigvalsh(C[64:,64:]))-Sg(eigvalsh(C))
        r=review[(name,T)]
        good=(raw==r[0])and(bnd==r[1])and abs(tx-r[2])<1e-5 and abs(MI-r[3])<1e-5; okd&=good
        print(f' {name:19s} T={T:<3} raw {raw:3d}|{r[0]:3d} boundary {bnd:2d}|{r[1]:2d} TrX {tx:.6f}|{r[2]:.6f}
             MI {MI:.6f}|{r[3]:.6f} {"MATCH" if good else "DIFFER"}')
        G=np.sort(np.abs(1-2*nu)); Gb=np.sort(np.sqrt(np.clip(1-4*lam,0,1)))
        if T==0:
            dg=np.max(np.abs(G-Gb)); okc&=(raw==bnd) and dg<1e-8
            print(f' (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = {dg:.1e}')
        else:
            print(f' warm: levels inside the window by raw Gamma {raw}, by boundary Gamma_b {bnd}; innermost |
                 Gamma| {G[0]:.3f}, innermost |Gamma_b| {Gb[0]:.3f}')
print('(c) T=0 boundary census = original census on the disk:', 'PASS' if okc else 'FAIL')
print('(d) review table reproduced:', 'PASS' if okd else 'DIFFER (see rows)')
# (e) opened matter: biorthogonal C of the non-Hermitian SSH chain
def bands(v,w,u,k):
    Hk=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(Hk); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]; L=np.linalg.inv(R).conj().T
```

```
    return ev,R,L
def corr(v,w,u,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N; C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=bands(v,w,u,k); P=np.outer(R[:,0],L[:,0].conj()); ph=np.exp(1j*k*np.arange(Lc))
        C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
oke=True
print('(e) opened chain, full ring of 200 cells (C is the whole system), block of 40 cells')
for v,u,tag in [(0.3,0.5,'opened, inside the disk'),(0.5,0.5,'opened, critical')]:
    C=corr(v,1.0,u,200,200); idem=np.max(np.abs(C@C-C)); ie=ident_err(C,80)
    CA,CAB,CBA=blocks(C,80); pure=np.max(np.abs(CAB@CBA-(CA-CA@CA)))
    oke&=idem<1e-8 and ie<1e-8 and pure<1e-8
    print(f' v={v} u={u} ({tag}): |C^2 - C| = {idem:.1e}, identity error {ie:.1e}, |X_A - C_A(1-C_A)| = {pure
        :.1e}')
print('(e) opened matter keeps the boundary identity:', 'PASS' if oke else 'FAIL')
```

Output:

```
(a) block identity, worst error on 32 random Hermitian contractions: 2.78e-16 PASS
(b) disconnected C=0.3 I, region 32: raw count 32, boundary count 0, Tr X_A 0.000 PASS
(d) ring N=128 antiperiodic, mu=0, region 64, eps=0.1 [ours | 3 Nov review]
    uniform T=0 raw 2| 2 boundary 2| 2 TrX 0.605657|0.605657 MI 3.923714|3.923714 MATCH
        (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = 4.0e-15
    uniform T=0.5 raw 24| 24 boundary 0| 0 TrX 0.170835|0.170835 MI 0.949562|0.949562 MATCH
        warm: levels inside the window by raw Gamma 24, by boundary Gamma_b 0; innermost |Gamma| 0.047,
            innermost |Gamma_b| 0.817
    uniform T=2 raw 64| 64 boundary 0| 0 TrX 0.027779|0.027779 MI 0.116547|0.116547 MATCH
        warm: levels inside the window by raw Gamma 64, by boundary Gamma_b 0; innermost |Gamma| 0.012,
            innermost |Gamma_b| 0.972
    gapped alternating T=0 raw 0| 0 boundary 0| 0 TrX 0.029017|0.029017 MI 0.345837|0.345837 MATCH
        (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = 4.2e-15
    gapped alternating T=0.5 raw 10| 10 boundary 0| 0 TrX 0.024935|0.024935 MI 0.188891|0.188891 MATCH
        warm: levels inside the window by raw Gamma 10, by boundary Gamma_b 0; innermost |Gamma| 0.763,
            innermost |Gamma_b| 0.980
    gapped alternating T=2 raw 64| 64 boundary 0| 0 TrX 0.006484|0.006484 MI 0.028357|0.028357 MATCH
        warm: levels inside the window by raw Gamma 64, by boundary Gamma_b 0; innermost |Gamma| 0.246,
            innermost |Gamma_b| 0.994
(c) T=0 boundary census = original census on the disk: PASS
(d) review table reproduced: PASS
(e) opened chain, full ring of 200 cells (C is the whole system), block of 40 cells
    v=0.3 u=0.5 (opened, inside the disk): |C^2 - C| = 9.0e-15, identity error 1.8e-16, |X_A - C_A(1-C_A)| =
        3.8e-15
    v=0.5 u=0.5 (opened, critical): |C^2 - C| = 2.8e-14, identity error 6.9e-15, |X_A - C_A(1-C_A)| = 1.3e-14
(e) opened matter keeps the boundary identity: PASS
```

s08_gas_carries_the_peel.py

```
# SCRIPT: MATTER-DISK-S08 a gas carries the VOTKP peel coordinate in density-free combinations (helium)
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# VOTKP v2.9 gives helium the ladder I0 = 24.587389011 eV, I1 = 54.4177655282 eV and the peel coordinate
# Gamma0 = (I1 - I0)/(I1 + I0). In an ideal Saha gas (ground-state weights U = 1, 2, 1) the populations
# n0, n1, n2 change enormously with temperature and density. Each adjacent pair is a point on the disk:
# odds z_q = n_q/n_(q+1), Gamma_pop,q = tanh(ln z_q / 2) (section 1), and half ionization is the centre.
# Two combinations lose the electron density and thermal wavelength exactly:
# ln(n0 n2 / n1^2) = -(I1 - I0)/kT - ln 4 (the second difference of the ladder)
# ln(n0 / (n2 n_e^2 lambda^6)) = (I0 + I1)/kT - ln 4 (the whole peel, both electrons)
# so their slopes in beta = 1/kT give D = I1 - I0 and B = I0 + I1, and D/B is Gamma0 at any density.
# KILL (written before the run):
# (a) the populations at 1e22 m^-3 miss the 3 Nov review table (10, 20, 50 kK) by more than 1e-6 percent,
# or the 20 kK density sweep misses 0.5680% (1e20) and 87.6482% (1e24) at the printed digits;
# (b) Gamma0 recovered from the gas differs from 0.377575066984 by more than 1e-9 at any density 1e18..1e26;
# (c) an independent solver (cubic in n_e) disagrees with the root finder by more than 1e-12.
import numpy as np
from scipy.optimize import brentq
```

```
h=6.62607015e-34; kB=1.380649e-23; me=9.1093837015e-31; eV=1.602176634e-19
I0,I1=24.587389011,54.4177655282; U=(1.0,2.0,1.0)
G0=(I1-I0)/(I1+I0)
def saha(T,n):
    kT=kB*T; lam3=(h/np.sqrt(2*np.pi*me*kT))**3
    S1=2*(U[1]/U[0])/lam3*np.exp(-I0*eV/kT); S2=2*(U[2]/U[1])/lam3*np.exp(-I1*eV/kT)
    s1,s2=S1/n,S2/n # charge neutrality in x = ne/n: x(x^2 + s1 x + s1 s2) = s1 x + 2 s1 s2, root in (0, 2)
    g=lambda x: x*(x*x+s1*x+s1*s2)-(s1*x+2*s1*s2)
    ne=n*brentq(g,0.0,2.0,xtol=1e-300,rtol=1e-15,maxiter=2000)
    r1=S1/ne; r2=r1*S2/ne; Z=1+r1+r2
    return np.array([1,r1,r2])/Z, ne, lam3
def saha_cubic(T,n):
    kT=kB*T; lam3=(h/np.sqrt(2*np.pi*me*kT))**3
    S1=2*(U[1]/U[0])/lam3*np.exp(-I0*eV/kT); S2=2*(U[2]/U[1])/lam3*np.exp(-I1*eV/kT)
    # ne(ne^2 + S1 ne + S1 S2) = n(S1 ne + 2 S1 S2) -> ne^3 + S1 ne^2 + (S1 S2 - n S1) ne - 2 n S1 S2 = 0
    rts=np.roots([1.0,S1/n,(S1*S2-n*S1)/n**2,-2*S1*S2/n**2])*n
    ne=max(r.real for r in rts if abs(r.imag)<1e-9*abs(r) and r.real>0)
    r1=S1/ne; r2=r1*S2/ne; Z=1+r1+r2; return np.array([1,r1,r2])/Z
print(f'helium ladder (VOTKP v2.9): I0={I0} eV, I1={I1} eV, Gamma0={G0:.12f}')
rev={10000:(99.937400,0.062600,0.000000),20000:(28.978059,71.021940,0.000001)
     ,50000:(0.000873,17.146019,82.853108)}
oka=True; okc=True
print('(a) populations at n = 1e22 m^-3, percent [ours | review], and the disk point of each pair')
for T,r in rev.items():
    p,ne,_=saha(T,1e22); pc=saha_cubic(T,1e22); okc&=np.max(np.abs(p-pc))<1e-12
    good=all(abs(100*p[i]-r[i])<1e-6+5e-7 for i in range(3)); oka&=good
    Gp=[np.tanh(0.5*np.log(p[q]/p[q+1])) if p[q+1]>0 else 1.0 for q in range(2)]
    print(f' T={T:6d} K He {100*p[0]:10.6f}|{r[0]:10.6f} He+ {100*p[1]:10.6f}|{r[1]:10.6f} He2+ {100*p[2]:10.6
        f}|{r[2]:10.6f} '
         f'Gamma_pop = ({Gp[0]:+.4f}, {Gp[1]:+.4f}) {"MATCH" if good else "DIFFER"}')
for n,target in [(1e20,0.5680),(1e24,87.6482)]:
    p,_,_=saha(20000,n); good=abs(100*p[0]-target)<5e-5; oka&=good
    print(f' T=20000 K n={n:.0e}: neutral {100*p[0]:.4f}% | review {target}% {"MATCH" if good else "DIFFER"}')
print('(a)', 'PASS' if oka else 'DIFFER'); print('(c) cubic solver agrees with the root finder:', 'PASS' if
    okc else 'FAIL')
# (b) recover Gamma0 from the gas alone at two temperatures, across densities
okb=True; worst=0
print('(b) Gamma0 read back from the gas (two temperatures 30 kK and 32 kK), by density')
for n in [1e18,1e20,1e22,1e24,1e26]:
    vals=[]
    for T in (30000,32000):
        p,ne,lam3=saha(T,n); beta=eV/(kB*T) # beta in 1/eV
        vals.append((beta,np.log(p[0]*p[2]/p[1]**2),np.log(p[0]/(p[2]*ne**2*lam3**2))))
    (b1,a1,c1),(b2,a2,c2)=vals
    D=-(a2-a1)/(b2-b1); B=(c2-c1)/(b2-b1); Gr=D/B; worst=max(worst,abs(Gr-G0))
    p,_,_=saha(30000,n)
    print(f' n={n:.0e}: neutral fraction at 30 kK {100*p[0]:9.5f}% D={D:.9f} eV B={B:.9f} eV Gamma0={Gr:.12f
        }')
okb=worst<1e-9
print(f'(b) worst |Gamma0(gas) - Gamma0(VOTKP)| = {worst:.1e}', 'PASS' if okb else 'FAIL')
```

Output:

```
helium ladder (VOTKP v2.9): I0=24.587389011 eV, I1=54.4177655282 eV, Gamma0=0.377575066984
(a) populations at n = 1e22 m^-3, percent [ours | review], and the disk point of each pair
   T= 10000 K He 99.937400| 99.937400 He+ 0.062600| 0.062600 He2+ 0.000000| 0.000000 Gamma_pop = (+0.9987,
       +1.0000) MATCH
   T= 20000 K He 28.978059| 28.978059 He+ 71.021940| 71.021940 He2+ 0.000001| 0.000001 Gamma_pop = (-0.4204,
       +1.0000) MATCH
   T= 50000 K He 0.000873| 0.000873 He+ 17.146019| 17.146019 He2+ 82.853108| 82.853108 Gamma_pop = (-0.9999,
       -0.6571) MATCH
   T=20000 K n=1e+20: neutral 0.5680% | review 0.568% MATCH
   T=20000 K n=1e+24: neutral 87.6482% | review 87.6482% MATCH
(a) PASS
(c) cubic solver agrees with the root finder: PASS
(b) Gamma0 read back from the gas (two temperatures 30 kK and 32 kK), by density
   n=1e+18: neutral fraction at 30 kK 0.00001% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
   n=1e+20: neutral fraction at 30 kK 0.00267% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
```

```
   n=1e+22: neutral fraction at 30 kK 0.26765% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
   n=1e+24: neutral fraction at 30 kK 18.06496% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
   n=1e+26: neutral fraction at 30 kK 82.49125% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
(b) worst |Gamma0(gas) - Gamma0(VOTKP)| = 2.2e-16 PASS
```

s09_check_gpt_formal.py

```
# SCRIPT: MATTER-DISK-S09 independent checks of the GPT formal edition's new claims
import numpy as np, importlib.util, io, contextlib, sys
sys.path.insert(0,'/home/claude/votkp-lean/matter-disk/scripts')
spec=importlib.util.spec_from_file_location('s06','/home/claude/votkp-lean/matter-disk/scripts/
    s06_second_model_exit_law.py')
src=open('/home/claude/votkp-lean/matter-disk/scripts/s06_second_model_exit_law.py').read().split('# check the
     general Petermann formula')[0]
ns={}; exec(src,ns)
# (1) endpoint filter sensitivity of the massive exit (claim: 0.2716 at 1e-8, 0.16917 at 1e-9, 0.12279 at 1e
    -12)
def exit_g(t1,m,d):
   rot=lambda g: ns['rotation'](t1,m,g,d=d)[0]
   lo,hi,g=0.0,None,0.05
   while hi is None and g<4:
       if rot(g)>=0.5: hi=g
       else: lo=g; g*=1.4
   for _ in range(28):
       mid=(lo+hi)/2
       if rot(mid)<0.5: lo=mid
       else: hi=mid
   return (lo+hi)/2
print('(1) massive band t1=0.5 m=0.4, exit gamma by endpoint filter:')
for d in [1e-6,1e-8,1e-9,1e-10,1e-12]:
   print(f' filter {d:.0e}: exit gamma {exit_g(0.5,0.4,d):.9f}')
print(' massless t1=0.5 m=0, filters 1e-8 and 1e-12:', f'{exit_g(0.5,0.0,1e-8):.9f}', f'{exit_g(0.5,0.0,1e-12)
    :.9f}')
# (2) P3: w=0, v^2=2u^2 -> <K>=2 yet compressed projector idempotent (eigenvalues 0,1)
lib=open('/home/claude/votkp-lean/matter-disk/scripts/nhssh_lib_and_s03a_occupations.py').read().split('w
    =1.0')[0]; ns2={}; exec(lib,ns2)
u=0.5; v=np.sqrt(2)*u
C=ns2['corr'](v,0.0,u,400,40); nu=np.linalg.eigvals(C)
print('(2) P3 w=0, v^2=2u^2: max distance of block eigenvalues from {0,1} =',f'{np.max(np.minimum(abs(nu),abs
    (1-nu))):.1e}',' band <K> = 1+u^2/(E0 Epi) =',1+u**2/(v**2-u**2))
# (3) D5 idempotent
C=np.array([[2,1],[-2,-1]]); print('(3) D5: C^2==C', np.allclose(C@C,C), ' Y_A =',C[0,1]*C[1,0], ' A-A^2
    =',2-4)
# (4) B13: 4 Tr X_A <= I_G on random Gaussian covariances; sharp family
rng=np.random.default_rng(3); h=lambda x: -np.sum(x*np.log(x)+(1-x)*np.log(1-x))
worst=np.inf
for _ in range(200):
   n=10; M=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(M); ev=rng.uniform(0.02,0.98,n)
   C=U@np.diag(ev)@U.conj().T; A=C[:4,:4]; D=C[4:,4:]; B=C[:4,4:]
   IG=h(np.linalg.eigvalsh(A))+h(np.linalg.eigvalsh(D))-h(ev); worst=min(worst,IG-4*np.trace(B@B.conj().T).
       real)
print('(4) B13 lower bound: min of I_G - 4 Tr X_A over 200 random cases =',f'{worst:.3e}','(>=0 required)')
# (5) G9 partition bias: U1 = 2+4exp(-10eV/kT), naive inversion at 30/32 kK
kB=8.617333262e-5; I0,I1=24.587389011,54.4177655282
def logs(T,U1):
   b=1/(kB*T); return b, -b*(I1-I0)+np.log(1*1/U1**2), b*(I0+I1)+np.log(1/(4*1)) # L_q and L_B with U0=U2=1
U1=lambda T: 2+4*np.exp(-10/(kB*T))
(b1,a1,c1),(b2,a2,c2)=logs(30000,U1(30000)),logs(32000,U1(32000))
Dn=-(a2-a1)/(b2-b1); Bn=(c2-c1)/(b2-b1)
print('(5) G9 naive Gamma with synthetic U1(T):',f'{Dn/Bn:.14f}','(claim 0.36614943350748); exact',f'{(I1-I0)
    /(I1+I0):.14f}')
```

Output:

```
(1) massive band t1=0.5 m=0.4, exit gamma by endpoint filter:
```

```
   filter 1e-06: exit gamma 0.271636928
   filter 1e-08: exit gamma 0.271636928
   filter 1e-09: exit gamma 0.169169918
   filter 1e-10: exit gamma 0.169169918
   filter 1e-12: exit gamma 0.122721579
   massless t1=0.5 m=0, filters 1e-8 and 1e-12: 0.474341648 0.474123961
(2) P3 w=0, v^2=2u^2: max distance of block eigenvalues from {0,1} = 1.0e-14 ; band <K> = 1+u^2/(E0 Epi) =
    1.9999999999999996
(3) D5: C^2==C True Y_A = -2 A-A^2 = -2
(4) B13 lower bound: min of I_G - 4 Tr X_A over 200 random cases = 2.025e-03 (>=0 required)
(5) G9 naive Gamma with synthetic U1(T): 0.36614943350815 (claim 0.36614943350748); exact 0.37757506698375
```

s10_nuclear_gaps_his_script.py

```
# SCRIPT: SNELL2-MASS-GAPS-ON-THE-SMITH-CHART (exact placement; exploration)
# NUBASE2020 (via pynucastro) mass excesses. The peel ratio of Theorem 8.1, z = S(next)/S(first),
# with S a separation energy. Every bound peel has z > 0 and sits inside the disk; a negative
# separation energy gives z < 0 and |Gamma| > 1: the point leaves the disk.
import os, numpy as np, pynucastro.nucdata as nd
F = os.path.dirname(nd.__file__) + '/AtomicMassEvaluation/nubase_4.mas20.txt'
ME = {}; HL = {}
for line in open(F):
    if not line[:3].strip().isdigit(): continue
    A = int(line[0:3]); zc = line[4:8]
    if zc[3] != '0': continue
    Z = int(zc[:3]); s = line[18:31].strip()
    if not s or s.startswith('*'): continue
    ME[(A, Z)] = float(s.replace('#', '')); HL[(A, Z)] = line[69:80].strip()
n, p = 8071.31806, 7288.97061 # keV, neutron and 1H mass excesses
Sn = lambda A, Z: ME[(A - 1, Z)] + n - ME[(A, Z)]
Sp = lambda A, Z: ME[(A - 1, Z - 1)] + p - ME[(A, Z)]
Sa = lambda A, Z: ME[(A - 4, Z - 2)] + ME[(4, 2)] - ME[(A, Z)]
G = lambda z: (z - 1) / (z + 1)
cases = [('5He: remove n, then n from 4He', Sn(5, 2), Sn(4, 2), HL[(5, 2)]),
        ('5Li: remove p, then p from 4He', Sp(5, 3), Sp(4, 2), HL[(5, 3)]),
        ('8Be: remove alpha (12C ladder: 12C -> 8Be -> 4He)', Sa(8, 4), Sa(12, 6), HL[(8, 4)]),
        ('7Li (bound, for contrast): remove n, then n from 6Li', Sn(7, 3), Sn(6, 3), HL[(7, 3)]),
        ('4He (bound): remove n, then p from 3He', Sn(4, 2), Sp(3, 2), HL[(4, 2)])]
print(f'{"case":52s} {"S first (keV)":>14s} {"S next (keV)":>13s} {"|Gamma|":>8s} half-life (NUBASE)')
for lab, s1, s2, hl in cases:
    if lab.startswith('8Be'): z = s1 / s2 # 12C ladder: first step 12C->8Be costs s2, next step 8Be->2 alpha
        costs s1
    else: z = s2 / s1
    print(f'{lab:52s} {s1:14.1f} {s2:13.1f} {abs(G(z)):8.3f} {hl}')
```

Output:

```
case S first (keV) S next (keV) |Gamma| half-life (NUBASE)
5He: remove n, then n from 4He -734.8 20577.6 1.074 602 ys
5Li: remove p, then p from 4He -1966.1 19813.9 1.220 370 ys
8Be: remove alpha (12C ladder: 12C -> 8Be -> 4He) -91.8 7366.6 1.025 81.9 as
7Li (bound, for contrast): remove n, then n from 6Li 7251.1 5664.4 0.123 stbl
4He (bound): remove n, then p from 3He 20577.6 5493.5 0.579 stbl
```

s11_exit_quarter_turn.py

```
# SCRIPT: MATTER-DISK-S11 is the disk exit the quarter-turn of the author's offset?
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# The Offset Belongs to the Boundary gives the endpoint offset of a gapped chain in closed form,
# 2 artanh(v/sqrt(E_min E_max)). Continued to the opened SSH chain it predicts a complex level offset
# whose imaginary part is Im tau_inf = 2 arctan(u/sqrt(E(0)E(pi))). Then <K> = 1 + u^2/(E(0)E(pi))
# = sec^2(Im tau_inf/2) exactly, so <K> = 2 is the same statement as Im tau_inf = pi/2: a quarter turn.
# If the census rotation (max over block levels of |Im Log((1-nu)/nu)|/pi) is this offset, then the gap
```

```
# between measured rotation and 2 arctan(u/sqrt(E0 Epi))/pi must shrink as the block grows.
# KILL (written before the run): at u = 0.3, 0.5, 0.7 and v at 60%, 80%, 95% of the way to the exit,
# the gap at block 160 is not smaller than at block 20, or exceeds 2e-3 at block 160.
# OPPOSITE: the same comparison against the plain Hermitian-form guess 2 arctan(u/E(pi))/pi must NOT pass.
import numpy as np
src=open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]; ns={}; exec(src,ns)
w=1.0
def rot(v,u,Lc,N=4000,d=1e-9):
    nu=np.linalg.eigvals(ns['corr'](v,w,u,N,Lc)); nu=nu[(abs(nu)>d)&(abs(1-nu)>d)]
    return np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi
def vexit(u): # E(0)E(pi)=u^2 with E0^2=(v+w)^2-u^2, Epi^2=(w-v)^2-u^2, v<w-u
    from scipy.optimize import brentq
    return brentq(lambda v: np.sqrt(((v+w)**2-u**2)*((w-v)**2-u**2))-u**2, 0.0, w-u-1e-12)
ok=True; okopp=True
for u in [0.3,0.5,0.7]:
    ve=vexit(u)
    for f in [0.6,0.8,0.95]:
        v=f*ve; E0=np.sqrt((v+w)**2-u**2); Ep=np.sqrt((w-v)**2-u**2)
        pred=2*np.arctan(u/np.sqrt(E0*Ep))/np.pi; opp=2*np.arctan(u/Ep)/np.pi
        gaps=[]
        for Lc in [20,40,80,160]:
            gaps.append(rot(v,u,Lc)-pred)
        g=np.abs(gaps); good=(g[-1]<g[0]) and g[-1]<2e-3; ok&=good
        okopp&= not (abs(gaps[-1]+pred-opp)<2e-3)
        print(f'u={u} v={v:.4f} pred={pred:.5f} opposite={opp:.5f} gaps by block 20,40,80,160: '+' '.join(f'{x
            :+.2e}' for x in gaps)+(' PASS' if good else ' FAIL'))
print('VERDICT:','census rotation converges to the offset quarter-turn form' if ok else 'KILLED at the stated
    bar')
print('OPPOSITE (Hermitian-form guess must fail):','fails as required' if okopp else 'ALSO PASSES - test does
    not discriminate')
```

Output:

```
u=0.3 v=0.4171 pred=0.22032 opposite=0.34418 gaps by block 20,40,80,160: +9.81e-03 +9.81e-03 +9.81e-03 +9.81e
    -03 FAIL
u=0.3 v=0.5562 pred=0.25556 opposite=0.47249 gaps by block 20,40,80,160: +2.13e-02 +2.13e-02 +2.13e-02 +2.13e
    -02 FAIL
u=0.3 v=0.6604 pred=0.33867 opposite=0.68960 gaps by block 20,40,80,160: +3.35e-02 +3.35e-02 +3.35e-02 +3.35e
    -02 FAIL
u=0.5 v=0.2809 pred=0.36265 opposite=0.48949 gaps by block 20,40,80,160: +4.90e-03 +4.90e-03 +4.90e-03 +4.90e
    -03 FAIL
u=0.5 v=0.3746 pred=0.39764 opposite=0.58975 gaps by block 20,40,80,160: +8.02e-03 +8.02e-03 +8.02e-03 +8.02e
    -03 FAIL
u=0.5 v=0.4448 pred=0.45733 opposite=0.71372 gaps by block 20,40,80,160: +6.25e-03 +6.25e-03 +6.25e-03 +6.25e
    -03 FAIL
u=0.7 v=0.0492 pred=0.49587 opposite=0.52679 gaps by block 20,40,80,160: +9.73e-06 +9.73e-06 +9.73e-06 +9.73e
    -06 FAIL
u=0.7 v=0.0656 pred=0.49765 opposite=0.53908 gaps by block 20,40,80,160: +9.93e-06 +9.93e-06 +9.93e-06 +9.93e
    -06 FAIL
u=0.7 v=0.0779 pred=0.49936 opposite=0.54877 gaps by block 20,40,80,160: +3.87e-06 +3.87e-06 +3.87e-06 +3.87e
    -06 FAIL
VERDICT: KILLED at the stated bar
OPPOSITE (Hermitian-form guess must fail): fails as required
```

s12_nuclear_rim_identity.py

```
# SCRIPT: MATTER-DISK-S12 the distance of an unbound nucleus outside the rim, as an exact identity
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# VOTKP 15.4 observes that the distance outside the rim is "close to twice the decay energy over the
# neighbouring step". With z = S_next/S_first, S_first = -s < 0 and a = S_next/s > 1:
# Gamma = (z-1)/(z+1) = (a+1)/(a-1), so |Gamma| - 1 = 2/(a-1) = 2s/(S_next - s) exactly; for s << S_next
# this is 2s/S_next, the author's "twice the decay energy over the neighbouring step".
# z = -1 (the Cayley pole) is S_next = -S_first, i.e. S_first + S_next = 0.
# KILL (written before the run): the identity misses by more than 1e-12 on any of the three gap nuclei,
# using the NUBASE2020 separation energies printed by the author's own script (s10).
```

```
rows=[('5He',-734.8,20577.6,1.074),('5Li',-1966.1,19813.9,1.220),('8Be',-91.8,7366.6,1.025)]
ok=True
for k,s1,s2,printed in rows:
    z=s2/s1; G=abs((z-1)/(z+1)); s=-s1
    ident=2*s/(s2-s); approx=2*s/s2
    ok&=abs((G-1)-ident)<1e-12
    print(f'{k}: |Gamma|={G:.6f} (VOTKP {printed}) |Gamma|-1={G-1:.6f} 2s/(S_next-s)={ident:.6f} 2s/S_next={
        approx:.6f}')
print('VERDICT:','exact identity holds' if ok else 'KILLED')
```

Output:

```
5He: |Gamma|=1.074062 (VOTKP 1.074) |Gamma|-1=0.074062 2s/(S_next-s)=0.074062 2s/S_next=0.071417
5Li: |Gamma|=1.220318 (VOTKP 1.22) |Gamma|-1=0.220318 2s/(S_next-s)=0.220318 2s/S_next=0.198457
8Be: |Gamma|=1.025238 (VOTKP 1.025) |Gamma|-1=0.025238 2s/(S_next-s)=0.025238 2s/S_next=0.024923
VERDICT: exact identity holds
```

s13_v4_bridges.py

```
# SCRIPT: s13_v4_bridges
# Checks every new statement that enters Matter, Then On Down v4.
# Each block states what it proves and its kill line, written before the run.
#
# K1 joint offset example: two projectors with equal census, equal offset and
# equal generator coupling saturate tau'^2 = 4 g S while their offsets move
# in opposite directions. KILL: saturation error > 1e-10 anywhere, or the
# bound written with the cross weight Tr X in place of E_H is NOT violated
# for omega >= 1 (then the two objects could be the same and the text is wrong).
# K2 metric signature of the opened SSH plane: det Q = -u^2 w^2 cos^2 k/(16 s^3).
# Symbolic identity in sympy, plus 2000 random eigenvector evaluations.
# KILL: sympy residual not identically 0, or scaled numeric error > 1e-9.
# K3 branch of the continued endpoint offset: tau_end = 2 artanh(i g) = 2 i arctan g,
# Gamma_end = i g, Kbar = 1 + g^2 = sec^2(Im tau_end / 2). KILL: any error > 1e-12,
# or |Gamma_end| != 1 at g = 1, or tau_end not finite at g = 1.
# K4 fermion record of a Hawking pair: tau = omega/T, Gamma = tanh(omega/2T) for every
# pair phase (partial trace of the two-mode pure state). KILL: error > 1e-12.
# K5 a step in spectral gap reflects with the peel coordinate:
# Gamma = (Y1-Y2)/(Y1+Y2), Y = sqrt(g_par), g_par ~ 1/Delta^2 => tanh(ln(D2/D1)/2).
# Checked against a direct transfer-matrix solution of the 1D wave equation on a step.
# KILL: error > 1e-10.
# K6 complement parity: nu -> 1-nu sends Gamma, tau to minus themselves and keeps nu(1-nu);
# P -> I-P, V,W -> -V,-W conjugates Q = Tr(P V W): metric even, curvature odd.
# KILL: error > 1e-12 on 200 random cases.
# K7 resolvent under adjoint: R_{H*}(z) = R_H(conj z)^*, and R_{H*}(z) != -R_H(z) in general.
# KILL: first identity error > 1e-12, or the minus-sign law holds on the random control.
# K8 angular chain ANG1-ANG4 (L = -i k M - kappa (I - P0)):
# ANG1 Z_b(0+) = 2k/pi; ANG2 isolated pole s* = k cot(k/kappa) - kappa for kappa > 2k/pi;
# ANG3 first stress return at order l-2 with coefficient prod c / (l-2)!;
# ANG4 E_- = k^2 kappa^2 / 3 for every N >= 2, Henrici departure at N = 200, 400.
# KILL: any analytic identity off by > 1e-10, or E_- depends on N.
import json, math
import numpy as np
from scipy.linalg import expm, eigvals
from scipy.integrate import quad
import sympy as sp

out = {}
rng = np.random.default_rng(20261104)
sx = np.array([[0, 1], [1, 0]], complex)
sy = np.array([[0, -1j], [1j, 0]])
sz = np.array([[1, 0], [0, -1]], complex)
I2 = np.eye(2)

# K1 ------------------------------------------------------------------------
sat = 0.0
```

```
viol = []
for om in [0.3, 1.0, 2.0]:
    H = om * sx
    for sgn in [1, -1]:
        P0 = (I2 + sgn * sy) / 2
        for t in np.linspace(0.0, 0.6, 31):
            U = expm(-1j * H * t)
            P = U @ P0 @ U.conj().T
            Pd = -1j * (H @ P - P @ H)
            nu = P[0, 0].real
            g = 0.5 * np.linalg.norm(Pd) ** 2
            S = 1 / (nu * (1 - nu))
            taud = -Pd[0, 0].real * S
            sat = max(sat, abs(taud ** 2 / (4 * g * S) - 1))
            EH = np.linalg.norm(P @ H @ (I2 - P)) ** 2
            TrX = nu * (1 - nu)
            if t == 0.0:
                viol.append([om, sgn, abs(taud), 2 * math.sqrt(EH * S), 2 * math.sqrt(TrX * S)])
assert sat < 1e-10
assert all(r[2] > r[4] + 1e-9 for r in viol if r[0] >= 1.0)
out['K1_joint_offset'] = {'max_saturation_error': sat, 'at_t0_[omega,sign,|taudot|,bound_EH,bound_TrX]': viol}

# K2 ----------------------------------------------------------------------
k, v, u, w = sp.symbols('k v u w', real=True)
d = sp.Matrix([v + w * sp.cos(k), w * sp.sin(k), sp.I * u])
s = (d.T * d)[0]
dk, dv = d.diff(k), d.diff(v)
def bil(a, b):
    return (a.T * b)[0]
def Qab(a, b):
    return (s * bil(a, b) - bil(d, a) * bil(d, b)) / (4 * s ** 2)
Qm = sp.Matrix([[Qab(dk, dk), Qab(dk, dv)], [Qab(dv, dk), Qab(dv, dv)]])
res = sp.simplify(sp.expand(Qm.det() * 16 * s ** 3 + u ** 2 * w ** 2 * sp.cos(k) ** 2))
assert res == 0
worst = 0.0
for _ in range(2000):
    vv, uu, kk, ww = rng.uniform(.05, 2), rng.uniform(.01, .8), rng.uniform(0, 2 * np.pi), rng.uniform(.3,
        1.5)
    E2 = vv * vv + ww * ww + 2 * vv * ww * np.cos(kk) - uu * uu
    if abs(E2) < .03:
        continue
    h = np.array([[1j * uu, vv + ww * np.exp(-1j * kk)], [vv + ww * np.exp(1j * kk), -1j * uu]])
    ev, R = np.linalg.eig(h)
    L = np.linalg.inv(R).conj().T
    Dk = np.array([[0, -1j * ww * np.exp(-1j * kk)], [1j * ww * np.exp(1j * kk), 0]])
    Dv = np.array([[0, 1], [1, 0]], complex)
    def Q(V):
        return ((L[:, 1].conj() @ V @ R[:, 0]) * (L[:, 0].conj() @ V @ R[:, 1]) / (ev[1] - ev[0]) ** 2).real
    a, c = Q(Dk), Q(Dv)
    b = Q((Dk + Dv) / np.sqrt(2)) - (a + c) / 2
    det = a * c - b * b
    exp_ = -uu * uu * ww * ww * np.cos(kk) ** 2 / (16 * E2 ** 3)
    worst = max(worst, abs(det - exp_) / max(1, abs(exp_)))
assert worst < 1e-9
grid = [(kk, vv) for kk in np.linspace(0, 2 * np.pi, 401)[:-1] for vv in np.linspace(0, 2, 400)]
lor = sum(1 for kk, vv in grid if (vv * vv + 1 + 2 * vv * np.cos(kk) - .25) > 0 and abs(np.cos(kk)) > 1e-12)
out['K2_metric_signature'] = {'sympy_residual': str(res), 'random_cases_max_scaled_error': worst,
                              'Lorentzian_fraction_u0.5_w1_grid_k0..2pi_v0..2': lor / len(grid)}

# K3 ----------------------------------------------------------------------
e3 = 0.0
for g in np.linspace(0.05, 3, 60):
    te = 2 * np.arctanh(1j * g)
    e3 = max(e3, abs(te - 2j * math.atan(g)), abs(np.tanh(te / 2) - 1j * g),
             abs((1 + g * g) - 1 / math.cos(te.imag / 2) ** 2))
t1 = 2 * np.arctanh(1j * 1.0)
assert e3 < 1e-12 and np.isfinite(t1) and abs(abs(np.tanh(t1 / 2)) - 1) < 1e-15
```

```
out['K3_branch'] = {'max_error': e3, 'tau_end_at_gamma1': [t1.real, t1.imag], 'Gamma_end_at_gamma1': [0.0,
     1.0]}

# K4 ----------------------------------------------------------------------
e4 = 0.0
for x in [0.1, 0.5, 1.0, 2.0, 5.0, 9.0]:
    r = math.atan(math.exp(-x / 2))
    for phi in np.linspace(0, 2 * np.pi, 7):
        psi = np.zeros(4, complex); psi[0] = math.cos(r); psi[3] = np.exp(1j * phi) * math.sin(r)
        rho = np.outer(psi, psi.conj()).reshape(2, 2, 2, 2)
        rout = np.einsum('ajbj->ab', rho)
        C = rout[1, 1].real
        tau = math.log((1 - C) / C)
        e4 = max(e4, abs(tau - x), abs((1 - 2 * C) - math.tanh(x / 2)))
assert e4 < 1e-12
out['K4_fermion_record'] = {'max_error': e4, 'frequencies_omega_over_T': [0.1, 0.5, 1, 2, 5, 9], 'phases': 7}

# K5 ----------------------------------------------------------------------
# 1D wave u'' + omega^2 g u = 0 with index sqrt(g); g = c / Delta^2. Step at y = 0.
e5 = 0.0
for _ in range(200):
    D1, D2 = rng.uniform(.2, 5, 2)
    om = rng.uniform(.5, 4)
    n1, n2 = 1 / D1, 1 / D2
    k1, k2 = om * n1, om * n2
    # incident e^{ik1 y} + r e^{-ik1 y} for y<0, t e^{ik2 y} for y>0; continuity of u and u'
    A = np.array([[1, -1], [-1j * k1, -1j * k2]])
    rhs = np.array([-1, -1j * k1])
    r_, t_ = np.linalg.solve(A, rhs)
    e5 = max(e5, abs(r_ - math.tanh(0.5 * math.log(D2 / D1))))
assert e5 < 1e-10
out['K5_gap_step_reflection'] = {'random_steps': 200, 'max_error': e5}

# K6 ----------------------------------------------------------------------
e6 = 0.0
for _ in range(200):
    nu = rng.uniform(.01, .99)
    G, Gc = 1 - 2 * nu, 1 - 2 * (1 - nu)
    ta, tc = math.log((1 - nu) / nu), math.log(nu / (1 - nu))
    e6 = max(e6, abs(G + Gc), abs(ta + tc), abs(nu * (1 - nu) - (1 - nu) * nu))
    n, rk = 5, 2
    Z = rng.normal(size=(n, n)) + 1j * rng.normal(size=(n, n))
    Qm_, _ = np.linalg.qr(Z)
    P = Qm_[:, :rk] @ Qm_[:, :rk].conj().T
    def tangent():
        Hh = rng.normal(size=(n, n)) + 1j * rng.normal(size=(n, n)); Hh = Hh + Hh.conj().T
        return 1j * (Hh @ P - P @ Hh)
    V, W = tangent(), tangent()
    Q1 = np.trace(P @ V @ W)
    Q2 = np.trace((np.eye(n) - P) @ (-V) @ (-W))
    e6 = max(e6, abs(Q2 - np.conj(Q1)), abs(np.trace(V @ W) - 2 * Q1.real))
assert e6 < 1e-12
out['K6_complement_parity'] = {'cases': 200, 'max_error': e6}

# K7 ----------------------------------------------------------------------
e7, minus_law = 0.0, 0.0
for _ in range(50):
    Hh = rng.normal(size=(4, 4)) + 1j * rng.normal(size=(4, 4))
    z = complex(rng.normal(), rng.normal())
    R1 = np.linalg.inv(z * np.eye(4) - Hh.conj().T)
    R2 = np.linalg.inv(np.conj(z) * np.eye(4) - Hh).conj().T
    e7 = max(e7, np.abs(R1 - R2).max())
    minus_law = max(minus_law, np.abs(R1 + np.linalg.inv(z * np.eye(4) - Hh)).max())
assert e7 < 1e-12 and minus_law > 1e-3
out['K7_resolvent_adjoint'] = {'max_error_adjoint_law': e7, 'min_violation_of_minus_law_max_over_cases':
     minus_law}

# K8 ----------------------------------------------------------------------
```

```
def cvec(n):
    return np.array([(l + 1) / math.sqrt((2 * l + 1) * (2 * l + 3)) for l in range(n - 1)])
ang = {}
kk = 1.0
eA1 = max(abs(quad(lambda mu: s_ / (s_ * s_ + kk * kk * mu * mu), -1, 1)[0] / 2 - math.atan(kk / s_) / kk) for
      s_ in [.01, .1, .5, 2])
ang['ANG1'] = {'angular_integral_error': eA1, 'Z_b_0plus': 2 * kk / math.pi,
               'Z_b_small_s': [[s_, 1 / (math.atan(kk / s_) / kk) - s_] for s_ in [1e-2, 1e-4, 1e-6]]}
poles = []
for kap in [.64, .7, .916, 1, 2]:
    p = kk / math.tan(kk / kap) - kap
    g0 = math.atan(kk / (p + kap)) / kk
    poles.append([kap, p, abs(1 - kap * g0)])
    assert p > -kap and abs(1 - kap * g0) < 1e-12
ang['ANG2'] = {'onset_kappa': 2 * kk / math.pi, 'poles_[kappa,s*,residual]': poles}
n = 16; c = cvec(n); M = np.diag(c, 1) + np.diag(c, -1); L = -1j * kk * M
ret = []
for l in [3, 4, 5, 8]:
    for q in range(l - 2):
        assert abs(np.linalg.matrix_power(L, q)[2, l]) < 1e-15
    exact = np.prod(c[2:l]) / math.factorial(l - 2)
    calc = abs(np.linalg.matrix_power(L, l - 2)[2, l]) / math.factorial(l - 2)
    assert abs(calc - exact) < 1e-14
    for q in range(l):
        assert abs(np.linalg.matrix_power(L, q)[0, l]) < 1e-15
    ret.append([l, l - 2, exact, abs(np.linalg.matrix_power(L, l)[0, l])])
ang['ANG3'] = {'[l, stress order, coefficient, density coefficient at order l]': ret}
rows = []
for n in [2, 5, 20, 100, 200, 400]:
    c = cvec(n); M = np.diag(c, 1) + np.diag(c, -1)
    for kap in [1.0, 0.5]:
        L = -1j * kk * M - kap * np.diag([0] + [1] * (n - 1)); A_ = L.conj().T
        Em = np.linalg.norm(L @ A_ - A_ @ L, 'fro') ** 2 / 8
        Ep = np.linalg.norm(L @ L - A_ @ A_, 'fro') ** 2 / 8
        dep = np.linalg.norm(L, 'fro') ** 2 - np.sum(abs(eigvals(L)) ** 2)
        assert abs(Em - kk * kk * kap * kap / 3) < 1e-10
        rows.append([n, kap, Em, Ep, float(dep)])
ang['ANG4'] = {'[N, kappa, E_minus, E_plus, departure^2]': rows}
out['K8_angular_chain'] = ang
out['status'] = 'PASS'
print(json.dumps(out, indent=1))
```

Output:


```
{
 "K1_joint_offset": {
  "max_saturation_error": 1.4055423491754482e-13,
  "at_t0_[omega,sign,|taudot|,bound_EH,bound_TrX]": [
   [
    0.3,
    1,
    1.2,
    1.2,
    2.0
   ],
   [
    0.3,
    -1,
    1.2,
    1.2,
    2.0
   ],
   [
    1.0,
    1,
    4.0,
    4.0,
    2.0
```

```
  ],
  [
   1.0,
   -1,
   4.0,
   4.0,
   2.0
  ],
  [
   2.0,
   1,
   8.0,
   8.0,
   2.0
  ],
  [
   2.0,
   -1,
   8.0,
   8.0,
   2.0
  ]
 ]
},
"K2_metric_signature": {
 "sympy_residual": "0",
 "random_cases_max_scaled_error": 4.120103029277773e-14,
 "Lorentzian_fraction_u0.5_w1_grid_k0..2pi_v0..2": 0.9304125
},
"K3_branch": {
 "max_error": 3.552713678800501e-15,
 "tau_end_at_gamma1": [
  0.0,
  1.5707963267948966
 ],
 "Gamma_end_at_gamma1": [
  0.0,
  1.0
 ]
},
"K4_fermion_record": {
 "max_error": 4.996003610813204e-16,
 "frequencies_omega_over_T": [
  0.1,
  0.5,
  1,
  2,
  5,
  9
 ],
 "phases": 7
},
"K5_gap_step_reflection": {
 "random_steps": 200,
 "max_error": 3.3306690738754696e-16
},
"K6_complement_parity": {
 "cases": 200,
 "max_error": 1.4501059790511035e-14
},
"K7_resolvent_adjoint": {
 "max_error_adjoint_law": 1.025259555530695e-14,
 "min_violation_of_minus_law_max_over_cases": 18.88337148491665
},
"K8_angular_chain": {
 "ANG1": {
  "angular_integral_error": 1.199040866595169e-14,
  "Z_b_0plus": 0.6366197723675814,
```

```
 "Z_b_small_s": [
  [
   0.01,
   0.6306984494256005
  ],
  [
   0.0001,
   0.6365603034211902
  ],
  [
   1e-06,
   0.6366191776525739
  ]
 ]
},
"ANG2": {
 "onset_kappa": 0.6366197723675814,
 "poles_[kappa,s*,residual]": [
  [
   0.64,
   -0.6317034828571313,
   0.0
  ],
  [
   0.7,
   -0.5568083071568761,
   0.0
  ],
  [
   0.916,
   -0.39654109846835683,
   1.1102230246251565e-16
  ],
  [
   1,
   -0.35790738406566935,
   0.0
  ],
  [
   2,
   -0.169512278287548,
   0.0
  ]
 ]
},
"ANG3": {
 "[l, stress order, coefficient, density coefficient at order l]": [
  [
   3,
   1,
   0.50709255283711,
   0.1511857892036909
  ],
  [
   4,
   2,
   0.12777531299998796,
   0.07619047619047618
  ],
  [
   5,
   3,
   0.021403170236928313,
   0.0382871548670176
  ],
  [
   8,
   6,
```

```
    2.2474017904030684e-05,
    0.004824329449052155
   ]
  ]
 },
 "ANG4": {
  "[N, kappa, E_minus, E_plus, departure^2]": [
   [
    2,
    1.0,
    0.3333333333333334,
    0.3333333333333334,
    1.0
   ],
   [
    2,
    0.5,
    0.08333333333333336,
    0.08333333333333336,
    0.2499999999999999
   ],
   [
    5,
    1.0,
    0.3333333333333334,
    3.444444444444444,
    1.0811427111292051
   ],
   [
    5,
    0.5,
    0.08333333333333336,
    0.861111111111111,
    0.3761404942669939
   ],
   [
    20,
    1.0,
    0.3333333333333334,
    18.487179487179482,
    1.1556239260556325
   ],
   [
    20,
    0.5,
    0.08333333333333336,
    4.6217948717948705,
    0.46440205623208897
   ],
   [
    100,
    1.0,
    0.3333333333333334,
    98.49748743718595,
    1.1715348878590248
   ],
   [
    100,
    0.5,
    0.08333333333333336,
    24.624371859296488,
    0.49256778221931086
   ],
   [
    200,
    1.0,
    0.3333333333333334,
    198.49874686716794,
```

```
    1.1734883039923147
   ],
   [
    200,
    0.5,
    0.08333333333333336,
    49.624686716791985,
    0.4962630462953257
   ],
   [
    400,
    1.0,
    0.3333333333333334,
    398.49937421777236,
    1.1744621662352301
   ],
   [
    400,
    0.5,
    0.08333333333333336,
    99.62484355444309,
    0.49812625235233554
   ]
  ]
 }
},
"status": "PASS"
}
```

s14_v4_coverage.py

```
# SCRIPT: s14_v4_coverage
# Merge check: every numbered result, figure and script of the sources must appear in v4.
# Each manifest row is (source, item, a distinctive string that only that item's content carries).
# KILL: any row missing from the v4 source text. A missing row is a compression and must be restored.
import glob, re, sys
tex = ''.join(open(f).read() for f in sorted(glob.glob('../paper/v4/*.tex')))
flat = re.sub(r'\s+', ' ', tex)
M = [
 # v3 (matter_then_on_down_v3): body results, appendix items, data rows
 ('v3', 'zero and span', r'K\mapsto cK'), ('v3', 'census formula eq', r'|\partial A|\,\frac{c}{\pi^2}'),
 ('v3', 'c=1 coefficient', '0.202692'), ('v3', 'Ising 41.8x', '41.8'), ('v3', 'rails spacing', r'4\pi K(\tilde
      u)/K'),
 ('v3', 'bridge prop', r'\int_{-\infty}^{\infty}H(\nu(\varepsilon))'), ('v3', 'offset log-odds', 'P(N_A=0)'),
 ('v3', 'census 0.16I example', '0.16I'), ('v3', 'defect split', 'C_{AB}C_{BA}'), ('v3', 'Gamma_b modulus', r
     '\sqrt{1-4\lambda}'),
 ('v3', 'speed limit', r'\dot\tau^2\le4gS'), ('v3', 'warm ring counts', '2 to 24 to 64'), ('v3', 'TrX warm',
     '0.170835'),
 ('v3', 'gas density-free', '0.37757506698375'), ('v3', 'partition bias', '0.36614943350748'),
 ('v3', 'peel add thm', r'u\oplus v=(u+v)/(1+uv)'), ('v3', 'Ising ladder residuals', r'8.0\times10^{-6}'),
 ('v3', 'helium thresholds', '54.4177655282'), ('v3', 'binding sum', '79.0051545392'), ('v3', 'ladder
     reconstruction', r'\prod_{j<q}(1+\Gamma_j)/(1-\Gamma_j)'),
 ('v3', 'quantum dots', 'quantum-dot'), ('v3', 'sodium 0.9938', '0.9938'), ('v3', 'Thomas-Fermi', 'Thomas'),
 ('v3', 'nuclear table 5He', '734.8'), ('v3', 'nuclear 5Li', '1966.1'), ('v3', 'nuclear 8Be', '7366.6'), ('v3
     ', 'half lives', '81.9 as'),
 ('v3', 'rim distance', r'\frac{2s}{S_{\mathrm{next}}-s}'), ('v3', 'rim exact values', '0.220318'), ('v3', '
     small-s values', '0.198457'),
 ('v3', 'resonance caution', 'is not a literal atomic or nuclear Smith reflection'), ('v3', 'half-turn 4000
     draws', '4000'),
 ('v3', 'z=1-i gamma', r'z=1-i\gamma'), ('v3', 'cone Q>=0', r'M_{+-}M_{-+}'), ('v3', 'biorthogonal idempotent
     ', r'3\times10^{-14}'),
 ('v3', 'metric 93%', '93'), ('v3', 'Renyi -1.79', '-1.79'), ('v3', 'exit 0.468', '0.468'), ('v3', 'Petermann
     K', '1+u^2/E(k)^2'),
 ('v3', 'average formula', r'\sqrt{A^2-B^2}'), ('v3', 'predictions', '0.7474014'), ('v3', 'massive 1.015-1.022
      limit', 'mass term moves the exit'),
```

```
 ('v3', 'filter table', '0.271636929'), ('v3', 'w=0 counterexample', 'v^2=2u^2'), ('v3', 'quarter turn sec^2',
      r'\sec^2'),
 ('v3', 's11 rotation gap', r'3.4\times10^{-2}'), ('v3', 'open list', 'placing nuclear resonances'),
 ('v3', 'A.1', 'A.1 Zero and span'), ('v3', 'A.2', 'A.2 The census coefficient'), ('v3', 'A.3', 'A.3 The
     bridge'), ('v3', 'A.4', 'A.4 The offset'),
 ('v3', 'A.5', 'A.5 Defect split'), ('v3', 'A.6', 'A.6 The boundary coordinate'), ('v3', 'A.7', 'A.7 The
     Offset Speed Limit'), ('v3', 'A.8', 'A.8 Peel steps'),
 ('v3', 'A.9', 'A.9 Scale blindness'), ('v3', 'A.10', 'A.10 The last peel step'), ('v3', 'A.11', 'A.11 Ideal
     Saha'), ('v3', 'A.12', 'A.12 Outside the rim'),
 ('v3', 'A.13', 'A.13 No cone'), ('v3', 'A.14', 'A.14 The Petermann'), ('v3', 'A.15', 'A.15 The quarter-turn')
     , ('v3', 'Unsold', 'Ann.\\ Phys.\\ 387'),
 ('v3', 'Lean MatterDisk', 'MatterDisk.lean'), ('v3', 'fig02', 'fig02_boundary_and_gas'), ('v3', 'fig03 (
     signed rebuild)', 'fig03v4_one_chart'),
] + [('v3', 'script ' + s, s) for s in ['s01_bridge', 's02_ladder', 'nhssh_lib_and_s03a', 's03b_renyi', '
    s03c_metric_raw', 's03d_metric_rank', 's04a_rotation', 's04b_rotation', 's05a_exit', 's05b_predict', '
    s06_second_model', 's07_boundary', 's08_gas', 's09_check', 's10_nuclear', 's11_exit', 's12_nuclear']] + [
 # formal edition (Matter_Gases_VOTKP_Formal_Edition_20261103)
 ('gpt', 'B1 defect identity', 'F_A=T_A+X_A'), ('gpt', 'B2 bounds', r'\|B_C\|\le\tfrac12'), ('gpt', 'B3 two-
     sided', r'\tfrac14\min(a,b)'),
 ('gpt', 'B4 comparison', r'N^\partial_\varepsilon\le N^{\mathrm{raw}}_\varepsilon'), ('gpt', 'B5 trace
     estimate', r'\theta N^\partial_\varepsilon\le\Tr X_A'),
 ('gpt', 'B6 local unitary', 'U_A\\oplus U_B'), ('gpt', 'B7 stability', 'certified'), ('gpt', 'B8 near
     projector', 'varepsilon_\\eta'),
 ('gpt', 'B9 extension', r'\sqrt{A-A^2}'), ('gpt', 'B10 variance', r'\mathrm{Cov}(N_A,N_B)=-\Tr X_A'), ('gpt',
      'B11 classical', r'\rho_{\mathrm{cl}}'),
 ('gpt', 'B11 entanglement', '|1100'), ('gpt', 'B12 matched boundary', r'C_\pm=\tfrac12I\pm r\sigma_y'), ('gpt
     ', 'B13 info bound', r'4\Tr X_A\le I_G(A:B)'),
 ('gpt', 'B13 upper', r'\delta(1-\delta)'), ('gpt', 'B13 census bound', r'I_G(A:B)/(4\varepsilon(1-\varepsilon
     ))'),
 ('gpt', 'D1 coordinate', r'\Gamma_b\le1-2\varepsilon'), ('gpt', 'D2 back to rim', 'disconnected control'), ('
     gpt', 'D3 idempotent', 'Y_A=A-A^2'),
 ('gpt', 'D4 radial disk', r'|\lambda-\tfrac14|\le r^2/4'), ('gpt', 'D5 C=[[2,1],[-2,-1]]', r'2&1\\-2&-1'), ('
     gpt', 'D5 embedding', r'\mathcal P(A)'),
 ('gpt', 'D6 branch', r'\cos(\Ima W)>0'), ('gpt', 'D7 Holder', r'2\sqrt{|\lambda-\tilde\lambda|}'), ('gpt', '
     D7 nonnormal J', r'\pm\sqrt\delta'),
 ('gpt', 'G1 anchor', 'anchored coordinate equivalence'), ('gpt', 'G2 exp family', r'd\mu/dy=-V'), ('gpt', 'G3
      unique', 'a unique neutral gas'),
 ('gpt', 'G3a helium cubic', r'u^3+\alpha u^2'), ('gpt', 'G4 density', r'\frac{\mu}{\mu+V}'), ('gpt', 'G5
     coefficient', 'C_h=\\mathrm{Cov}_p(q,h_q)'),
 ('gpt', 'G6 temperature', r'\tfrac32q+E_q/k_BT'), ('gpt', 'G7 scaling', '(ab)^{3/2}'), ('gpt', 'G7a
     insufficiency', 'what the atom alone cannot know'),
 ('gpt', 'neutral fractions', '0.289781'), ('gpt', 'G8 inverse', r'(Z-1-r)D_r'), ('gpt', 'G9 partition', r'D_
     {\mathrm{app}}'), ('gpt', 'G9 conditioning', r'(\eta_1+\eta_2)/|\Delta\beta|'),
 ('gpt', 'G9 bias value', '0.0114256'), ('gpt', 'R1 reduced', 'hidden source'), ('gpt', 'R2 channel measure',
     r'd\mathsf M'),
 ('gpt', 'R3 witness', '(3,7,1,5)/16'), ('gpt', 'R3 resolvents', '5z^2+3z-9'), ('gpt', 'R4 kernel', r'J_\pm=\
     pm i/\sqrt2'),
 ('gpt', 'M1 moments', r'\delta(-1)^mm!h^m'), ('gpt', 'M2/readout reflected', '(5,1,7,3)/16'), ('gpt', 'K1', r
     '48av_0^3'),
 ('gpt', 'K2', r'\sin^3(kv_0t)'), ('gpt', 'K3', r'96a\eta\rho_*k^3v_0^3'), ('gpt', 'K4 reset', r'\mathcal E_s
     '), ('gpt', 'K null table', 'placement matters'),
 ('gpt', 'P1', r'\tfrac12(1+\sum|d_j|^2/|E|^2)'), ('gpt', 'P2', 'E(0)E(\\pi)=u^2'), ('gpt', 'P3', 'direct sum
     of identical cell projectors'),
 ('gpt', '9.4 energy origin', r'-\tfrac14,\tfrac34'), ('gpt', '9.4 negative census coefficient', r'a\log\xi+O
     (1)'),
 ('gpt', '10.1 condition numbers', '575'), ('gpt', '10.1 residual', r'4.4\times10^{-15}'), ('gpt', 'Lean 34',
     '34 positive declarations'),
 ('gpt', 'Lean (7,11,5,9)', '(7,11,5,9)/32'), ('gpt', 'Peschel', 'Peschel'), ('gpt', 'Wolf et al', 'Hastings')
     ,
] + [('gpt', 'script ' + s, s) for s in ['verify_formal_gas', 'adversarial_checks', '
    gaussian_information_checks', 'incoming_exit_filter', 'incoming_exit_residuals']] + [
 # reconciled review strengthening and angular cards
 ('review', 'metric det', r'\det Q=-\frac{u^2w^2\cos^2k}{16\,s^3}'), ('review', 'branch Gamma_end=i', r'\
     Gamma_{\mathrm{end}}=i'),
 ('review', 'complement parity', 'complement parity'), ('review', 'resolvent adjoint', r'R_{H^*}(z)=R_H(\bar z
     )^*'),
 ('review', 'Hermitian tangent counterexample', r'\sigma_z+i\theta\sigma_x'), ('review', 'Gaussian hypothesis
     on offset', 'quasi-free'),
```

```
 ('review', 'ANG1', '2k/\\pi'), ('review', 'ANG2', r'k\cot(k/\kappa)-\kappa'), ('review', 'ANG3 coefficient',
     '0.507092553'), ('review', 'ANG4 Henrici', '1.174462'),
 ('review', 'ANG4 E-', r'k^2\kappa^2/3'), ('review', 'cross weight vs E_H', 'cross weight would understate'),
 # Beyond the Event Horizon ed. 10 bridges
 ('bh', 'gap step', r'\tanh\Big(\tfrac12\ln\frac{\Delta_2}{\Delta_1}\Big)'), ('bh', 'Clairaut step', 'Clairaut
     '), ('bh', 'Schur-Cayley', 'Schur'),
 ('bh', 'membrane resistivity', '376.73'), ('bh', 'fermion record', r'\tanh\frac{\hbar\omega}{2k_BT_H}'), ('bh
     ', 'off-diagonal block family', 'PHQHP'),
 ('bh', 'spectral vs geometric walls', 'Two kinds of wall'), ('bh', 'polar mirror', 'perfect mirror'), ('bh',
     'ringdown 391', '391'),
] + [('new', 'figure ' + f, f) for f in ['fig04_reflection_chart', 'fig05_metric_signature', '
    fig06_records_and_futures', 'fig07_exit_and_filter', 'fig08_angular_chain', 's13_v4_bridges']]
miss = [(a, b) for a, b, c in M if re.sub(r'\s+', ' ', c) not in flat]
by = {}
for a, b, c in M: by.setdefault(a, [0, 0]); by[a][0] += 1; by[a][1] += (a, b) not in miss
for k, (n, ok) in by.items(): print(f'{k:7s} {ok}/{n} present')
for a, b in miss: print('MISSING', a, b)
print('v4 source characters', len(tex), ' v3 source characters', len(open('../paper/matter_then_on_down_v3.tex
    ').read()))
print('PASS' if not miss else 'FAIL')
```

Output:

```
v3 83/83 present
gpt 67/67 present
review 12/12 present
bh 9/9 present
new 6/6 present
v4 source characters 110959 v3 source characters 41821
PASS
```

verify_formal_gas.py

```
#!/usr/bin/env python3
"""Prospective finite-model controls for G1-G9 and R1-R4.

Analytic proofs are in the companion notes. Exact-arithmetic checks and
floating-point implementation controls are explicitly distinguished below.
No observations are generated or fitted here.
"""
from pathlib import Path
from fractions import Fraction as F
import hashlib
import json
import math

import numpy as np
from scipy.constants import h, m_e, k, electron_volt
from scipy.linalg import expm
from scipy.optimize import brentq
from scipy.special import logsumexp

HERE = Path(__file__).resolve().parent
SEED = 20261103
rng = np.random.default_rng(SEED)
checks = {}


def solve(log_s, density):
    log_s = np.asarray(log_s, dtype=float)
    z = len(log_s)
    charges = np.arange(z + 1, dtype=float)
    log_a = np.r_[0.0, np.cumsum(log_s)]
    def state(y):
        lp = log_a - charges * y
```

```
        lp -= logsumexp(lp)
        p = np.exp(lp)
        mean = float(p @ charges)
        variance = float(p @ ((charges - mean) ** 2))
        return lp, p, mean, variance
    def residual(y):
        mean = state(y)[2]
        return y - math.log(density) - math.log(mean)
    y = brentq(residual, math.log(density) - 500.0,
               math.log(z * density), xtol=5e-14,
               rtol=4*np.finfo(float).eps)
    lp, p, mean, variance = state(y)
    return dict(log_x=y, x=math.exp(y), p=p, lp=lp, mean=mean,
                variance=variance, q=charges,
                neutrality=abs(math.exp(y)/density-mean),
                normalization=abs(float(p.sum())-1.0))


def cov(state, values):
    return float(state['p'] @ ((state['q']-state['mean']) * values))


def poly_mul(a, b):
    out = [F(0)] * (len(a)+len(b)-1)
    for i, ai in enumerate(a):
        for j, bj in enumerate(b):
            out[i+j] += ai*bj
    return out


# G1: exact rational reconstruction, no tolerances.
for _ in range(100):
    size = int(rng.integers(1, 9))
    ladder = [F(int(rng.integers(1, 40)), int(rng.integers(1, 20))) for _ in range(size)]
    gamma = [(b-a)/(b+a) for a, b in zip(ladder, ladder[1:])]
    reconstructed = [ladder[0]]
    for g in gamma:
        reconstructed.append(reconstructed[-1]*(1+g)/(1-g))
    assert reconstructed == ladder
    scaled = [F(7, 3)*a for a in ladder]
    assert gamma == [(b-a)/(b+a) for a, b in zip(scaled, scaled[1:])]
checks['G1_exact_rational_roundtrip_and_scale_100'] = True

# G3: exact positive polynomial root manufactured at x=1.
manufactured_errors = []
for z in range(1, 9):
    s = [F(q+2, q+1) for q in range(z)]
    a = [F(1)]
    for sq in s:
        a.append(a[-1]*sq)
    mean = sum(F(q)*aq for q, aq in enumerate(a))/sum(a)
    density = 1/mean
    polynomial_at_one = sum(a)-density*sum(F(q)*aq for q, aq in enumerate(a))
    assert polynomial_at_one == 0
    solution = solve(np.log([float(v) for v in s]), float(density))
    manufactured_errors.append(abs(solution['x']-1.0))
checks['G3_exact_manufactured_neutrality_polynomials_8'] = max(manufactured_errors) < 1e-11

# G3-G7: random finite charge systems and their sensitivity controls.
step = 1e-5
density_errors = []
coefficient_errors = []
scale_errors = []
neutrality_errors = []
normalization_errors = []
density_signs = []
coefficient_signs = []
tail_signs = []
```

```
for _ in range(64):
    z = int(rng.integers(1, 9))
    logs = rng.uniform(-1.5, 1.5, z)
    density = math.exp(float(rng.uniform(-1, 1)))
    base = solve(logs, density)
    m, v = base['mean'], base['variance']
    minus = solve(logs, density*math.exp(-step))
    plus = solve(logs, density*math.exp(step))
    observed = (plus['log_x']-minus['log_x'])/(2*step)
    density_errors.append(abs(observed-m/(m+v)))
    den_up = solve(logs, density*1.01)
    density_signs.append(den_up['x'] > base['x'] and den_up['mean'] < m)
    tail_signs.extend(float(den_up['p'][r:].sum()) < float(base['p'][r:].sum())
                  for r in range(1, z+1))
    r = int(rng.integers(0, z))
    indicator = (base['q'] > r).astype(float)
    derivative = cov(base, indicator)/(m+v)
    lm, lp = logs.copy(), logs.copy()
    lm[r] -= step
    lp[r] += step
    observed = (solve(lp, density)['log_x']-solve(lm, density)['log_x'])/(2*step)
    coefficient_errors.append(abs(observed-derivative))
    up = logs.copy()
    up[r] += math.log(1.01)
    coefficient_signs.append(solve(up, density)['mean'] > m)
    for factor in (0.1, 0.5, 2.0, 10.0):
        scaled = solve(logs+math.log(factor), density*factor)
        scale_errors.append(float(np.max(abs(scaled['p']-base['p']))))
    coefficient_signs.append(solve(logs+math.log(1.01), density)['mean'] > m)
    neutrality_errors.append(base['neutrality'])
    normalization_errors.append(base['normalization'])
checks['G3_neutrality_and_normalization_64'] = max(neutrality_errors+normalization_errors) < 1e-11
checks['G4_density_elasticity'] = max(density_errors) < 2e-7
checks['G4_density_stochastic_order'] = all(density_signs+tail_signs)
checks['G5_single_coefficient_response'] = max(coefficient_errors) < 2e-7 and all(coefficient_signs)
checks['G7_coefficient_density_scale_covariance'] = max(scale_errors) < 1e-11

# G6: dimensionless fixed-partition temperature responses (k_B=1).
temperature_errors = []
temperature_signs = []
dictionary_errors = []
for _ in range(32):
    z = int(rng.integers(1, 9))
    thresholds = rng.uniform(0.2, 8, z)
    partitions = np.exp(rng.uniform(-0.5, 0.5, z+1))
    temperature = math.exp(float(rng.uniform(-0.5, 2)))
    density = math.exp(float(rng.uniform(-1, 1)))
    def logs_at(t, energies=thresholds, mass=1.0):
        return math.log(2)+1.5*math.log(mass*t)+np.diff(np.log(partitions))-energies/t
    base = solve(logs_at(temperature), density)
    cumulative = np.r_[0.0, np.cumsum(thresholds)]
    hv = 1.5*base['q']+cumulative/temperature
    predicted = cov(base, hv)/(base['mean']+base['variance'])
    observed = (solve(logs_at(temperature*math.exp(step)), density)['log_x']-
                solve(logs_at(temperature*math.exp(-step)), density)['log_x'])/(2*step)
    temperature_errors.append(abs(observed-predicted))
    temperature_signs.append(solve(logs_at(1.01*temperature), density)['mean'] > base['mean'])
    for factor in (0.5, 2.0):
        fixed_mass = solve(logs_at(factor*temperature, factor*thresholds), factor**1.5*density)
        changed_mass = solve(logs_at(factor*temperature, factor*thresholds, factor), factor**3*density)
        dictionary_errors.extend([float(np.max(abs(fixed_mass['p']-base['p']))),
                                  float(np.max(abs(changed_mass['p']-base['p'])))])
checks['G6_fixed_partition_temperature_response'] = max(temperature_errors) < 2e-7 and all(temperature_signs)
checks['G7_both_physical_scaling_dictionaries'] = max(dictionary_errors) < 1e-11

# R3: exact spectral moment and rational-resolvent checks.
energies = [F(-3), F(-1), F(1), F(3)]
weights = [[F(1, 4)]*4, [F(v, 16) for v in (3, 7, 1, 5)]]
```

```
moments = [[sum(w*e**r for w, e in zip(ws, energies)) for r in range(4)] for ws in weights]
assert moments == [[F(1), F(0), F(5), F(0)], [F(1), F(0), F(5), F(3)]]
denominator = [F(1)]
for e in energies:
    denominator = poly_mul(denominator, [-e, F(1)])
numerators = []
for ws in weights:
    num = [F(0)]*4
    for j, w in enumerate(ws):
        prod = [F(1)]
        for kidx, e in enumerate(energies):
            if kidx != j:
                prod = poly_mul(prod, [-e, F(1)])
        num = [a+w*b for a, b in zip(num, prod)]
    numerators.append(num)
assert denominator == [F(9), F(0), F(-10), F(0), F(1)]
assert numerators == [[F(0), F(-5), F(0), F(1)], [F(3), F(-5), F(0), F(1)]]
checks['R3_exact_moments_and_resolvent_polynomials'] = True
hamiltonian_errors = []
survival = []
third_hidden_moments = []
matrix_records = []
for ws in weights:
    s = np.sqrt(np.array([float(v) for v in ws]))
    e0 = np.array([1.0, 0.0, 0.0, 0.0])
    v = e0-s
    r = np.eye(4)-np.outer(v, v)/(1-s[0])
    hh = r @ np.diag([float(e) for e in energies]) @ r.T
    coupling, hidden = hh[0, 1:], hh[1:, 1:]
    hamiltonian_errors.extend([float(np.linalg.norm(r.T@r-np.eye(4))),
                               float(np.max(abs(np.linalg.eigvalsh(hh)-np.array(energies, dtype=float)))),
                               abs(hh[0, 0]), abs(float(coupling@coupling)-5)])
    survival.append(float(abs(expm(-1j*hh*math.pi/4)[0, 0])**2))
    third_hidden_moments.append(float(coupling@hidden@coupling))
    matrix_records.append(hh.tolist())
checks['R3_constructive_equal_spectrum_equal_footprint'] = max(hamiltonian_errors) < 2e-12
checks['R3_later_survival_0_versus_1_over_32'] = max(abs(survival[0]), abs(survival[1]-1/32)) < 2e-12
checks['R3_hidden_first_channel_moment_0_versus_3'] = max(abs(third_hidden_moments[0]), abs(
    third_hidden_moments[1]-3)) < 2e-12

# R4: the same kernel requires the hidden-source record as well.
hh = np.array([[0.0, 1.0], [1.0, 0.0]])
uu = expm(-1j*hh*math.pi/4)
source_survival = [float(abs((uu@np.array([1, sign*1j])/math.sqrt(2))[0])**2)
                   for sign in (1, -1)]
checks['R4_same_kernel_different_hidden_source'] = max(abs(source_survival[0]-1), abs(source_survival[1])) < 2
    e-12

# G8-G9: actual §8 helium values, followed by a synthetic partition opposite.
helium_i = np.array([24.587389011, 54.4177655282])
true_gamma = float(np.diff(helium_i)[0]/helium_i.sum())


def combinations(temperature, density, threshold, partitions, physical=False):
    z = len(threshold)
    if physical:
        beta = electron_volt/(k*temperature)
        log_lam3 = 3*math.log(h)-1.5*math.log(2*math.pi*m_e*k*temperature)
    else:
        beta = 1/temperature
        log_lam3 = -1.5*math.log(temperature)
    lu = np.log(partitions)
    logs = math.log(2)-log_lam3+np.diff(lu)-beta*threshold
    st = solve(logs, density)
    lp = st['lp']
    ld = lp[:-2]+lp[2:]-2*lp[1:-1]
    lb = lp[0]-lp[-1]-z*(st['log_x']+log_lam3)
    gd = lu[:-2]+lu[2:]-2*lu[1:-1]
```

```
    gb = lu[0]-lu[-1]-z*math.log(2)
    return dict(beta=beta, ld=ld, lb=lb, gd=gd, gb=gb)


def invert(pair, corrected):
    aa, bb = pair
    db = bb['beta']-aa['beta']
    dd = -((bb['ld']-(bb['gd'] if corrected else 0))-
           (aa['ld']-(aa['gd'] if corrected else 0)))/db
    total = ((bb['lb']-(bb['gb'] if corrected else 0))-
             (aa['lb']-(aa['gb'] if corrected else 0)))/db
    z = len(dd)+1
    i0 = (total-sum((z-1-r)*value for r, value in enumerate(dd)))/z
    ladder = np.r_[i0, i0+np.cumsum(dd)]
    gamma = np.diff(ladder)/(ladder[1:]+ladder[:-1])
    return ladder, gamma, dd, total


helium_records = []
for density in (1e18, 1e20, 1e22, 1e24, 1e26):
    for excited in (False, True):
        pair = []
        for temperature in (30000.0, 32000.0):
            middle = 2+4*math.exp(-10*electron_volt/(k*temperature)) if excited else 2
            pair.append(combinations(temperature, density, helium_i,
                                     np.array([1.0, middle, 1.0]), physical=True))
        naive = invert(pair, False)
        corrected = invert(pair, True)
        helium_records.append(dict(density_m3=density, synthetic_excited_partition=excited,
                                   naive_gamma=float(naive[1][0]), corrected_gamma=float(corrected[1][0]),
                                   naive_difference_eV=float(naive[2][0]), naive_sum_eV=float(naive[3]),
                                   naive_gamma_bias=float(naive[1][0]-true_gamma),
                                   corrected_gamma_error=float(abs(corrected[1][0]-true_gamma))))
fixed = [r for r in helium_records if not r['synthetic_excited_partition']]
excited = [r for r in helium_records if r['synthetic_excited_partition']]
checks['G8_incoming_section8_fixed_partition_inverse'] = max(abs(r['naive_gamma_bias']) for r in fixed) < 1e
    -10
checks['G9_excited_partition_naive_slope_false_control'] = min(abs(r['naive_gamma_bias']) for r in excited) >
    1e-3
checks['G9_known_partition_correction_restores_inverse'] = max(r['corrected_gamma_error'] for r in excited) <
    1e-10

general_inverse_errors = []
for _ in range(16):
    z = int(rng.integers(2, 9))
    threshold = rng.uniform(0.5, 4, z)
    partitions = np.exp(rng.uniform(-0.5, 0.5, z+1))
    pair = [combinations(t, math.exp(float(rng.uniform(-1, 1))), threshold, partitions)
            for t in (1.3, 2.7)]
    recovered = invert(pair, False)[0]
    general_inverse_errors.append(float(np.max(abs(recovered-threshold))))
checks['G8_general_ladder_inverse_16_with_changed_density'] = max(general_inverse_errors) < 1e-9

result = {
    'status': 'PASS' if all(checks.values()) else 'FAIL',
    'scope': 'Analytic theorem implementation controls and synthetic counterexamples; no new experimental
        observations',
    'seed': SEED,
    'protocol_sha256': hashlib.sha256((HERE/'PREREGISTERED_FORMAL_CONTROLS.md').read_bytes()).hexdigest(),
    'inverse_addendum_sha256': hashlib.sha256((HERE/'PREREGISTRATION_ADDENDUM_GAS_PEEL.md').read_bytes()).
        hexdigest(),
    'checks': {name: bool(value) for name, value in checks.items()},
    'max_errors': {
        'manufactured_root': max(manufactured_errors), 'charge_neutrality': max(neutrality_errors),
        'normalization': max(normalization_errors), 'density_elasticity_finite_difference': max(density_errors)
            ,
        'coefficient_elasticity_finite_difference': max(coefficient_errors),
        'temperature_elasticity_finite_difference': max(temperature_errors),
```

```
      'coefficient_density_scaling': max(scale_errors), 'physical_scaling_dictionaries': max(
          dictionary_errors),
      'hamiltonian_structure': max(hamiltonian_errors), 'general_inverse_thresholds': max(
          general_inverse_errors)},
   'spectral_witness': {'energies': [-3,-1,1,3], 'weights': [[str(x) for x in w] for w in weights],
      'moments_orders_0_to_3': [[str(x) for x in row] for row in moments],
      'resolvent_denominator_ascending': [str(x) for x in denominator],
      'resolvent_numerators_ascending': [[str(x) for x in row] for row in numerators],
      'hamiltonians_numerical_evaluations_of_exact_householder_formula': matrix_records,
      'time': math.pi/4, 'survival_probabilities': survival,
      'hidden_first_channel_moments': third_hidden_moments},
   'same_kernel_hidden_source_survival': source_survival,
   'gas_peel_inverse': {'helium_thresholds_eV': helium_i.tolist(), 'input_grades': ['E','T'],
                        'input_gamma': true_gamma, 'synthetic_partition_rule': 'U=(1,2+4 exp(-10 eV/kT),1)',
                        'records': helium_records},
}
(HERE/'formal_gas_results.json').write_text(json.dumps(result, indent=2)+'\n')
print(json.dumps(result, indent=2))
assert all(checks.values()), checks
```

Output:


```
{
  "status": "PASS",
  "scope": "Analytic theorem implementation controls and synthetic counterexamples; no new experimental
      observations",
  "seed": 20261103,
  "protocol_sha256": "37c25d1a25436b08af18adffc0317100c7188e955db839e30758ac95ce16e2ed",
  "inverse_addendum_sha256": "2f0552f669e4bb21aaadfed8207f59b299246bcfb4800df89ebed30e451de97c",
  "checks": {
    "G1_exact_rational_roundtrip_and_scale_100": true,
    "G3_exact_manufactured_neutrality_polynomials_8": true,
    "G3_neutrality_and_normalization_64": true,
    "G4_density_elasticity": true,
    "G4_density_stochastic_order": true,
    "G5_single_coefficient_response": true,
    "G7_coefficient_density_scale_covariance": true,
    "G6_fixed_partition_temperature_response": true,
    "G7_both_physical_scaling_dictionaries": true,
    "R3_exact_moments_and_resolvent_polynomials": true,
    "R3_constructive_equal_spectrum_equal_footprint": true,
    "R3_later_survival_0_versus_1_over_32": true,
    "R3_hidden_first_channel_moment_0_versus_3": true,
    "R4_same_kernel_different_hidden_source": true,
    "G8_incoming_section8_fixed_partition_inverse": true,
    "G9_excited_partition_naive_slope_false_control": true,
    "G9_known_partition_correction_restores_inverse": true,
    "G8_general_ladder_inverse_16_with_changed_density": true
  },
  "max_errors": {
    "manufactured_root": 1.0658141036401503e-14,
    "charge_neutrality": 4.39648317751562e-14,
    "normalization": 3.3306690738754696e-16,
    "density_elasticity_finite_difference": 1.4381607016389353e-11,
    "coefficient_elasticity_finite_difference": 1.823957651581054e-11,
    "temperature_elasticity_finite_difference": 2.01392680310164e-10,
    "coefficient_density_scaling": 3.885780586188048e-16,
    "physical_scaling_dictionaries": 1.6653345369377348e-15,
    "hamiltonian_structure": 8.881784197001252e-16,
    "general_inverse_thresholds": 6.661338147750939e-15
  },
  "spectral_witness": {
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      -1,
      1,
      3
    ],
```

```
  "weights": [
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    ],
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      "1/16",
      "5/16"
    ]
  ],
  "moments_orders_0_to_3": [
    [
      "1",
      "0",
      "5",
      "0"
    ],
    [
      "1",
      "0",
      "5",
      "3"
    ]
  ],
  "resolvent_denominator_ascending": [
    "9",
    "0",
    "-10",
    "0",
    "1"
  ],
  "resolvent_numerators_ascending": [
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      "-5",
      "0",
      "1"
    ],
    [
      "3",
      "-5",
      "0",
      "1"
    ]
  ],
  "hamiltonians_numerical_evaluations_of_exact_householder_formula": [
    [
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        -2.0,
        -1.0,
        0.0
      ],
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        0.0,
        0.0,
        -1.0
      ],
      [
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        0.0,
        0.0,
        -2.0
```

```
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        -2.0,
        0.0
      ]
    ],
    [
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        -0.32278095559281783,
        0.3962738302020131
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        -1.7665659842461605
      ],
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        0.7013958484311376,
        -1.1606705372795723
      ],
      [
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        -1.7665659842461605,
        -1.160670537279572,
        -0.6976556838323428
      ]
    ]
  ],
  "time": 0.7853981633974483,
  "survival_probabilities": [
    1.232595164407831e-32,
    0.03125000000000001
  ],
  "hidden_first_channel_moments": [
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    3.0
  ]
},
"same_kernel_hidden_source_survival": [
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  6.1629758220391534e-33
],
"gas_peel_inverse": {
  "helium_thresholds_eV": [
    24.587389011,
    54.4177655282
  ],
  "input_grades": [
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    "T"
  ],
  "input_gamma": 0.37757506698375054,
  "synthetic_partition_rule": "U=(1,2+4 exp(-10 eV/kT),1)",
  "records": [
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      "density_m3": 1e+18,
      "synthetic_excited_partition": false,
      "naive_gamma": 0.3775750669837528,
      "corrected_gamma": 0.3775750669837528,
      "naive_difference_eV": 29.83037651720008,
      "naive_sum_eV": 79.00515453919972,
```

```
  "naive_gamma_bias": 2.275957200481571e-15,
  "corrected_gamma_error": 2.275957200481571e-15
},
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  "corrected_gamma": 0.37757506698376214,
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  "naive_sum_eV": 79.00515453919972,
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  "corrected_gamma_error": 1.1601830607332886e-14
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  "corrected_gamma": 0.3775750669837528,
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  "naive_sum_eV": 79.00515453919972,
  "naive_gamma_bias": 2.275957200481571e-15,
  "corrected_gamma_error": 2.275957200481571e-15
},
{
  "density_m3": 1e+20,
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  "naive_gamma": 0.3661494335074752,
  "corrected_gamma": 0.37757506698376214,
  "naive_difference_eV": 28.927692578698508,
  "naive_sum_eV": 79.00515453919972,
  "naive_gamma_bias": -0.011425633476275343,
  "corrected_gamma_error": 1.1601830607332886e-14
},
{
  "density_m3": 1e+22,
  "synthetic_excited_partition": false,
  "naive_gamma": 0.3775750669837528,
  "corrected_gamma": 0.3775750669837528,
  "naive_difference_eV": 29.83037651720008,
  "naive_sum_eV": 79.00515453919972,
  "naive_gamma_bias": 2.275957200481571e-15,
  "corrected_gamma_error": 2.275957200481571e-15
},
{
  "density_m3": 1e+22,
  "synthetic_excited_partition": true,
  "naive_gamma": 0.3661494335074752,
  "corrected_gamma": 0.37757506698376214,
  "naive_difference_eV": 28.927692578698508,
  "naive_sum_eV": 79.00515453919972,
  "naive_gamma_bias": -0.011425633476275343,
  "corrected_gamma_error": 1.1601830607332886e-14
},
{
  "density_m3": 1e+24,
  "synthetic_excited_partition": false,
  "naive_gamma": 0.37757506698375376,
  "corrected_gamma": 0.37757506698375376,
  "naive_difference_eV": 29.830376517200154,
  "naive_sum_eV": 79.00515453919972,
  "naive_gamma_bias": 3.219646771412954e-15,
  "corrected_gamma_error": 3.219646771412954e-15
},
{
  "density_m3": 1e+24,
  "synthetic_excited_partition": true,
  "naive_gamma": 0.3661494335074752,
  "corrected_gamma": 0.37757506698376214,
```

```
      "naive_difference_eV": 28.927692578698508,
      "naive_sum_eV": 79.00515453919972,
      "naive_gamma_bias": -0.011425633476275343,
      "corrected_gamma_error": 1.1601830607332886e-14
    },
    {
      "density_m3": 1e+26,
      "synthetic_excited_partition": false,
      "naive_gamma": 0.37757506698375465,
      "corrected_gamma": 0.37757506698375465,
      "naive_difference_eV": 29.830376517200225,
      "naive_sum_eV": 79.00515453919972,
      "naive_gamma_bias": 4.107825191113079e-15,
      "corrected_gamma_error": 4.107825191113079e-15
    },
    {
      "density_m3": 1e+26,
      "synthetic_excited_partition": true,
      "naive_gamma": 0.3661494335074779,
      "corrected_gamma": 0.37757506698376486,
      "naive_difference_eV": 28.92769257869873,
      "naive_sum_eV": 79.00515453919972,
      "naive_gamma_bias": -0.011425633476272623,
      "corrected_gamma_error": 1.432187701766452e-14
    }
  ]
 }
}
```

adversarial_checks.py

```
"""Independent finite checks for the formal matter/gas review; see protocol."""
from pathlib import Path
import hashlib
import json
import math
import platform

import numpy as np
from scipy.linalg import expm

ROOT = Path(__file__).resolve().parent
RNG = np.random.default_rng(2026110307)
TOL = 2e-12


def norm(a):
    return float(np.linalg.norm(a, 2))


def contraction(n):
    a = RNG.normal(size=(n,n)) + 1j * RNG.normal(size=(n,n))
    u, _ = np.linalg.qr(a)
    return (u * RNG.uniform(0, 1, n)) @ u.conj().T


def cross(c, k):
    b = c[:k, k:]
    return b @ b.conj().T


def eig(a):
    return np.linalg.eigvalsh((a + a.conj().T) / 2)


rows = []
```

```
stable_cases = 0
for case in range(128):
    n, k = 12, 5
    c = contraction(n)
    other = contraction(n)
    alpha = float(RNG.uniform(1e-5, .03))
    d = (1-alpha)*c + alpha*other
    delta = norm(c-d)
    xc, xd = cross(c, k), cross(d, k)
    change = norm(xc-xd)
    assert change <= delta + TOL
    ecs, eds = eig(xc), eig(xd)
    for theta in (.002, .01, .03, .06, .09, .13, .2):
        margin = float(np.min(np.abs(ecs-theta)))
        if margin > delta + 1e-11:
            assert np.count_nonzero(ecs >= theta) == np.count_nonzero(eds >= theta)
            stable_cases += 1
    rows.append(dict(delta=delta, gram_change=change, ratio=change/delta))

near_rows = []
theta = .09
for case in range(64):
    n, k = 12, 5
    z = RNG.normal(size=(n,n)) + 1j*RNG.normal(size=(n,n))
    u, _ = np.linalg.qr(z)
    p = u[:, :6] @ u[:, :6].conj().T
    alpha = float(RNG.uniform(.0001, .15))
    c = (1-alpha)*p + alpha*contraction(n)
    a, x = c[:k,:k], cross(c,k)
    f = a-a@a
    defect = c-c@c
    tau = norm(defect)
    ordering = [float(eig(f-x).min()), float(eig(x+tau*np.eye(k)-f).min())]
    assert min(ordering) >= -TOL
    ex, ef = eig(x), eig(f)
    margin = min(float(np.min(np.abs(ex-theta))),
                 float(np.min(np.abs(ef-theta))),
                 float(np.min(np.abs(ef-(theta+tau)))))
    if margin > 1e-11:
        lhs = int(np.count_nonzero(ef >= theta+tau))
        middle = int(np.count_nonzero(ex >= theta))
        rhs = int(np.count_nonzero(ef >= theta))
        assert lhs <= middle <= rhs
    near_rows.append(dict(tau=tau, ordering_min=min(ordering), margin=margin))

sharp = []
for eps in (1e-2, 1e-4, 1e-6):
    c = np.array([[.5,.5],[.5,.5]])
    d = np.array([[.5,.5-eps],[.5-eps,.5]])
    ratio = norm(cross(c,1)-cross(d,1))/norm(c-d)
    assert abs(ratio-(1-eps)) < 1e-9
    sharp.append(dict(epsilon=eps, ratio=ratio, exact_ratio=1-eps))

boundary = []
for eps in (1e-3,1e-6,1e-9):
    low = np.array([[.5,.3-eps],[.3-eps,.5]])
    high = np.array([[.5,.3+eps],[.3+eps,.5]])
    counts = [int(np.count_nonzero(eig(cross(c,1)) >= theta)) for c in (low,high)]
    assert counts == [0,1]
    boundary.append(dict(epsilon=eps, input_distance=norm(low-high), counts=counts))

# Fock basis integer encoding, mode0 most-significant bit: |n0 n1 n2 n3>.
modes = 4
annihilators = []
for j in range(modes):
    aj = np.zeros((2**modes,2**modes),complex)
    mask = 1 << (modes-1-j)
    for s in range(2**modes):
```

```
        if s & mask:
            parity = sum((s >> (modes-1-q)) & 1 for q in range(j))
            aj[s ^ mask,s] = (-1)**parity
    annihilators.append(aj)
psi = np.zeros(16,complex)
psi[12] = psi[3] = 1/math.sqrt(2)
expect = lambda op: complex(np.vdot(psi,op@psi))
c = np.array([[expect(annihilators[j].conj().T @ annihilators[i])
               for j in range(modes)] for i in range(modes)])
ns = [a.conj().T@a for a in annihilators]
na, nb = ns[0]+ns[1], ns[2]+ns[3]
cov = (expect(na@nb)-expect(na)*expect(nb)).real
schmidt = np.linalg.svd(psi.reshape(4,4),compute_uv=False)
probs = schmidt[schmidt>1e-12]**2
entropy = -float(np.sum(probs*np.log(probs)))
assert np.max(np.abs(c-.5*np.eye(4))) < TOL
assert abs(cov+1)<TOL and abs(entropy-math.log(2))<TOL

spectrum = np.array([-3.,-1.,1.,3.])
weights = [np.full(4,.25),np.array([3,7,1,5])/16]
hs = []
spectral_rows = []
for ix,w in enumerate(weights):
    e0 = np.array([1.,0.,0.,0.])
    v = np.sqrt(w)
    u = (e0-v)/np.linalg.norm(e0-v)
    r = np.eye(4)-2*np.outer(u,u)
    h = r @ np.diag(spectrum) @ r.T
    hs.append(h)
    coupling2 = float(h[0,1:]@h[1:,0])
    amplitude = complex(expm(-1j*math.pi*h/4)[0,0])
    assert abs(h[0,0])<TOL and abs(coupling2-5)<TOL
    assert np.max(np.abs(np.linalg.eigvalsh(h)-spectrum))<TOL
    for z in (.2+.7j, 2+1j, -4+.2j):
        g = np.linalg.inv(z*np.eye(4)-h)[0,0]
        formula = (z**3-5*z+3*ix)/(z**4-10*z*z+9)
        assert abs(g-formula)<TOL
    probability = abs(amplitude)**2
    assert abs(probability-(0 if ix==0 else 1/32))<TOL
    spectral_rows.append(dict(weights=w.tolist(),mean=float(w@spectrum),
                          second_moment=float(w@(spectrum**2)),
                          third_moment=float(w@(spectrum**3)),
                          coupling_square=coupling2,
                          survival_probability=probability))

report = {
    "status":"PASS",
    "protocol_sha256":hashlib.sha256((ROOT/'ADVERSARIAL_CHECK_PROTOCOL.md').read_bytes()).hexdigest(),
    "python":platform.python_version(),
    "scope":"Finite implementation checks and explicit counterexamples; not substitutes for analytic proofs.",
    "perturbation_pairs":len(rows),
    "max_random_lipschitz_ratio":max(r['ratio'] for r in rows),
    "certified_margin_cases":stable_cases,
    "near_projector_cases":len(near_rows),
    "minimum_ordering_eigenvalue":min(r['ordering_min'] for r in near_rows),
    "smallest_near_threshold_margin":min(r['margin'] for r in near_rows),
    "sharp_lipschitz_family":sharp,
    "discontinuous_threshold_family":boundary,
    "nonGaussian_counterexample":{
        "covariance_error":float(np.max(np.abs(c-.5*np.eye(4)))),
        "cross_gram_trace":float(np.trace(cross(c,2)).real),
        "number_covariance":cov,
        "bipartite_entropy":entropy,
        "fixed_total_number":float(expect(na+nb).real)},
    "full_spectrum_response_witness":spectral_rows,
}
(ROOT/'adversarial_results.json').write_text(json.dumps(report,indent=2)+'\n')
print(json.dumps(report,indent=2))
```

Output:


```
{
  "status": "PASS",
  "protocol_sha256": "8637b0fc68f65f3a0b6efde5f1d63bfd6df02d8fa90a4b7f9cde151bb9aa5590",
  "python": "3.13.18",
  "scope": "Finite implementation checks and explicit counterexamples; not substitutes for analytic proofs.",
  "perturbation_pairs": 128,
  "max_random_lipschitz_ratio": 0.6042534415334075,
  "certified_margin_cases": 455,
  "near_projector_cases": 64,
  "minimum_ordering_eigenvalue": 0.00025492682239148677,
  "smallest_near_threshold_margin": 4.9588167872274824e-05,
  "sharp_lipschitz_family": [
    {
      "epsilon": 0.01,
      "ratio": 0.9900000000000011,
      "exact_ratio": 0.99
    },
    {
      "epsilon": 0.0001,
      "ratio": 0.9999000000000526,
      "exact_ratio": 0.9999
    },
    {
      "epsilon": 1e-06,
      "ratio": 0.9999989999943661,
      "exact_ratio": 0.999999
    }
  ],
  "discontinuous_threshold_family": [
    {
      "epsilon": 0.001,
      "input_distance": 0.0020000000000000018,
      "counts": [
        0,
        1
      ]
    },
    {
      "epsilon": 1e-06,
      "input_distance": 1.999999999946489e-06,
      "counts": [
        0,
        1
      ]
    },
    {
      "epsilon": 1e-09,
      "input_distance": 2.0000000544584395e-09,
      "counts": [
        0,
        1
      ]
    }
  ],
  "nonGaussian_counterexample": {
    "covariance_error": 1.1102230246251565e-16,
    "cross_gram_trace": 0.0,
    "number_covariance": -0.9999999999999996,
    "bipartite_entropy": 0.6931471805599454,
    "fixed_total_number": 1.9999999999999996
  },
  "full_spectrum_response_witness": [
    {
      "weights": [
        0.25,
        0.25,
```

```
      0.25,
      0.25
    ],
    "mean": 0.0,
    "second_moment": 5.0,
    "third_moment": 0.0,
    "coupling_square": 5.0,
    "survival_probability": 1.232595164407831e-32
  },
  {
    "weights": [
      0.1875,
      0.4375,
      0.0625,
      0.3125
    ],
    "mean": 0.0,
    "second_moment": 5.0,
    "third_moment": 3.0,
    "coupling_square": 5.000000000000001,
    "survival_probability": 0.03125000000000002
  }
 ]
}
```

gaussian_information_checks.py

```
"""Independent finite checks for B13, with explicit hypothesis/limit controls."""
from pathlib import Path
import json,hashlib
import numpy as np
from scipy.special import xlogy

ROOT=Path(__file__).resolve().parent
rng=np.random.default_rng(2026110313)
def h(x):
    x=np.clip(x,0,1)
    return -xlogy(x,x)-xlogy(1-x,1-x)
def ent(c):return float(np.sum(h(np.linalg.eigvalsh(c))))
rows=[]
for delta in (.05,.15,.25,.45):
    for sample in range(16):
        z=rng.normal(size=(12,12))+1j*rng.normal(size=(12,12))
        u,_=np.linalg.qr(z)
        c=(u*rng.uniform(delta,1-delta,12))@u.conj().T
        a,d,b=c[:5,:5],c[5:,5:],c[:5,5:]
        x=b@b.conj().T;tr=float(np.trace(x).real)
        info=ent(a)+ent(d)-ent(c)
        count=int(np.count_nonzero(np.linalg.eigvalsh(x)>=.09))
        lower=info-4*tr;upper=tr/(delta*(1-delta))-info
        assert lower>=-2e-12 and upper>=-2e-12
        assert count<=info/.36+2e-12
        rows.append(dict(delta=delta,information=info,cross_trace=tr,
                         lower_margin=lower,upper_margin=upper,census=count))
sharp=[]
for r in (.01,.001,.0001):
    info=float(2*np.log(2)-2*h(.5+r));ratio=info/r**2
    assert abs(ratio-4)<.002
    sharp.append(dict(r=r,ratio=ratio))
pure=[]
for p in (.01,.0001,.000001):
    pure.append(dict(p=p,cross_trace=p*(1-p),information=float(2*h(p)),
                     ratio=float(2*h(p)/(p*(1-p)))))
assert all(pure[i+1]['ratio']>pure[i]['ratio'] for i in range(2))
out={'status':'PASS','class':'Gauge-invariant Gaussian covariance implementation checks',
     'natural_log_units':True,'cases':len(rows),
```

```
    'minimum_lower_bound_margin':min(x['lower_margin'] for x in rows),
    'minimum_upper_bound_margin':min(x['upper_margin'] for x in rows),
    'sharp_lower_family':sharp,'pure_upper_counterfamily':pure,
    'protocol_sha256':hashlib.sha256((ROOT/'GAUSSIAN_INFORMATION_PROTOCOL.md').read_bytes()).hexdigest()}
(ROOT/'gaussian_information_results.json').write_text(json.dumps(out,indent=2)+'\n')
print(json.dumps(out,indent=2))
```

Output:


```
{
  "status": "PASS",
  "class": "Gauge-invariant Gaussian covariance implementation checks",
  "natural_log_units": true,
  "cases": 64,
  "minimum_lower_bound_margin": 9.114944760997903e-06,
  "minimum_upper_bound_margin": 5.852263636260853e-05,
  "sharp_lower_family": [
    {
      "r": 0.01,
      "ratio": 4.000266709340838
    },
    {
      "r": 0.001,
      "ratio": 4.000002666648683
    },
    {
      "r": 0.0001,
      "ratio": 4.000000020099037
    }
  ],
  "pure_upper_counterfamily": [
    {
      "p": 0.01,
      "cross_trace": 0.0099,
      "information": 0.11200306870969469,
      "ratio": 11.313441283807544
    },
    {
      "p": 0.0001,
      "cross_trace": 9.999000000000001e-05,
      "information": 0.0020420580740618645,
      "ratio": 20.422623002918936
    },
    {
      "p": 1e-06,
      "cross_trace": 9.99999e-07,
      "information": 2.9631020115985722e-05,
      "ratio": 29.631049747035473
    }
  ],
  "protocol_sha256": "cffe0462b988385eb4c6761d707866cd2dcba1390650f81e26d008bae118207a"
}
```


incoming_exit_filter.py

```
"""Independent endpoint-filter audit of preserved incoming matter-disk scripts."""
from pathlib import Path
import json, math, hashlib
import numpy as np
from scipy.linalg import eigvals

ROOT=Path(__file__).resolve().parent
N=2000;L=40;t1=.5
k=2*np.pi*(np.arange(N)+.5)/N
lags=np.arange(-(L-1),L)
```

```
ph=np.exp(1j*np.outer(lags,k))/N
disp=np.arange(L)[:,None]-np.arange(L)[None,:]+L-1

def band(m,g):
    dz=m+1j*g
    q=t1+np.exp(1j*k)
    E=np.sqrt(abs(q)**2+dz*dz+0j)
    P=np.zeros((N,2,2),complex)
    P[:,0,0]=(1-dz/E)/2;P[:,1,1]=(1+dz/E)/2
    P[:,0,1]=-q.conj()/(2*E);P[:,1,0]=-q/(2*E)
    return P,E

def spectrum(m,g):
    P,E=band(m,g)
    coeff=np.einsum('lk,kab->lab',ph,P,optimize=True)
    c=coeff[disp].transpose(0,2,1,3).reshape(2*L,2*L)
    return eigvals(c,check_finite=False)

def rotation(m,g,cut):
    nu=spectrum(m,g)
    nu=nu[(np.abs(nu)>cut)&(np.abs(1-nu)>cut)]
    return float(np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi) if len(nu) else 0.

def exit_at(m,cut):
    lo=0.;g=.05
    while g<.8:
        if rotation(m,g,cut)>=.5:
            hi=g;break
        lo=g;g*=1.4
    else:
        return None
    for _ in range(26):
        mid=(lo+hi)/2
        if rotation(m,mid,cut)<.5:lo=mid
        else:hi=mid
    g=(lo+hi)/2
    _,E=band(m,g)
    dnorm=(t1+np.cos(k))**2+np.sin(k)**2+abs(m+1j*g)**2
    K=float(np.mean((1+dnorm/abs(E)**2)/2))
    return dict(m=m,cut=cut,exit_gamma=g,mean_Petermann=K)

# Independent eigensolver cross-check for projector selection.
errors=[]
for m,g in ((.4,.1),(.4,.2),(0.,.4)):
    P,E=band(m,g)
    for ix in (0,137,999,1567):
        q=t1+np.exp(1j*k[ix]);h=np.array([[m+1j*g,q.conjugate()],[q,-m-1j*g]])
        e,R=np.linalg.eig(h);order=np.argsort(e.real);R=R[:,order]
        inverse=np.linalg.inv(R)
        errors.append(float(np.max(abs(P[ix]-np.outer(R[:,0],inverse[0,:])))))
assert max(errors)<1e-12
rows=[]
for m in (.4,0.):
    for cut in (1e-6,1e-8,1e-9,1e-10,1e-12):
        row=exit_at(m,cut);rows.append(row);print(row,flush=True)
oblique=np.array([[2.,1.],[-2.,-1.]])
assert np.array_equal(oblique@oblique,oblique)
report={'classification':'Endpoint-filter sensitivity, not a new physical measurement',
       'N':N,'region_cells':L,'t1':t1,'max_projector_crosscheck_error':max(errors),
       'rows':rows,'oblique_projector':{'C':oblique.tolist(),'X_A':-2.,'Gamma_b':3.},
       'protocol_sha256':hashlib.sha256((ROOT/'INCOMING_EXIT_FILTER_PROTOCOL.md').read_bytes()).hexdigest()}
(ROOT/'incoming_exit_filter_results.json').write_text(json.dumps(report,indent=2)+'\n')
```

Output:

```
{'m': 0.4, 'cut': 1e-06, 'exit_gamma': 0.2716369288976192, 'mean_Petermann': 1.0579728462043967}
{'m': 0.4, 'cut': 1e-08, 'exit_gamma': 0.2716369288976192, 'mean_Petermann': 1.0579728462043967}
{'m': 0.4, 'cut': 1e-09, 'exit_gamma': 0.16916993247628215, 'mean_Petermann': 1.0223524908929857}
```

```
{'m': 0.4, 'cut': 1e-10, 'exit_gamma': 0.16916993247628215, 'mean_Petermann': 1.0223524908929857}
{'m': 0.4, 'cut': 1e-12, 'exit_gamma': 0.1227913656055927, 'mean_Petermann': 1.0117452195516827}
{'m': 0.0, 'cut': 1e-06, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-08, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-09, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-10, 'exit_gamma': 0.4743414673329256, 'mean_Petermann': 1.9999957440096103}
{'m': 0.0, 'cut': 1e-12, 'exit_gamma': 0.47422795577696786, 'mean_Petermann': 1.9973451918175165}
```

incoming_exit_residuals.py

```
"""Post-sensitivity diagnostics at recorded exits; residuals are not certified bounds."""
from pathlib import Path
import json
import numpy as np
from scipy.linalg import eig

ROOT=Path(__file__).resolve().parent
source=json.loads((ROOT/'incoming_exit_filter_results.json').read_text())
N,L,t1=source['N'],source['region_cells'],source['t1']
k=2*np.pi*(np.arange(N)+.5)/N
lags=np.arange(-(L-1),L)
ph=np.exp(1j*np.outer(lags,k))/N
disp=np.arange(L)[:,None]-np.arange(L)[None,:]+L-1
out=[]
for row in source['rows']:
    m,g=row['m'],row['exit_gamma']
    q=t1+np.exp(1j*k);dz=m+1j*g;E=np.sqrt(abs(q)**2+dz*dz+0j)
    P=np.zeros((N,2,2),complex)
    P[:,0,0]=(1-dz/E)/2;P[:,1,1]=(1+dz/E)/2
    P[:,0,1]=-q.conj()/(2*E);P[:,1,0]=-q/(2*E)
    coef=np.einsum('lk,kab->lab',ph,P,optimize=True)
    c=coef[disp].transpose(0,2,1,3).reshape(2*L,2*L)
    vals,V=eig(c,check_finite=False)
    cnorm=float(np.linalg.norm(c,2))
    residual=float(np.max(np.linalg.norm(c@V-V*vals,axis=0))/max(1.,cnorm))
    keep=(abs(vals)>row['cut'])&(abs(1-vals)>row['cut'])
    out.append(dict(m=m,cut=row['cut'],exit_gamma=g,
                    normalized_max_eigenpair_residual=residual,
                    eigenvector_condition_number=float(np.linalg.cond(V)),
                    retained_eigenvalues=int(keep.sum()),compression_norm=cnorm))
report={'classification':'Diagnostic added after observing endpoint-filter sensitivity; not a preregistered
     new decision test',
       'caveat':'Small residuals in a nonnormal eigenproblem do not certify small eigenvalue errors or a
            continuum-limit threshold.',
       'maximum_normalized_eigenpair_residual':max(x['normalized_max_eigenpair_residual'] for x in out),
       'rows':out}
(ROOT/'incoming_exit_residuals.json').write_text(json.dumps(report,indent=2)+'\n')
print(json.dumps(report,indent=2))
```

Output:

```
{
  "classification": "Diagnostic added after observing endpoint-filter sensitivity; not a preregistered new
       decision test",
  "caveat": "Small residuals in a nonnormal eigenproblem do not certify small eigenvalue errors or a continuum
       -limit threshold.",
  "maximum_normalized_eigenpair_residual": 4.406378670976857e-15,
  "rows": [
    {
      "m": 0.4,
      "cut": 1e-06,
      "exit_gamma": 0.2716369288976192,
      "normalized_max_eigenpair_residual": 3.182735498977942e-15,
      "eigenvector_condition_number": 135.08185523602376,
      "retained_eigenvalues": 6,
```

```
  "compression_norm": 1.0507874021697408
},
{
  "m": 0.4,
  "cut": 1e-08,
  "exit_gamma": 0.2716369288976192,
  "normalized_max_eigenpair_residual": 3.182735498977942e-15,
  "eigenvector_condition_number": 135.08185523602376,
  "retained_eigenvalues": 8,
  "compression_norm": 1.0507874021697408
},
{
  "m": 0.4,
  "cut": 1e-09,
  "exit_gamma": 0.16916993247628215,
  "normalized_max_eigenpair_residual": 4.322929209051072e-15,
  "eigenvector_condition_number": 574.9088072508466,
  "retained_eigenvalues": 10,
  "compression_norm": 1.0207050696097535
},
{
  "m": 0.4,
  "cut": 1e-10,
  "exit_gamma": 0.16916993247628215,
  "normalized_max_eigenpair_residual": 4.322929209051072e-15,
  "eigenvector_condition_number": 574.9088072508466,
  "retained_eigenvalues": 10,
  "compression_norm": 1.0207050696097535
},
{
  "m": 0.4,
  "cut": 1e-12,
  "exit_gamma": 0.1227913656055927,
  "normalized_max_eigenpair_residual": 4.406378670976857e-15,
  "eigenvector_condition_number": 351.93147861449006,
  "retained_eigenvalues": 14,
  "compression_norm": 1.0110432285326036
},
{
  "m": 0.0,
  "cut": 1e-06,
  "exit_gamma": 0.4743416490950296,
  "normalized_max_eigenpair_residual": 1.4423660472602578e-15,
  "eigenvector_condition_number": 145.42622327140032,
  "retained_eigenvalues": 10,
  "compression_norm": 3.041024145848934
},
{
  "m": 0.0,
  "cut": 1e-08,
  "exit_gamma": 0.4743416490950296,
  "normalized_max_eigenpair_residual": 1.4423660472602578e-15,
  "eigenvector_condition_number": 145.42622327140032,
  "retained_eigenvalues": 14,
  "compression_norm": 3.041024145848934
},
{
  "m": 0.0,
  "cut": 1e-09,
  "exit_gamma": 0.4743416490950296,
  "normalized_max_eigenpair_residual": 1.4423660472602578e-15,
  "eigenvector_condition_number": 145.42622327140032,
  "retained_eigenvalues": 14,
  "compression_norm": 3.041024145848934
},
{
  "m": 0.0,
  "cut": 1e-10,
```

```
    "exit_gamma": 0.4743414673329256,
    "normalized_max_eigenpair_residual": 1.2382937166700015e-15,
    "eigenvector_condition_number": 162.37158134723273,
    "retained_eigenvalues": 18,
    "compression_norm": 3.04101441355121
  },
  {
    "m": 0.0,
    "cut": 1e-12,
    "exit_gamma": 0.47422795577696786,
    "normalized_max_eigenpair_residual": 1.8840893883918226e-15,
    "eigenvector_condition_number": 330.2849945171612,
    "retained_eigenvalues": 22,
    "compression_norm": 3.0349556693697504
  }
 ]
}
```


fig02_boundary_and_gas.py

```
# SCRIPT: MATTER-DISK-FIG02 warm census on the disk (raw vs boundary) and the helium gas on the disk
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
import importlib.util, sys, io, contextlib
def load(name):
    spec=importlib.util.spec_from_file_location(name,name+'.py'); m=importlib.util.module_from_spec(spec)
    with contextlib.redirect_stdout(io.StringIO()): spec.loader.exec_module(m)
    return m
b=load('s07_boundary_census_disk'); g=load('s08_gas_carries_the_peel')
fig,ax=plt.subplots(1,2,figsize=(12,5.2))
H=b.ring(1,1); cols={0:'#1b6f8a',0.5:'#c08a2e',2:'#7b5ea7'}
for i,T in enumerate([0,0.5,2]):
    C=b.thermal(H,T); CA,CAB,CBA=b.blocks(C,64); nu=np.linalg.eigvalsh(CA); lam=np.linalg.eigvalsh((CAB@CBA+(
        CAB@CBA).T)/2)
    G=np.abs(1-2*nu); Gb=np.sqrt(np.clip(1-4*lam,0,1))
    ax[0].scatter(G,np.full_like(G,i+0.15),s=14,color=cols[T],marker='o',label=f'T={T}: raw |Γ| = |1-2ν|')
    ax[0].scatter(Gb,np.full_like(Gb,i-0.15),s=14,color=cols[T],marker='x',label=f'T={T}: boundary |Γ_b|')
ax[0].axvline(0.8,color='0.6',ls=':',lw=1)
ax[0].axvspan(0,0.8,color='0.93',zorder=0)
ax[0].set_xlim(0,1.02); ax[0].set_yticks([0,1,2]); ax[0].set_yticklabels(['T = 0','T = 0.5','T = 2'])
ax[0].set_xlabel('distance from the disk centre |Γ| (rim = 1)'); ax[0].set_title('Heat pulls raw levels inward
    ; the boundary census sends them back to the rim\n(uniform ring, 128 sites, half-ring region, shaded =
    count window ε= 0.1)',fontsize=10)
ax[0].legend(fontsize=7,ncol=3,loc='upper center',bbox_to_anchor=(0.5,-0.13))
Ts=np.linspace(8000,80000,300)
for n,ls in [(1e18,':'),(1e22,'-'),(1e26,'--')]:
    G0s=[];G1s=[]
    for T in Ts:
        p,_,_=g.saha(T,n); G0s.append(np.tanh(0.5*np.log(max(p[0],1e-300)/max(p[1],1e-300)))); G1s.append(np.
            tanh(0.5*np.log(max(p[1],1e-300)/max(p[2],1e-300))))
    ax[1].plot(Ts/1000,G0s,color='#1b6f8a',ls=ls,label=f'He ↔He$^{+}$ pair, n = {n:.0e} m$^{-3}$')
    ax[1].plot(Ts/1000,G1s,color='#c08a2e',ls=ls,label=f'He$^{+}$ ↔He$^{2+}$ pair, n = {n:.0e} m$^{-3}$')
ax[1].axhline(g.G0,color='#7b5ea7',lw=2.2,label=f'$\Gamma_0$ read back from the gas = {g.G0:.6f} at every density')
ax[1].axhline(0,color='0.6',lw=0.8)
ax[1].set_xlabel('temperature (thousand K)'); ax[1].set_ylabel('population disk point Γ_pop = tanh($\frac{1}{2}$ ln n_q/
    n_(q+1))')
ax[1].set_title('Helium (VOTKP ladder): the gas state sweeps the disk,\nthe peel coordinate $\Gamma_0$stays put',
    fontsize=10)
ax[1].legend(fontsize=7,loc='lower left'); ax[1].set_ylim(-1.05,1.05)
plt.tight_layout(); plt.savefig('../figures/fig02_boundary_and_gas.png',dpi=150)
```

Output:

```
(this script prints nothing; it writes ../figures/fig02_boundary_and_gas.png, unchanged on rerun)

real 0m2.170s
```

```
user 0m1.608s
sys 0m0.159s
```

fig03v4_one_chart.py

```
# SCRIPT: MATTER-DISK-FIG03V4 one Smith disk, every scale, every point at its true signed value.
# Real ratios sit ON the real axis (Im Gamma = 0); labels are moved off the axis with leader lines only.
# Nuclear ratios use VOTKP 15.4's order z = S_next / S_first: bound nuclei give z < 1, so Gamma < 0.
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
ns = {}; exec(open('s02_ladder.py').read().split('op=lambda')[0], ns)
ssh = {}; exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0], ssh)
G = lambda z: (z - 1) / (z + 1)
fig, ax = plt.subplots(figsize=(8.8, 8.8))
t = np.linspace(0, 2 * np.pi, 400)
ax.plot(np.cos(t), np.sin(t), color='0.2', lw=1.5)
for r in (0.25, 0.5, 0.75): ax.plot(r * np.cos(t), r * np.sin(t), color='0.9', lw=0.7)
ax.axhline(0, color='0.75', lw=0.8); ax.axvline(0, color='0.88', lw=0.7)
# matter: Ising census ladder (c = 1/2), both signs of the level energy
xi, em = ns['ladder'](1.02, 8192); lev = em[::2][:5]
Gl = np.concatenate([np.tanh(lev / 2), -np.tanh(lev / 2)])
ax.scatter(Gl, 0 * Gl, s=52, color='#1b6f8a', zorder=6, label='matter: Ising census levels (c = 1/2), $\\Gamma
    =\\tanh(\\varepsilon/2)$, both signs')
# atom: sodium thresholds, NIST ASD (eV), composite peel from the first removal
IE = np.array([5.139076, 47.28636, 71.6200, 98.936, 138.39, 172.18, 208.50, 264.18, 299.856, 1465.091,
    1648.659])
walk = np.tanh(0.5 * np.log(IE / IE[0]))
ax.scatter(walk, 0 * walk, s=30, marker='o', facecolor='white', edgecolor='#c08a2e', lw=1.6, zorder=7, label='
    atom: sodium peel walk $\\tanh(\\frac{1}{2}\\ln I_k/I_1)$ (NIST)')
ax.scatter([0.377575], [0], marker='D', s=80, color='#c08a2e', edgecolor='k', zorder=8, label='atom: helium $
    \\Gamma_0=0.377575$')
# nucleus, z = S_next / S_first (keV, NUBASE2020 via VOTKP 15.4)
nuc = {'$^7$Li': (7251.1, 5664.4), '$^4$He': (20577.6, 5493.5), '$^5$He': (-734.8, 20577.6), '$^5$Li':
    (-1966.1, 19813.9), '$^8$Be': (-91.8, 7366.6)}
lab_y = {'$^7$Li': -0.16, '$^4$He': -0.16, '$^5$He': -0.42, '$^5$Li': -0.16, '$^8$Be': -0.30}
lab_x = {'$^7$Li': -0.06, '$^4$He': -0.06, '$^5$He': -0.02, '$^5$Li': -0.06, '$^8$Be': -0.20}
for k, (s1, s2) in nuc.items():
    g = G(s2 / s1)
    inside = abs(g) < 1
    ax.scatter([g], [0], marker='s', s=70, color='#7b5ea7' if inside else 'white', edgecolor='#7b5ea7', lw=2,
        zorder=9)
    ax.annotate(f'{k} {g:+.3f}', (g, 0), (g + lab_x[k], lab_y[k]), fontsize=9, color='#5a3f85',
            arrowprops=dict(arrowstyle='-', color='#7b5ea7', lw=0.7))
ax.scatter([], [], marker='s', color='#7b5ea7', label='nucleus: bound inside ($\\Gamma<0$), unbound outside ($
    \\Gamma>1$)')
# opened matter, u = 0.5, before and past the exit
for v, c, lab in [(0.30, '#2a9d55', 'opened matter, gapped: complex $\\Gamma$ inside'), (0.49, '#d1495b', '
    opened matter past the exit: $\\Gamma$ leaves the disk')]:
    nu = np.linalg.eigvals(ssh['corr'](v, 1.0, 0.5, 2000, 40)); nu = nu[(abs(nu) > 1e-9) & (abs(1 - nu) > 1e
        -9)]
    g = 1 - 2 * nu; g = g[abs(g) < 1.6]
    ax.scatter(g.real, g.imag, s=14, color=c, alpha=0.85, zorder=5, label=lab)
ax.text(0, 1.06, 'rim $|\\Gamma|=1$: rapidity turned a quarter ($|{\\rm Im}\\,W|=\\pi/2$)', ha='center',
    fontsize=9.5, color='0.3')
ax.text(0.02, 0.04, 'centre: $z=1$', fontsize=8.5, color='0.35')
ax.set_xlim(-1.3, 1.34); ax.set_ylim(-1.34, 1.3); ax.set_aspect('equal')
ax.set_xlabel('Re $\\Gamma$'); ax.set_ylabel('Im $\\Gamma$')
ax.set_title('Matter, then on down: one chart, signed values', fontsize=13)
ax.legend(fontsize=7.6, loc='lower left', framealpha=0.95)
plt.tight_layout(); plt.savefig('../figures/fig03v4_one_chart.png', dpi=170)
print('sodium walk end', walk[-1]); print({k: round(G(b / a), 6) for k, (a, b) in nuc.items()})
```

Output:

```
sodium walk end 0.9937851227733561
{'$^7$Li': -0.122852, '$^4$He': -0.578576, '$^5$He': 1.074062, '$^5$Li': 1.220318, '$^8$Be': 1.025238}
```

fig04_reflection_chart.py

```
# SCRIPT: MATTER-DISK-FIG04 the disk as a reflection chart.
# Left: one Smith disk carrying the boundary readings that Beyond the Event Horizon (ed. 10) places on it:
# centre = matched load (S_b = 0), rim = total reflection, fermion record Gamma = tanh(omega/2T),
# gap-step reflection Gamma = tanh(ln(D2/D1)/2). Different normalised impedances on one chart, not one object.
# Right: the single curve Gamma = tanh(eps/2) with four readings of eps marked at real values.
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
fig, (a, b) = plt.subplots(1, 2, figsize=(13.2, 6.2), gridspec_kw={'width_ratios': [1, 1.15]})
t = np.linspace(0, 2 * np.pi, 400)
a.fill(np.cos(t), np.sin(t), color='#f3efe6', zorder=0)
a.plot(np.cos(t), np.sin(t), color='0.15', lw=2.2)
for r in (0.33, 0.66): a.plot(r * np.cos(t), r * np.sin(t), color='0.86', lw=0.7)
a.axhline(0, color='0.7', lw=0.8)
a.scatter([0], [0], s=150, marker='o', facecolor='white', edgecolor='k', lw=2, zorder=8)
a.annotate('centre: matched load\n$Z_b=I$, $S_b=0$ returns nothing\n(horizon surface resistance = $Z_0$)', (0,
     0), (-0.05, -0.62), fontsize=9,
         arrowprops=dict(arrowstyle='-', color='0.3', lw=0.8))
x = np.array([0.25, 0.5, 1, 2, 4, 8]); G = np.tanh(x / 2)
a.scatter(G, 0 * G, s=60, color='#c0392b', zorder=7, label='fermion record of a Hawking pair: $\\Gamma=\\tanh
    (\\hbar\\omega/2k_BT_H)$')
for xi, gi, dx in zip(x[[0, 2, 3, 5]], G[[0, 2, 3, 5]], [-0.02, -0.01, -0.02, 0.03]):
   a.annotate(f'{xi:g}', (gi, 0), (gi + dx, 0.07), fontsize=8, color='#c0392b')
a.text(0.42, 0.16, '$\\hbar\\omega/k_BT_H$', fontsize=9, color='#c0392b')
r = np.array([1 / 8, 1 / 4, 1 / 2, 3 / 4]); Gs = np.tanh(0.5 * np.log(r))
a.scatter(Gs, 0 * Gs, s=60, marker='^', color='#1f5f6b', zorder=7, label='step in spectral gap: $\\Gamma=\\
    tanh(\\frac{1}{2}\\ln\\Delta_2/\\Delta_1)$')
for ri, gi in zip(r, Gs):
   a.annotate(f'{ri:g}', (gi, 0), (gi - 0.03, -0.11), fontsize=8, color='#1f5f6b')
a.text(-0.75, -0.24, '$\\Delta_2/\\Delta_1$', fontsize=9, color='#1f5f6b')
th = np.linspace(0.62 * np.pi, 0.95 * np.pi, 60)
a.plot(np.cos(th), np.sin(th), color='#7b5ea7', lw=5, alpha=0.75, solid_capstyle='round')
a.annotate('rim $|\\Gamma|=1$: total reflection\n(past a geodesic turning point;\npolar stratum = perfect
    mirror)', (np.cos(0.8 * np.pi), np.sin(0.8 * np.pi)), (-1.28, 1.05), fontsize=9,
         color='#5a3f85', arrowprops=dict(arrowstyle='-', color='#7b5ea7', lw=0.8))
a.text(0.0, -1.16, 'inside: passive, $\\|S_b\\|\\leq 1$', ha='center', fontsize=9, color='0.3')
a.set_xlim(-1.35, 1.35); a.set_ylim(-1.75, 1.45); a.set_aspect('equal'); a.axis('off')
a.legend(fontsize=8, loc='lower center', bbox_to_anchor=(0.5, 0.0), framealpha=0.95)
a.set_title('One disk, three boundary readings', fontsize=12)
e = np.linspace(-6, 6, 400)
b.plot(e, np.tanh(e / 2), color='0.15', lw=2)
b.axhline(1, color='0.7', ls='--', lw=0.8); b.axhline(-1, color='0.7', ls='--', lw=0.8); b.axhline(0, color
    ='0.85', lw=0.8); b.axvline(0, color='0.85', lw=0.8)
pts = [(np.log((1 - 0.2) / 0.2), 'occupation odds: $\\varepsilon=\\ln((1-\\nu)/\\nu)$, $\\nu=0.2$', '#1b6f8a')
    ,
     (np.log(54.4177655282 / 24.587389011), 'helium thresholds: $\\varepsilon=\\ln I_1/I_0$', '#c08a2e'),
     (np.log(2.0), 'gap step: $\\varepsilon=\\ln\\Delta_2/\\Delta_1$, ratio 2', '#1f5f6b'),
     (1.0, 'Hawking fermion record: $\\varepsilon=\\hbar\\omega/k_BT_H=1$', '#c0392b')]
pts = sorted(pts, key=lambda q: -q[0] if False else -np.tanh(q[0] / 2))
for i, (ei, lab, c) in enumerate(pts):
   gi = np.tanh(ei / 2)
   b.scatter([ei], [gi], s=70, color=c, zorder=6, edgecolor='k', lw=0.6)
   b.annotate(f'{lab}\n$\\Gamma={gi:.4f}$', (ei, gi), (1.9, 0.12 - 0.3 * i), fontsize=8.5, color=c, va='
       center',
            arrowprops=dict(arrowstyle='-', color=c, lw=0.7, relpos=(0, 0.5)))
b.set_xlabel('rapidity $\\varepsilon=\\ln z$'); b.set_ylabel('$\\Gamma$')
b.set_title('$\\Gamma=\\frac{z-1}{z+1}=\\tanh\\frac{\\varepsilon}{2}$: one function, every boundary ratio',
    fontsize=12)
b.set_xlim(-6, 6); b.set_ylim(-1.25, 1.25)
plt.tight_layout(); plt.savefig('../figures/fig04_reflection_chart.png', dpi=170)
print('Hawking points', dict(zip(x.tolist(), np.round(G, 6).tolist())))
print('gap points', dict(zip(r.tolist(), np.round(Gs, 6).tolist())))
print('readings', [(round(p[0], 6), round(float(np.tanh(p[0] / 2)), 6)) for p in pts])
```

Output:

```
Hawking points {0.25: 0.124353, 0.5: 0.244919, 1.0: 0.462117, 2.0: 0.761594, 4.0: 0.964028, 8.0: 0.999329}
gap points {0.125: -0.777778, 0.25: -0.6, 0.5: -0.333333, 0.75: -0.142857}
```

```
readings [(np.float64(1.386294), 0.6), (1.0, 0.462117), (np.float64(0.794457), 0.377575), (np.float64
    (0.693147), 0.333333)]
```

fig05_metric_signature.py

```
# SCRIPT: MATTER-DISK-FIG05 the opened plane's metric signature, drawn from the exact determinant
# det Q = -u^2 w^2 cos^2 k / (16 s^3), s = E^2 = v^2 + w^2 + 2 v w cos k - u^2 (u = 0.5, w = 1).
# Orange: s > 0, det < 0, Lorentzian. Blue: s < 0, negative definite. Dashed: cos k = 0, degenerate.
# Black: s = 0, where the formula is singular (exceptional points). Stars: critical points k = pi, v = w -+ u.
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap
u, w = 0.5, 1.0
K, V = np.meshgrid(np.linspace(0, 2 * np.pi, 800), np.linspace(0, 2, 600))
s = V ** 2 + w ** 2 + 2 * V * w * np.cos(K) - u ** 2
sign = np.where(s > 0, 1, 0)
fig, ax = plt.subplots(figsize=(8.6, 5.6))
ax.pcolormesh(K, V, sign, cmap=ListedColormap(['#4fa3c7', '#e07b39']), shading='auto')
ax.contour(K, V, s, levels=[0], colors='k', linewidths=1.6)
for kk in (np.pi / 2, 3 * np.pi / 2): ax.axvline(kk, color='k', ls='--', lw=1)
ax.scatter([np.pi, np.pi], [w - u, w + u], marker='*', s=260, color='k', zorder=5)
ax.text(0.15, 1.85, 'Lorentzian: $s>0$, $\\det Q<0$', fontsize=10, color='white', weight='bold')
ax.text(np.pi - 0.42, 1.0, 'negative\ndefinite\n$s<0$', fontsize=9, color='white', weight='bold', ha='center')
ax.text(np.pi / 2 + 0.05, 0.08, '$\\cos k=0$: degenerate', fontsize=8.5, color='k', rotation=90, va='bottom')
ax.set_xlabel('momentum $k$'); ax.set_ylabel('hopping $v$ ($w=1$, $u=0.5$)')
ax.set_title('$\\det Q=-\\frac{u^2w^2\\cos^2k}{16\\,s^3}$, $s=E(k)^2$', fontsize=13)
plt.tight_layout(); plt.savefig('../figures/fig05_metric_signature.png', dpi=170)
cells = np.mean((s > 0) & (np.abs(np.cos(K)) > 1e-9))
print('Lorentzian fraction of this grid', round(float(cells), 4))
```

Output:

```
Lorentzian fraction of this grid 0.9355
```

fig06_records_and_futures.py

```
# SCRIPT: MATTER-DISK-FIG06 equal present records, different futures.
# Left: R3 / M1. Weights w0 = (1,1,1,1)/4 and w1 = (3,7,1,5)/16 on levels (-3,-1,1,3) share moments 0,1,2;
# retained survival |sum w e^{-i lambda t}|^2 differs; the reflected weights (5,1,7,3)/16 give the same
# intensity as w1 at every t (intensity-only readout erases the distinction).
# Right: K2 / K4. Equal density, flow and stress; free streaming separates density as -16 a eta rho sin^3(k v0
     t)
# at sin(kx) = 1; a pre-release velocity reset multiplies the contrast by exp(-nu s).
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
lam = np.array([-3, -1, 1, 3.])
W = {'uniform $(1,1,1,1)/4$': np.ones(4) / 4, '$(3,7,1,5)/16$': np.array([3, 7, 1, 5]) / 16, 'reflected $
    (5,1,7,3)/16$': np.array([5, 1, 7, 3]) / 16}
t = np.linspace(0, np.pi, 600)
fig, (a, b) = plt.subplots(1, 2, figsize=(13, 4.9))
sty = ['-', '-', ':']; col = ['#1b6f8a', '#c0392b', '0.1']
for (k, wv), s_, c in zip(W.items(), sty, col):
    amp = (wv[None, :] * np.exp(-1j * lam[None, :] * t[:, None])).sum(1)
    a.plot(t, abs(amp) ** 2, s_, color=c, lw=2 if s_ == '-' else 2.4, label=k)
    print(k, 'moments', [float((wv * lam ** m).sum()) for m in range(4)])
a.axvline(np.pi / 4, color='0.6', lw=0.8); a.text(np.pi / 4 + 0.03, 0.9, '$t=\\pi/4$: $0$ vs $1/32$', fontsize
    =9)
a.set_xlabel('$t$'); a.set_ylabel('retained survival $|a(t)|^2$')
a.set_title('Same spectrum, same $A$, same $BB^\\dagger=5$: different survival', fontsize=11.5)
a.legend(fontsize=9)
aa, eta, rho = 1 / 24, 0.5, 1.0
for ns_, ls in [(0, '-'), (0.5, '--'), (1.5, ':')]:
    dn = -16 * aa * eta * rho * np.exp(-ns_) * np.sin(t) ** 3
    b.plot(t, dn, ls, color='#7b5ea7', lw=2, label=f'reset before release, $\\nu s={ns_:g}$')
b.axhline(0, color='0.7', lw=0.8)
```

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b.scatter([np.pi / 2], [-1 / 3], color='k', zorder=5); b.annotate('$\\Delta n=-1/3$', (np.pi / 2, -1 / 3), (np
    .pi / 2 + 0.45, -0.335), fontsize=10)
b.set_xlabel('$k v_0 t$'); b.set_ylabel('$\\Delta n$ at $\\sin kx=1$')
b.set_title('Same density, flow and stress: density splits as $\\sin^3$', fontsize=11.5)
b.legend(fontsize=9, loc='lower left')
plt.tight_layout(); plt.savefig('../figures/fig06_records_and_futures.png', dpi=170)
for k, wv in W.items():
    amp = (wv * np.exp(-1j * lam * np.pi / 4)).sum(); print(k, 'survival at pi/4', round(abs(amp) ** 2, 12))
```

Output:

```
uniform $(1,1,1,1)/4$ moments [1.0, 0.0, 5.0, 0.0]
$(3,7,1,5)/16$ moments [1.0, 0.0, 5.0, 3.0]
reflected $(5,1,7,3)/16$ moments [1.0, 0.0, 5.0, -3.0]
uniform $(1,1,1,1)/4$ survival at pi/4 0.0
$(3,7,1,5)/16$ survival at pi/4 0.03125
reflected $(5,1,7,3)/16$ survival at pi/4 0.03125
```

fig07_exit_and_filter.py

```
# SCRIPT: MATTER-DISK-FIG07 the exit line and the instrument that reads it.
# Left: band-averaged Petermann factor <K> = 1 + u^2/(E(0)E(pi)) along v at u = 0.5, w = 1, with the
# continued endpoint phase theta = 2 arctan(u/sqrt(E(0)E(pi))) on the right axis; <K> = 2 exactly where
# theta = pi/2. Measured census exit 0.468 (s05a) marked.
# Right: the exit returned by the endpoint filter (rerun of the formal edition's audit, g_incoming_exit_filter
    ).
import numpy as np, ast, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
u, w = 0.5, 1.0
v = np.linspace(0.0, 0.499, 600)
p = np.sqrt((v + w) ** 2 - u ** 2) * np.sqrt((v - w) ** 2 - u ** 2)
K = 1 + u * u / p; th = 2 * np.arctan(u / np.sqrt(p))
vex = np.sqrt(w * w + u * u - np.sqrt(4 * w * w * u * u + u ** 4 * 0 + 0) * 0) if False else None
# solve E(0)E(pi) = u^2 exactly: ((v+w)^2-u^2)((v-w)^2-u^2) = u^4
from scipy.optimize import brentq
f = lambda x: ((x + w) ** 2 - u ** 2) * ((x - w) ** 2 - u ** 2) - u ** 4
vex = brentq(f, 0.3, 0.4999)
fig, (a, b) = plt.subplots(1, 2, figsize=(13, 4.9))
a.plot(v, K, color='#c0392b', lw=2, label='$\\langle K\\rangle=1+u^2/E(0)E(\\pi)$')
a.axhline(2, color='0.5', ls='--', lw=1); a.axvline(vex, color='k', lw=1)
a.text(vex - 0.005, 3.3, f'$\\langle K\\rangle=2$ at $v={vex:.4f}$', ha='right', fontsize=9.5)
a.set_ylim(1, 4); a.set_xlabel('$v$ ($w=1$, $u=0.5$)'); a.set_ylabel('$\\langle K\\rangle$', color='#c0392b')
a2 = a.twinx(); a2.plot(v, th / np.pi, color='#1f5f6b', lw=2, ls='-.', label='$\\theta/\\pi$, $\\theta=2\\
    arctan(u/\\sqrt{E(0)E(\\pi)})$')
a2.axhline(0.5, color='#1f5f6b', lw=0.6, ls=':'); a2.set_ylabel('$\\theta/\\pi$', color='#1f5f6b'); a2.
    set_ylim(0, 1)
a.set_title('$\\langle K\\rangle=\\sec^2(\\theta/2)$: the exit line is a quarter turn', fontsize=11.5)
h1, l1 = a.get_legend_handles_labels(); h2, l2 = a2.get_legend_handles_labels(); a.legend(h1 + h2, l1 + l2,
    fontsize=8.5, loc='upper left')
rows = [ast.literal_eval(x) for x in open('../outputs/g_incoming_exit_filter.txt') if x.startswith('{')]
for m, c, lab in [(0.4, '#7b5ea7', 'massive $m=0.4$'), (0.0, '#c08a2e', 'massless $m=0$')]:
    rr = [r for r in rows if r['m'] == m]
    b.plot([r['cut'] for r in rr], [r['exit_gamma'] for r in rr], 'o-', color=c, lw=2, label=lab)
    for r in rr: b.annotate(f"{r['mean_Petermann']:.3f}", (r['cut'], r['exit_gamma']), (r['cut'] * 0.75, r['
        exit_gamma'] + 0.012), fontsize=7.5, color=c)
b.set_xscale('log'); b.set_ylim(0.1, 0.51); b.invert_xaxis(); b.set_xlabel('endpoint filter $\\delta$ (levels
    with $|\\nu|$ or $|1-\\nu|<\\delta$ dropped)')
b.set_ylabel('returned exit $\\gamma$'); b.set_title('The filter is part of the instrument (labels: $\\langle
    K\\rangle$ at exit)', fontsize=11.5)
b.legend(fontsize=9)
plt.tight_layout(); plt.savefig('../figures/fig07_exit_and_filter.png', dpi=170)
print('v at <K>=2', vex)
```

Output:

```
v at <K>=2 0.46821319246223553
```

fig08_angular_chain.py

```
# SCRIPT: MATTER-DISK-FIG08 the angular gas chain L = -i k M - kappa (I - P0), k = 1.
# Left (ANG1): reference impedance Z_b(s) = 1/g0(s) - s, g0 = arctan(k/s)/k, tends to 2k/pi as s -> 0+.
# Middle (ANG2): isolated real pole s* = k cot(k/kappa) - kappa, right of the cut at s = -kappa, born at kappa
      = 2k/pi.
# Right (ANG3): |(L^q)_{2,l}| is zero below q = l - 2: a planted moment reaches stress no sooner than order l
     - 2.
import numpy as np, math, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
k = 1.0
fig, (a, b, c) = plt.subplots(1, 3, figsize=(14.5, 4.4))
s = np.logspace(-4, 1.2, 400); Z = 1 / (np.arctan(k / s) / k) - s
a.semilogx(s, Z, color='#1b6f8a', lw=2); a.axhline(2 * k / np.pi, color='#c0392b', ls='--', lw=1)
a.text(2e-4, 2 / np.pi - 0.05, '$2k/\\pi=0.6366$', color='#c0392b', fontsize=10)
a.set_xlabel('$s$'); a.set_ylabel('$Z_b(s)$'); a.set_title('ANG1: free-streaming impedance', fontsize=11)
kap = np.linspace(2 / np.pi + 1e-4, 3, 400); ps = k / np.tan(k / kap) - kap
b.plot(kap, ps, color='#7b5ea7', lw=2, label='pole $s^*=k\\cot(k/\\kappa)-\\kappa$')
b.plot(kap, -kap, color='0.4', ls=':', lw=1.5, label='branch point $s=-\\kappa$')
b.axvline(2 / np.pi, color='#c0392b', ls='--', lw=1); b.text(2 / np.pi + 0.04, -2.6, '$\\kappa=2k/\\pi$',
     color='#c0392b')
b.set_xlabel('$\\kappa$'); b.set_ylabel('$s$'); b.set_title('ANG2: the pole leaves the cut at $\\kappa=2k/\\
     pi$', fontsize=11); b.legend(fontsize=8.5)
n = 20; cc = np.array([(l + 1) / math.sqrt((2 * l + 1) * (2 * l + 3)) for l in range(n - 1)])
M = np.diag(cc, 1) + np.diag(cc, -1); L = -1j * k * M
img = np.zeros((9, 7))
for qi, q in enumerate(range(9)):
    Lq = np.linalg.matrix_power(L, q)
    for li, l in enumerate(range(3, 10)):
        img[qi, li] = abs(Lq[2, l])
c.imshow(np.log10(img + 1e-18), origin='lower', cmap='magma', vmin=-6, vmax=0, aspect='auto', extent=[2.5,
     9.5, -0.5, 8.5])
c.plot(range(3, 10), [l - 2 for l in range(3, 10)], 'w--', lw=1.2)
c.set_xlabel('planted moment $l$'); c.set_ylabel('order $q$ of $L^q$'); c.set_title('ANG3: $|(L^q)_{2,l}|$,
     zero below $q=l-2$', fontsize=11)
plt.tight_layout(); plt.savefig('../figures/fig08_angular_chain.png', dpi=170)
print('Z_b(1e-4)', 1 / (math.atan(k / 1e-4) / k) - 1e-4, '2k/pi', 2 * k / math.pi)
```

Output:

```
Z_b(1e-4) 0.6365603034211902 2k/pi 0.6366197723675814
```